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A Quick Approach to Statistics

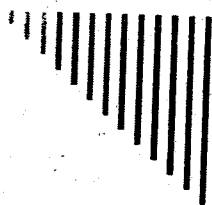
with Questions & Answers



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Chapter 1

Introductory Statistics

Basic Statistics

Statistics

Statistics is the collection of methods for planning experiments, obtaining data and then organizing, summarizing, presenting, analyzing, interpreting and drawing conclusions based on data.

Statistical Methods

Statistical methods are those ways that are used to collect, present, analyze, and interpret quantitative data.

Type of Statistics

There are two major types of Statistics: Descriptive Statistics and Inferential Statistics:

Descriptive Statistics

It consists of methods for organizing and summarizing information in a presentable and effective way.

Inferential Statistics

It consists of methods of drawing conclusions about a population based on information obtained from a sample of the population.

Data

A collection of facts from which conclusions may be drawn is referred as data.

Observation

Any sort of recording of information is called observation.

Types of Data

Generally, data can be classified by their nature and way of collection.

Types of Data (Nature)

Qualitative Data

Qualitative (or Categorical or Attribute) data can be separated into different categories that are distinguished by some non-numerical characteristics. For example, gender of person, blood type, and eye color etc.

Quantitative Data

Quantitative data consist of numbers representing counts or measurements such as number of patients in a hospital, ages of a group of persons, data about height and weight of individuals etc.

Quantitative data can be further classified into discrete and continuous data. All type of count data are referred as discrete data where measured data are referred as continuous data.

Discrete Data

Data obtained by categorizing subjects so that there is a distinct interval between any two possible values e.g., number of patients in a hospital and number of chairs in a room etc.

Continuous Data

Continuous data result from infinitely many possible values that can be associated with points on a continuous scale in such a way that there are no gaps or interruptions. For example, data about height and weight of individuals etc.

Types of Data (Collection)

Primary Data

The data collected directly from people and organizations via questionnaires or surveys before being analyzed to reach conclusions concerning the issues covered in the questionnaire or survey.

Secondary Data

The data that have undergone any sort of treatment by statistical methods. In other words, the data that have already been assembled, having been collected for some other purpose, are referred as secondary data. Sources include census reports, trade publications, and subscription services.

Scales of Measurement

Nominal Scale

In nominal scale is categorized by data that consists of names, labels or categories only. Such data cannot be arranged in an ordering scheme. For example, gender; "male" and "female", response; "yes" or "no", etc.

Ordinal Scale

The ordinal scale involves data that may be arranged in some order but differences between data values either cannot be determined or are meaningless. For example, in a sample of 36 stereo speakers, 12 were rated good, 16 were rated better and 8 were rated best.

Interval Scale

The interval scale is like the ordinal scale with the additional property that meaningful amount of differences between data can be determined. However, there is no inherent (natural) zero starting point. Interval scales take the notion of ranking items in order one step further, since the distance between adjacent points on the scale are equal. For instance, the Fahrenheit scale is an interval scale, since each degree is equal but there is no absolute zero point. This means that although we can add and subtract degrees (100° is 10° warmer than 90°), we cannot multiply values or create ratios (100° is not twice as warm as 50°). What is important in determining whether a scale is considered interval or not is the underlying intent regarding the equal intervals: although in an IQ scale, the intervals are not necessarily equal (e.g. the difference between 105 and 110 is not really the same as between 80 and 85), behavioral scientists are willing to assume that most of their measures are interval scales as this allows the calculation of averages – mode, median and mean –, the range and standard deviation.

Ratio Scale

The ratio scale is the interval scale modified to include the inherent zero starting point. For values at this level, differences and ratios are meaningful. Ratio scales are the most sophisticated of scales, since it incorporates all the characteristics of nominal, ordinal and interval scales. As a result, a large number of descriptive calculations are applicable such as when respondents are asked for their age, height, income etc.

Presentation of Data

Classification

A classification is the separation or ordering of objects into classes where classes are categories for grouping data.

Tabulation

Tabulation is placement of data into rows and columns with suitable heads and subheads.

Frequency

The frequency of a particular class is the number of original scores that fall into that class. Simply, frequency is the number of times that a repeated observation occurs.

Cumulative Frequency

The cumulative frequency for a class is the sum of the frequencies of that class and all the previous classes.

Relative Frequency

The relative frequency of a particular class can be found by dividing by the class frequency by the total of all frequencies.

Grouped data

The data presented in the form of a frequency distribution are called grouped data.

Frequency Distribution

The division of counts (frequencies) of number of scores that fall into each class (category) is called frequency distribution. In other words, a listing of classes and their frequencies is called frequency distribution. A table that represents classes along with their respective class frequencies is called frequency table.

Class Limits

The values or numbers specifying a class are called class limits. The smallest value specifying a class is called lower class limit while the largest value for specifying a class is called upper class limit.)

Class Boundaries

Class boundaries are the numbers used to separate classes, but without the gaps created by class limits. They are obtained by increasing the upper class limits and decreasing the lower class limits by the same amount so that there are no gaps between consecutive classes. These boundaries are also called precise limits or true limits.

Class Mark (Midpoints)

Class marks or midpoints of the classes are obtained by adding lower class limits to the corresponding upper class limits and dividing by 2.

Class Interval (Width)

Class interval or class width is the difference between two consecutive lower class limits or two consecutive lower class boundaries.

In the table below, class interval (width) = $196 - 191 = 201 - 196 = 5$.

Cholesterol Level (Class Limits)	Class Boundaries	Class Mark	Frequency	Cumulative Frequency	Relative Frequency
191 – 195	190.5 – 195.5	193	1	1	$1/25 = 0.04$
196 – 200	195.5 – 200.5	198	3	$3+1 = 4$	$3/25 = 0.12$
201 – 205	200.5 – 205.5	203	4	$4+4 = 8$	0.16
206 – 210	205.5 – 210.5	208	7	$7+8 = 15$	0.28
211 – 215	210.5 – 215.5	213	5	$5+15 = 20$	0.20
216 – 220	215.5 – 220.5	218	4	$4+20 = 24$	0.16
221 – 225	220.5 – 225.5	223	1	$1+24 = 25$	0.04
Total			25		1.00

Graph

A drawing representing the relationship between data sets is called the graph.

Histogram

A graph that displays the classes on the horizontal axis and the frequencies of the classes on the vertical axis is called histogram. The frequency of each class is represented by a vertical bar whose height is proportional to the frequency of that class.

Frequency Polygon

Polygon means closed shape. A frequency polygon is obtained by joining the mid points of the adjacent bars of histograms with straight lines and then joining the both ends with X -axis by assuming class frequencies zero at those points.

Frequency Curve

When frequency polygon is constructed for large numbers of observations and small class intervals, a smoothed curve can be approximated that is referred as frequency curve.

Cumulative Frequency Polygon (Ogive)

A cumulative frequency polygon or ogive is obtained by plotting cumulative frequencies of classes along Y -axis while upper class boundaries along X -axis.

Charts

It is the plotting data and showing results of a process over a period of time (day, month, etc.)

Diagram

A diagram is a simplified and structured visual representation of concepts, ideas, constructions, relations, statistical data, anatomy, etc. used in all aspects of human activities to visualize and clarify the topic.

Measures of Central Tendency

Central Tendency

The general level, characteristic, or typical value that is representative of the majority of cases is referred as central or average value in general. The tendency of observations to gather around the central part of data is called central tendency.

Measure of Central Tendency

A measure of central tendency is a value at the center or middle of a data set.

Arithmetic Mean

It is a type of average (measure of central tendency), which is defined as the sum of all the values in a set of numerical data divided by total number of observations in that data set. This is the most commonly used measure of central tendency and is simply called Mean. It is labeled as either μ (lowercase Greek letter "mu") to denote a population mean or $\bar{X}$ (X -bar) to denote a sample mean.

Weighted Mean

An average of means calculated by weighting each individual mean according to the number of data points that made up that individual mean.

Geometric Mean

A mean of n objects that is computed by taking the n -th root of the product of the n terms. A measure of the central tendency of a data set that minimizes the effects of extreme values.

Harmonic Mean

It is reciprocal of the mean of reciprocal values.

Median

The median of a set of scores is the middle value when the scores are arranged in order of increasing (or decreasing) magnitude. The median is often denoted by $\tilde{X}$ (X -tilde).

Mode

The mode is the value that has the largest frequency in a data set. When two scores occur with the same greatest frequency, each one is Mode and the data set is called bimodal. When a data set has more than two modes, it is called multimodal.

Empirical Relation among Mean, Median, and Mode

There is an empirical relation between mean, median and mode that is

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean.}$$

Quartile

A quartile is any of the 3 values which divide the sorted data set into 4 equal parts, so that each part represents 1/4th of the sample or population. It is denoted by Q_i ($i = 1, 2, 3$). The second quartile is, obviously, equal to median.

Deciles

A decile is any of the 9 values which divide the sorted data set into 10 equal parts, so that each part represents 1/10th of the sample or population. It is denoted by D_i ($i = 1, 2, \dots, 9$). The 5th decile is, obviously, equal to median.

Percentile

A percentile is any of the 99 values which divide the sorted data set into 100 equal parts, so that each part represents 1/100th of the sample or population. It is denoted by P_i ($i = 1, 2, \dots, 99$). The 50th percentile is, obviously, equal to median.

Measures of Dispersion

Dispersion

Dispersion indicates variation while a descriptive measure that indicates the amount of variation in a data set is called measure of dispersion.

Range

The range is the length of the smallest interval which contains all the data. It is defined to be the difference between the largest and the smallest value of a data set.

Mean Deviation

The absolute deviation of an element of a data set is the absolute difference between that element and a given point (usually mean or median). The average of such deviations is called mean deviation.

Quartile Deviation

Quartile deviation is the half of the difference between first and the third quartile.

Variance

A measure of the variation shown by a set of observations and defined as the mean of the squares deviations of all the observations from their mean. It is usually denoted by σ^2 (sigma-square) for population and s^2 for sample.

Standard Deviation

The positive square root of the variance (defined above).

Coefficient of Variation (CV)

It is used to describe the standard deviation relative to mean. It is expressed in percentage. It allows us to compare variability of data sets with different measurement units (such as centimeters versus minutes).

Z-Score (Standard Score)

It is the number of standard deviations that a given value x is above or below the mean. It is simply referred as the standardized variable.

Moments

A moment designates the power to which deviations are raised before averaging them. Moments determine the shape and location of a distribution.

Skewness

A distribution of data is symmetric if the left half of its histogram is roughly a mirror image of its right half. The lack of symmetry or departure from symmetry is called skewness. A skewed distribution extends more to one side than the other. If it has longer right tail, it is called positively skewed distribution. If it has longer left tail, it is called negatively skewed distribution.

It is important to note that

Mean = Median = Mode; Symmetric Distribution.

Mean $\geq$ Median $\geq$ Mode; Positively Skewed

Mean $\leq$ Median $\leq$ Mode; Negatively Skewed

A measure to describe the degree of skewness is called coefficient of skewness.

Kurtosis

Kurtosis describes the extent to which a frequency distribution of scores is bunched around the center or spread toward the endpoints. Simply speaking, it measures the degree of peakedness or flatness of a unimodal distribution.

A measure to describe the degree of kurtosis is called coefficient of kurtosis.

Questions-Answers

Q.1 *What is the main objective of Statistics while applying on numerical data?*

Ans. *The main objective of Statistics is summarization of numerical data.*

Q.2 *What is meant by interpretation of data?*

Ans. *Interpretation of the data means drawing of conclusions from the analysis and comparisons made with the data.*

Q.3 *Can accurate data be misused? Explain.*

Ans. *Certainly, any treatment of accurate data that are not comparable as if they were comparable is misused. Any example would be a comparison of the annual number of automobile accidents in the Punjab and Sindh to determine which province has the more careful or better driver. Also, presentation of only part of the facts could lead to the misuse.*

Q.4 *Men have more auto accidents than women; therefore women are better drivers than men. True or False? Explain.*

Ans. *This is false because more men than women drive cars and for the same span of time, say one year, the total hours driven by men are more than by the females; therefore, men would have more accidents even if they are equally good drivers. But if we take average number of road accidents per average number of driven hours for both the genders then the situation may be comparable.*

Q.5 *Why is the presentation of the data necessary?*

Ans. *Presentation of the data refers to a meaningful and orderly assembly and display of data so that the significant attributes of the data can be determined.*

Q.6 *Name three Pakistani Government agencies that are good source of the data.*

Ans. *Some that might be named are, Federal Bureau of statistics, Punjab Bureau of Statistics, Sindh Bureau of Statistics, NWFP Bureau of Statistics, Balochistan Bureau of Statistics, Pakistan Census Organization, Agriculture Census Organization etc.*

Q.7 *Why should an analyst know the accuracy of the data he plans to analyze?*

Ans. *If the analyst does not know the accuracy of his data he cannot trust any analysis and interpretation that are made with them.*

Q.8 *Why do we construct a histogram?*

Ans. *We construct histogram in order to assess the shape of the frequency distribution (to assess the degree of skewness and kurtosis)*

Q.9 *Discuss the advantages of the polygon over the histogram.*

Ans. *A polygon provides the true shape of a distribution better than a histogram. A polygon is easier to construct and two polygons can be compared on the same chart.*

Q.10 *In what ways a frequency polygon is different from a histogram?*

Ans. *Most commonly representations of the frequency distribution take the form of the frequency polygon, which is a line graph instead of the bar-type graph of the histogram. It sketches an outline of the data pattern more clearly.*

Q.11 *On what kind of data, the use of Arithmetic mean is most suitable?*

Ans. *If there is less variation in data (likely to be homogeneous) and the observations are equally weighted then Arithmetic Mean is the most suitable measure of central tendency. In other words, if all the observations have same importance or there are no extreme values (outliers), Arithmetic Mean is preferable.*

Q.12 *Give chief characteristics of arithmetic mean, median, mode, geometric mean and harmonic mean.*

Ans. *Some important characteristics of averages are:*

Arithmetic Mean:

- *Each value in the group to be averaged directly influences the magnitude of the arithmetic mean.*
- *The arithmetic mean is subject to algebraic manipulation; for the number of values from which it was computed will give their sum.*

- *The arithmetic mean can be computed even if all that is known is the total of a group of figures and the number of figures in the group.*

Median:

- *The median is a positional average, being the center figure in an arrayed list of figures. After it has been determined, its value cannot be changed, changing values of the other figures, unless the changes result in placing a new figure at the center of the list.*
- *The median cannot be treated algebraically like the arithmetic mean, geometric mean and harmonic mean.*

Mode:

- *The mode, being that figure which appears most often in a group, is not changed in value by changes in the other figures.*
- *The mode cannot be treated algebraically like the arithmetic mean, geometric mean and harmonic mean.*

Geometric Mean:

- *All of the values being averaged directly affect the magnitude of the geometric mean.*
- *The geometric mean is subject to algebraic manipulation.*
- *The geometric mean gives the large values less weight than arithmetic mean; therefore it will be smaller than the arithmetic mean of the same figures.*
- *The geometric mean will be zero if any of the values in the series to be averaged is zero.*
- *When negative values are used, the geometric mean does not exist.*

Harmonic Mean:

- *Each figure being averaged directly affects the magnitude of the harmonic mean.*
- *The harmonic mean gives less weight to the larger figures than the arithmetic and geometric means; therefore, it will be smaller than these means when all three are computed from the same figures.*
- *The harmonic mean of a set of ratios may be appropriately calculated if the numerators of the fractions from which the ratios were computed are the same.*

- If any of the values of the variable is zero, the harmonic mean cannot be determined.

Q.13 Under what general conditions each of the five averages the most appropriate to use for a given group of figures?

Ans. General conditions under which the five averages would be used are;

- Arithmetic mean used as an average when the group of figures is quite homogeneous.
- Median used as an average of data that are highly skewed.
- Mode used when it is a highly predominate figure in a group.
- Geometric Mean used primarily as an average of ratios of change.
- Harmonic Mean used when the numerators of the fractions used to compute the ratios were the same or nearly so.

Q.14 Why cannot the arithmetic mean be computed from an open-end frequency distribution?

Ans. An arithmetic mean cannot be computed from an open-end frequency distribution because the midpoints of the open-end classes cannot be determined but which are needed for the computation.

Q.15 Why an average computed from a frequency distribution is not exactly the same as that computed from the original figures used to construct the distribution?

Ans. The individual values are lost in the groups therefore an average computed from a frequency distribution is an estimate of the average from the same data in the ungrouped form. In this case, we assume that only the mid-points have the relevant class frequencies rather than original values so it makes difference in the average calculation for grouped and ungrouped data. This difference is called Grouping Error. However, in most cases the estimates are accurate enough to be most useful.

Q.16 When computed from the same data, which of the following averages will be the largest, second largest, and the smallest: arithmetic mean, geometric mean, and harmonic mean? Why?

Ans. The arithmetic mean would be the largest and the harmonic mean would be smallest. The geometric mean would be the smaller than the arithmetic mean because it gives less weight to the larger values. The harmonic mean, however, gives less weight to the larger values than does the geometric mean.

Q.17 Why is the median usually a better average for a highly skewed frequency distribution than the arithmetic mean?

Ans. The extremes in the highly skewed distribution will distort or bias an arithmetic mean in their direction. The median, however, being a positional average will not receive the same bias.

Q.18 In order to use the formula for the mode:

$$\text{Mode} = L + \frac{(f_m - f_o)}{(f_m - f_o) + (f_m + f_1)} \times h$$

Why must the interval of the modal class and the two adjacent classes be the same?

Ans. In the mode formula the h represents the interval, which is common to the modal class, and the classes that immediately precede and follow the modal class.

Q.19 If your chairman of the department asked you to plan an evening for the sons of the department employees and told you that the average ages of the sons was 14, what more would you want to know about the ages before you started planning?

Ans. One would need information on the variations in the ages of the sons. If their ages all are in the range 13-15, the entertainment plan would be entirely different than it would be if the ages were distributed from ages 6 to 18.

Q.20 A man who stated that he manufactured lid for the glass jars was asked what size lids he manufactured. He replied, "the typical size". To which average was he referring?

Ans. This would be the mode. The arithmetic mean size and median size may not fit any glass jar.

Q.21 Could a man six feet tall drown while crossing a river with an average depth of two feet?

Ans. Of course, the question was asked to get a little humor into a sometimes dry subject. The question should make a student realize how misleading an average can be when nothing is known about the values that makeup the average.

Q.22 In 2003, the mean (arithmetic) monthly income of the families in a particular city was Rs. 10,900. The median family income was Rs. 18,200. If you were planning a large sales campaign in that particular city for the product that would be purchased by families in the middle and upper-middle income groups, which of the two averages would be more useful? Why?

Ans. The median would be because one knows that one-half of the families had incomes of the Rs. 18,200 or more and would therefore have some estimate of the size of the market. The mean of Rs. 10,900 cannot indicate a market size at all because nothing is known about the values that are averaged.

Q.23 Why do we need measure of dispersion even when we have measure of central tendency?

Ans. The measure of central tendency, such as mean, median, mode etc., do not reveal the whole picture of the distribution of a data set. Two data sets with the same mean may have completely different spreads. Therefore we need measure of dispersion, in addition to measure of central tendency, in order to assess the clear picture of the data.

For example, consider two groups of students having marks in a class test as follows:

Group-I: 12, 5, 10, 8, 3, 22

Group-II: 10, 8, 9, 11, 12

Both groups have equal mean marks i.e., 10 but obviously, in Group-I, there is more variation in marks. In this situation, just mean is not enough to evaluate the group performance.

Q.24 For small sample size of 5 or less, which measure of dispersion is preferable; Range or Standard Deviation?

Ans. For small samples like of size 5, range and standard deviation give the same interpretation but due to easy computations, Range is preferred in such cases.

Q.25 Why is the measurement of the dispersion computed for a frequency distribution not equal to that computed for the original figures from which the distribution was constructed?

Ans. Due to same Grouping Error as discussed in answering for above Q-15.

Q.26 When computed from the same data, which will be larger, the mean deviation or the standard deviation? Why?

Ans. The standard deviation. Because it gives more weight to the large values.

Q.27 Which measure of the dispersion, is most widely used?

Ans. The standard deviation is used more than any other measure because of its application to the analysis of the sample data.

Q.28 How can we compare the variation of the two or more distributions if: (i). the arithmetic means of the distribution are the same? (ii). if they are different?

Ans. When the arithmetic means of the two or more distributions are the same, a comparison of absolute measures of dispersions will enable one to see which is the most varied. If the arithmetic means differ, measures of relative dispersion must be compared.

Q.29 What are the measures of relative dispersion? How are they used?

Ans. Measures of the relative dispersion are absolute measures of dispersion expressed as a fraction or percent of some base, usually the arithmetic mean. They are used to compare the variations of two or more sets of the data that are expressed in the different units or two or more sets of data expressed in the same units but which have different arithmetic means.

Q.30 Define mesokurtic, platykurtic, and leptokurtic.

Ans. Mesokurtic means middle peakedness (between flat and highly peaked, usually called normal), platykurtic means "flat topped" and leptokurtic means highly peaked.

Exercises

Exercise 1.1 (MCQs')

- Q.1** Statistics is used in the situations of
- (a) uncertainty
 - (b) dealing aggregate data
 - (c) variability
 - (d) All three (a), (b) & (c)
- Q.2** Indicate the following for what the type of the data described below is nominal:
- (a) Team scores in a cricket match.
 - (b) Daily temperatures in degree Celsius.
 - (c) Room numbers in the Holiday Inn hotel.
 - (d) Identification of the children who have chicken pox.
- Q.3** Indicate the discrete variable for the variables given below:
- (a) Batting average of the Pakistan cricket team.
 - (b) Number of children of each of 1,000 married graduates of a public university.
 - (c) Average heights of the students of the 1st year class.
 - (d) Daily hours of the sunshine during the period from September 21 to December 21.
- Q.4** Indicate for which of the following data, the shape of the curve will be normal?
- (a) A computer printout shows the current checking account balances for all the checking customers of the national bank.
 - (b) Diagnostic reading test scores are tabulated for all the low graders in a school of a district.
 - (c) Gifted students in a creative writing class are given the verbal subtest of an intelligent test.
 - (d) A nation wide mathematics exam is given to 1,000 students at a college with a selective admission policy.

- Q.5 Indicate for which of the following set of computations, the distribution to be approximately positively skewed?
- (a) Mean is 79.3, median is 75.4 and mode is 72.
 - (b) Mean is 25.6, median is 24.9 and mode is 24.
 - (c) Mean is 128.74, median is 12.68 and mode is 135.
 - (d) Mean is 50.3, median is 49.6 and mode is 50.

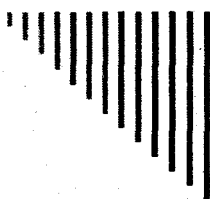
Exercise 1.2 (True/False)

Read the following statements carefully and indicate which statement is "True" or "False":

1. Any one who assembles numerical data is a Statistician.
2. As a general rule more detailed descriptions of data can be found in the primary source than in a secondary source.
3. Discrete data can be expressed only in the whole numbers.
4. Continuous data are just ratio-scale data.
5. The class width of a frequency distribution is of equal size.
6. The class intervals of the frequency distribution should ordinarily be equal to either 10 or 5.
7. The more classes in the frequency distribution the better the data are summarized.
8. Open-ended frequency distribution should always be avoided.
9. The sum of the class frequencies is equal to the number of observations made.
10. Relative frequencies can never be greater than 1.
11. Cumulative frequencies never decrease as the corresponding observed values increases.
12. A frequency polygon touches the horizontal axis just once.
13. If we arrange the observations in a data set from highest to lowest, the data point lying in the middle is the median of the data set.
14. The value most often repeated in the data set is called the arithmetic mean.
15. Extreme values in a data set have a strong effect on the median.

16. The difference between the highest and the lowest observations in a data set, called the quartile range.
17. The range is more stable measure of variability than the inter-quartile range.
18. The median is less influenced by chance than the mode because the median take into account the entire distribution.
19. In a symmetric unimodel distribution, the arithmetic mean is equal to the median.
20. The standard deviation is the square of the variance.
21. If the data of an experiment is measured in meters, then the standard deviation is reported as meters squared.
22. The variance, like the standard deviation, takes into account every observation in the data set.
23. The coefficient of variation is an absolute measure of dispersion.
24. When calculating the average rate of debt expansion for a company, the correct mean to use is the arithmetic mean.
25. The geometric mean is an average.
26. When there are a few extremely large items in the series being averaged the geometric mean will be the larger than the arithmetic mean.
27. The median can be estimated from a cumulative frequency curve.
28. The reciprocal of the harmonic mean is equal to the arithmetic mean.
29. In symmetrical distribution, the mean, median, and mode are equal; provided that the mode is a single number.
30. The 50th percentile is equal to the median only in the symmetrical samples.

31. For a positive skewed distribution, the mode must be smaller than the median.
32. The quartile deviation is a measure of a scatter of the first half the data.
33. A leptokurtic curve is more skewed than a platykurtic curve.
34. The mean deviation cannot be computed from an open-ended frequency distribution.
35. One cannot determine how much a distribution is varied by its standard deviation alone.
36. The coefficient of variation is a useful measure for comparing degree of skewness in two or more frequency distributions.
37. Histograms are better than polygons for comparing two frequency distributions on the same chart.
38. The frequency polygon is a special type of the time series line chart.
39. Charts should usually be higher than they are wide.
40. Diagrams are less accurate than tables.



Chapter 2

Basic Probability Theory

Basics in Probability

Experiment

The process of obtaining an observation is called an experiment. Hitting a target, checking the boiling point of a liquid, taking examination for a student, conducting interviews for some jobs, tossing of a coin, rolling of a die, hitting a ball of a batsman, sale of some products, chemical reaction of elements, are few examples of experiments.

Trial

A single performance of an experiment is called trial. If a batsman plays a single ball, if a bowler bowls, if a student solves a single question, single rolling of a die, all these are the examples of a trial.

Outcome

An outcome is the result of an experiment. Each possible distinct result of an experiment is referred as outcome. Hitting or not hitting a target, making some scores, leaving a ball or being out for a batsman, boiling of water on 100°C are the examples of outcomes relevant to the experiments discussed above.

Random Experiment

An experiment is called random experiment if its outcomes cannot be predicted in advance even if it is performed under similar conditions. Any random experiment has the following properties:

- (i) It has at least two outcomes.
- (ii) The number of all possible outcomes is known in advance.
- (iii) It can be repeated any number of times under similar conditions.

Among above examples of experiments, hitting a target, taking examination for a student, tossing of a coin, rolling of a die, hitting a ball of a batsman, sale of some product are examples of Random Experiments while checking the boiling point of a liquid and chemical reaction of elements will be nonrandom experiments.

Sample Space

The set of all possible outcomes of a random experiment is called a sample space. It is usually denoted by Ω (omega).

In the random experiment in which a student takes a examination, suppose the result of examination can be in form of grades, 'A', 'B', 'C', 'D' and 'F' then $\Omega = \{'A', 'B', 'C', 'D', 'F'\}$.

In rolling a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$.

Event

Any subset of sample space is called event. It is a collection of outcomes of an experiment. Events may be either simple or composite. Formally, if an event consists of single sample point of a sample space, it is called elementary or simple event and in case of two or more sample points, it is referred as composite event.

Simple Events: A student passes an examination, a batsman makes shot for six, a die show a number 2, etc.

Composite Event: A motor accident for rash driving and failure of brakes, a ball results in 1 run and a run-out during an over in a cricket match, etc.

Event Space

A set of all events relevant to a sample space is called event space. It is usually denoted by $\mathcal{A}$.

Equally Likely Events

Two events A_1 and A_2 relevant to a sample space are said to be equally likely if their probabilities of occurrences are equal. In other words, if event A_1 is as likely to occur as A_2 does then A_1 and A_2 are said to be equally likely events.

Mutually Exclusive or Disjoint Events

Two events A_1 and A_2 relevant to a sample space are said to be mutually exclusive or disjoint if they cannot occur simultaneously. Thus tossing of a coin is associated with mutually exclusive events. If the event 'head' occurs, the event 'tail' cannot occur at the same time. In above example of grading

of a student in an examination, if all the five grades refer to five different events then they can not occur simultaneously as a student can take, obviously any one of the grade, so these events are also disjoint.

Mathematically, if A_1 and A_2 are two mutually exclusive events then

$$P(A_1 \cap A_2) = 0.$$

Collectively Exhaustive Events

The events $A_1, A_2, \dots, A_k$ are said to be collectively exhaustive if all A_i 's ($i = 1, 2, \dots, k$) are Disjoint and $A_1 \cup A_2 \cup \dots A_k = \Omega$.

In above example of grades, all the five events are collectively exhaustive.

Complementary Events

Complementary event for an event A is the event that A does not occur. For event A it is denoted by A' or A^c . For example, passing of a student is complementary event to the failing of that student.

Independent Events

Two events A_1 and A_2 relevant to a sample space are said to be statistically independent if the occurrence of A_1 does not affect the probability of occurrence or non-occurrence of A_2 .

Symbolically,

$$P(A_1 \cap A_2) = P(A_1) \cdot P(A_2).$$

The passing (or failing) of one student is statistically independent to the passing or failing of other student(s) in the same examination, score on a current ball is statistically independent to the result of previous ball in a cricket match, are the examples of statistically independent events.

Probability

Probability may be defined as the likelihood of the occurrence of an event. A probability provides a quantitative description of the likely occurrence of a particular event. In other words, it is a numerical measure of uncertainty. Probability is conventionally expressed on a scale from 0 to 1; a rare event has a probability close to 0, a very common event has a probability close to 1.

Subjective Probability

A subjective probability describes an individual's personal judgment about how likely a particular event is to occur. It is not based on any precise computations but is often a reasonable assessment by a knowledgeable person. A person's subjective probability of an event describes his/her

degree of belief in the event. For example, a cricket expert says that there are more than 80% chances that the team A will win the tournament. A planning minister guesses that at least $3/4$ villages of the country will be supplied electricity by the end of next year.

Objective Probability

A probability that can be established theoretically or from historical data. The objective probability has following main approaches to define Probability:

- (i) *Classical (Priori) Definition of Probability*
- (ii) *Relative-Frequency Definition of Probability*
- (iii) *Axiomatic Definition of Probability*

Classical (Priori) Definition of Probability

If an experiment can produce n different mutually exclusive results all of which are equally likely, and if m of these results are considered favorable (or result in event A), then the probability of a favorable result (or the probability of event A) is m/n .

In tossing of coin, the probability of occurring 'head' or 'tail' and the probability of selection a female student among a group of 20 male and 30 female students, follow classical definition of probability.

Relative-Frequency Definition of Probability

This definition of probability states that if in n trials, an event A occurs n_A times, its probability, $P(A)$ is approximately n_A/n :

$$P(A) = n_A/n,$$

provided that n is sufficiently large and the ratio n_A/n is nearly constant as n increases.

The probability of having number 6 on an irregular die, the probability of survival of a 40 years old Hepatitis C patient, the probability for winning of a tournament by a team, are the examples of relative frequency definition of probability.

Axiomatic Definition of Probability

Let Ω be the Sample Space, the probability of an event A is, by definition, a number $P(A)$ assigned to A . This number satisfies the following three axioms:

- (i) $P(A)$ is a non-negative number; $P(A) \geq 0$.
- (ii) Probability of the event Ω (sure or certain event) is equal to 1;
 $P(\Omega) = 1$.

- (iii) If two events A_1 and A_2 have no common element (i.e. Mutually Exclusive) then

$$P(A_1 \cup A_2) = P(A_1) + P(A_2)$$

Conditional Probability

In many situations, once more information becomes available; we are able to revise our estimates for the probability of further outcomes or events happening. For example, suppose you go out for lunch at the same place and time every Friday and you are served lunch within 15 minutes with probability 0.9. However, given that you notice that the restaurant is exceptionally busy, the probability of being served lunch within 15 minutes may reduce to 0.7. This is the conditional probability of being served lunch within 15 minutes given that the restaurant is exceptionally busy. The effect of such information is to reduce the sample space by excluding some outcomes as being impossible which before receiving the information were believed possible.

Formally, the conditional probability of A given B is the probability of event A occurring, given that event B has already occurred. It can be found by dividing the probability of events A and B both occurring by the probability of event B as shown below

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

It is obvious to note that if A and B are statistically independent then

$$P(A|B) = P(A).$$

Some Probability Rules

Addition Rule of Probability

If A_1 and A_2 be any two events relevant to a sample space Ω , then

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$$

In words, the probability that either event A_1 or event A_2 or both events occur equals the probability that event A_1 occurs plus the probability that event A_2 occurs minus the probability that both occur.

Generally, for n events:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} P\left(\bigcap_{i=1}^n A_i\right)$$

Multiplication Rule of Probability

For any two events A_1 and A_2 relevant to a sample space Ω , the multiplication rule is a result used to determine the probability that two events, A_1 and A_2 , both occur. This rule follows the definition of conditional probability as

$$P(A_1 \cap A_2) = P(A_1) \cdot P(A_2 | A_1)$$

Generally, for n events:

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | A_1 \cap A_2) \dots \dots P(A_n | A_1 \cap A_2 \cap \dots \cap A_{n-1})$$

Law of Total Probability

Suppose, A_1 and A_2 be mutually exclusive and exhaustive events with non-zero probabilities then for any event B (with non-zero probability)

$$P(B) = P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2)$$

Generally, for n events:

$$P(B) = \sum_{i=1}^n P(A_i) \cdot P(B | A_i)$$

Baye's Theorem

Suppose, A_1 and A_2 be mutually exclusive and exhaustive events with non-zero probabilities then for any event B (with non-zero probability)

$$P(A_1 | B) = \frac{P(A_1) \cdot P(B | A_1)}{P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2)}$$

Generally, for n events:

$$P(A_i | B) = \frac{P(A_i) \cdot P(B | A_i)}{\sum_{i=1}^n P(A_i) \cdot P(B | A_i)}, \quad (i = 1, 2, \dots, n)$$

Counting Rules

Rule of Multiplication

If there are K procedures and i th procedure may be performed in n_i ways ($i = 1, 2, \dots, k$) then all the k procedures may be performed in $n_1 \times n_2 \times \dots \times n_k$ ways. For example, a person has 3 different pairs of shoes and 4 different pairs of socks then he may use all of these pairs in $3 \times 4 = 12$ different ways.

Rule of Addition

If there are K procedures and i th procedure may be performed in n_i ways ($i = 1, 2, \dots, k$) then the number of ways in which one can perform procedure 1 or procedure 2 ... or procedure k given by $n_1 + n_2 + \dots + n_k$ (assuming that one procedure can be performed one time or no two procedures can be performed together).

Suppose, a group of students is planning a trip and thinking about either bus or train to use for that. If there are 3 different routes available when using bus for the trip and 2 different routes for train then there are $3 + 2 = 5$ routes available for that trip.

Permutations

A permutation of n different objects taking r at a time ($0 \leq r \leq n$)

${}^n P_r = \frac{n!}{(n-r)!}$. In permutations, order of objects is important or meaningful.

For example, if one wants to calculate the ways in which 6 persons may be seated on a bench having a capacity of 4 seats then in this case, $n = 6$, $r = 4$

and answer is ${}^6 P_4 = \frac{6!}{(6-4)!} = 6 \times 5 \times 4 \times 3 = 360$.

Combinations

A combination of n different objects taking r at a time ($0 \leq r \leq n$)

${}^n C_r = \frac{n!}{r!(n-r)!}$. In combinations, order of objects is not meaningful.

To differentiate between the case of permutation and combination, we consider the example that out of 7 Statisticians a committee is to be formed

of 4 then in ${}^7C_4 = \binom{7}{4} = \frac{7!}{4!(7-4)!} = 35$ different ways the committee can

be formed. Now, if we specify the positions of committee members like, one president, one secretary, one treasurer and one speaker then this will be a case of permutation because here the positions (order) are meaningful and the number of ways in which such committee can be constituted is

$${}^7P_4 = \frac{7!}{(7-4)!} = 840.$$

Questions-Answers

Q.1 *Why do we use probability theory?*

Ans. *We use this theory for decision making in the situations of uncertainty so it becomes useful in nearly every phase of the human activity. There are uncertainties in every sphere of life; whenever we turn to predict the outcome of a cricket/ hockey match or a political election; dig for oil search for uranium deposits; experiment with the space ships; and even when we choose a school for our children, or buy a dress shirt. Much of usefulness of the Statistics arises from the ability to measure and to control risk and errors in the inferential and decision problem. This ability is embodied in the theory of the probability. Since the probability theory is at once the language and the measure of uncertain conditions/atmosphere. Thus, probability theory is properly considered as the foundation of statistics. Indeed no analytical statistics can be developed without this theory.*

Q.2 *Discuss the difference between the objective and subjective interpretation of the probability.*

Ans. *The objective interpretation of the probability is that a probability can be determined as a prior. A probability can also be determined on the basis of experience. For example, the probability of a defective item can be determined as after observing that 10 out of sample of 100 parts selected at random. On the other hand, the subjective interpretation of the probability is one of the personal beliefs. For example, if a manager strongly believes that the level of demand for a white-color Suzuki car will be high, he might assign the probability of 0.90 to this level.*

Q.3 *What is the probability that a professor will meet his next class? Is this a priori probability or posterior probability?*

Ans. *This answer will depend on our previous experience that is, on the frequency that professor has met his previous classes. This is the example of posterior probability.*

Q.4 *With respect to set theory, explain what is mean by Intersection and Union of two sets?*

Ans. *The Intersection of the elements of the sets A and B consists of elements common to both A and B.*

The Union of the elements of the set A and set B consists of the elements that belong to set A or set B or both A and B.

Q.5 *Can two mutually exclusive events be independent?*

Ans. *No. Two mutually exclusive events are also dependent events so they can never be independent.*

Q.6 *What is the difference between Sample Space and Event Space?*

Ans. *Sample space is the set of all possible outcomes of a random experiment while the event space is the set of all events associated with a sample space.*

Q.7 *What is the joint probability?*

Ans. *Joint probability is the probability of the occurrence of events of interest concurrently.*

Q.8 *Why conditional probability is different from common probability?*

Ans. *Conditional probability is a probability assigned to an event when a given set of Information related to that event of interest so it is, obviously, different from the common probability.*

Q.9 *In what sense do we say that a conditional probability is defined on reduced sample space?*

Ans. *Sometimes, we are provided with the additional information relating to the outcomes of the random experiment for this reason sample space is changed and reduced. The probability of any event derived from such a reduced sample space is called conditional probability.*

Q.10 *What is the major purpose (or advantage) of using the Baye's rule in decision making problem?*

Ans. *The advantage of this approach is that it allows the initial forecasts in the form of probabilities to be revised upward or downward as related additional information becomes available.*

Q.11 *What is the weakness of Baye's approach in decision making?*

Ans. *The weakness in this approach is that the prior probabilities represent the opinion of a panel of experts. Another panel would probably have given a different set of initial probabilities. That is, there is subjective judgment involved in determining the prior probabilities.*

Q.12 *What is the probability that a student will pass a course in probability? How did you arrive at your answer?*

Ans. *Any value between 0 and 1 is acceptable here. The student may use the ratio of those who passed this probability course to all of those who took the course in the last few years as estimates.*

Q.13 *What is the probability that the price of oil will higher next year than this year? Why?*

Ans. *Again, any value between 0 and 1 is acceptable here. Judging from recent past experience, the probability could be quite high such as 0.8, or even 1.*

Q.14 *An economist assessed the probability of continued unfavourable balance of trade for the Pakistan as 0.80, the probability of a recession with in the next 3-year as 0.50, and the probability of both of these events as 0.40. Does the economist believe that occurrence a recession is independent of an unfavorable balance of trade? Explain.*

Ans. *Yes, because the probability of both events occurring is assessed as equal to the product of the separate marginal probabilities;
 $P(A \text{ and } B) = P(A) \cdot P(B) = 0.80 \times 0.50 = 0.40$.*

Q.15 *Weather forecaster often gives probability for various events. Two such events are*

- (i) The probability of measurable precipitation on a day in June in your areas.*
- (ii) The probability that it will rain tomorrow in your area;*

Would you expect two different forecasters to give the same or different probabilities for (i) and (ii)?

Ans. *The probability of measurable precipitation on a day in June would be established from historical weather records. Two forecasters would give the same probability if they have consulted the same historical record.*

The probability of rain tomorrow would be based on a number of indicators and two forecasters might give differing weights to the different indicators. The probability is thus in part subjective and two forecasters could give different probabilities.

Exercises

Exercise 2.1 (MCQs')

- Q.1** A probability of 1 represents
- (a) Impossibility
 - (b) An improbable event
 - (c) A 50 - 50 chance
 - (d) Certainty
- Q.2** Three coins are tossed. What is probability that there will all be heads?
- (a) $1/8$
 - (b) $1/4$
 - (c) $1/3$
 - (d) $3/2$
- Q.3** A cricket team captain wins the toss for three consecutive matches. What is the probability that he will call correctly for the fourth match?
- (a) $1/16$
 - (b) $1/8$
 - (c) $1/4$
 - (d) $1/2$
- Q.4** Which one is not the characteristic of a random experiment:
- a) It has at least two outcomes.
 - b) The number of all possible outcomes are not known in advance.
 - c) The outcomes are not predictable in advance.
 - d) It can be repeated any number of times under similar conditions.
- Q.5** *A* and *B* are two mutually exclusive events. The probability of *A* happening is $1/4$. The probability of *B* happening is $1/3$. The probability of neither *A* nor *B* happening is?
- (a) $5/12$
 - (b) $1/2$

- (c) $3/4$
- (d) $11/12$

Q.6 A and B are two independent events. The probability of A is $1/4$ and B is $1/3$. The neither probability of A nor B is.

- (a) $5/12$
- (b) $1/3$
- (c) $3/4$
- (d) $11/12$

Q.7 The probability of an event happening is $1/3$. The probability of it not happening is?

- (a) -12
- (b) 0
- (c) $2/3$
- (d) 3

Q.8 If a letter is chosen at random from the 10 letters of the word STATISTICS, what is probability that it is a vowel?

- (a) 0.20
- (b) 0.23
- (c) 0.30
- (d) 0.40

Q.9 Given the following table, what is the probability of selecting a female university graduate student from this group?

	College graduate	University graduate	Total
Male	100	36	136
Female	120	44	164
Total	220	80	300

- (a) 0.120
- (b) 0.268
- (c) 0.265
- (d) 0.147

Q.10 Subjective probabilities are assigned to the events A and B which together comprise a sample space. Which of the following probability statements is not valid?

- (a) $P(A) = 0.7, P(B) = 0.5, P(A \text{ and } B) = 0.2$
- (b) $P(A) = 0.9, P(B) = 0.7, P(A \text{ and } B) = 0.6$
- (c) $P(A) = 0.4, P(B) = 0.3, P(A \text{ and } B) = 0.2$
- (d) $P(A) = 0.5, P(B) = 0.5, P(A \text{ and } B) = 0.0$

- Q.11** A personal manager selects an applicant at random from a large group for an interview. The probability of the applicant being male is 0.60. The probability of selecting an adult is 0.70. The probability of selecting an adult male is 0.45. Given that a male is selected, the probability that he is an adult is;
- (a) 0.27
 - (b) 0.42
 - (c) 0.64
 - (d) 0.75
- Q.12** Which of the following is collection of all mutually exclusive events representing a card randomly selected from a deck of ordinary playing cards?
- (a) King, Queen, Face.
 - (b) Heart, Diamond, Black, Red.
 - (c) 10, 7, jack.
 - (d) 10, red.
- Q.13** Which of the following are collectively exhaustive events representing a card randomly selected from a deck of ordinary playing cards?
- (a) Club, Spade, Heart, Diamond
 - (b) 7 of Spades, Spade, Club Red
 - (c) Face, 7, 9, 10 Jack
 - (d) Red, Spades
- Q.14** A conditional probability might be found in which of the following ways?
- (a) Multiplying together two conditional probabilities.
 - (b) Dividing a joint probability by the given event's probability.
 - (c) Applying the basic concepts of probability to the portion of the event space for which the condition holds.
 - (d) Finding the run frequency of times that the event in question occurs out of all those times when the given event occurs.
- Q.15** Indicate in which one of the following situations the events A and B are independent.
- (a) $P(A) = 0.6$; $P(B) = 0.3$; $P(A/B) = 0.6$
 - (b) $P(A) = 0.3$; $P(B) = 0.7$; $P(B/A) = 0.3$
 - (c) $P(A) = 0.3$; $P(B) = 0.3$; $P(A \text{ and } B) = 0.2$
 - (d) A and B are mutually exclusive; $P(A) = 0.1$; $P(B) = 0.2$

- Q.16** Which one of the following statements is false?
- (a) $P(A/B) = 0.3$, since A and B are independent and $P(A) = 0.3$
 - (b) $P(A/B) = 0.3$, since A and B are mutually exclusive.
 - (c) $P(B/A) = 0.4$, since A and B are independent and $P(B) = 0.4$
 - (d) $P(A/B) = 0$, since $P(A \text{ and } B) = 0$ and $P(B) = 0.7$
- Q.17** A marginal probability might be found by any but which one of the following?
- (a) Adding together appropriate joint probabilities.
 - (b) Subtracting or the sum of several marginal probabilities from 1.
 - (c) Dividing the size of the appropriate event set by the number of possible equally likely elementary events.
 - (d) Multiplying together all probabilities in the same column or row.
- Q.18** Which one of the following statement is not true?
- (a) Mutually exclusive events are statistically dependent.
 - (b) Complementary events have probabilities that sum to 1.
 - (c) Opposite events, are statistically independent.
 - (d) An experiment's elementary events are collectively exhaustive and mutually exclusive.
- Q.19** The number of ways to select 2 persons from 6, ignoring order of selection;
- (a) 64
 - (b) 15
 - (c) 36
 - (d) 12
- Q.20** A fair coin is tossed 50 times, the expected number of heads are;
- (a) 100
 - (b) 50
 - (c) 15
 - (d) None of these.

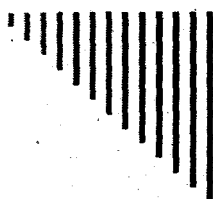
Exercise 2.2 (True/False)

Read the following statements carefully and indicate which statement is "True" or "False":

1. A 50-50 chance of the rain means that the probability of the rain is $1/2$.
2. For a random experiment, all the outcomes are known in advance.
3. A or B is an event occurring whenever A occurs alone, B occurs alone, or the both A and B do not occur.
4. If A and B is impossible, then the two events A , B are mutually exclusive.
5. If $P(A \text{ and } B) = 0$, then the two events; A and B must be collectively exhaustive.
6. Suppose two events A and B are mutually exclusive and also collectively exhaustive, this indicate that $P(A) = 1 - P(B)$.
7. Suppose that $P(A) = 0.8$, $P(B) = 0.2$, and $P(A \text{ and } B) = 0.16$. Then, A is independent of B .
8. A conditional probability is equal to the unconditional probability for the event A given B whenever A and B are mutually exclusive.
9. A marginal probability will always be equal to the sum of two or more joint probabilities.
10. The event space is the set of all possible sample points of a sample space.
11. The event space corresponding to the union of the events A , B represents all elementary events of A and of B , but not events common to both.

12. If the probability of success is 0.4 than the probability of failure is 0.6.
13. Joint events are those that are not mutually exclusive.
14. If three events, A , B , and C are independent, the probability that A , B , and C will happen at the same time is $P(A) \cdot P(B) \cdot P(C)$.
15. If A and B are mutually exclusive events then the $P(A \text{ or } B)$ equals the $P(A) + P(B) - P(A \cap B)$.
16. When events A and event B are independent, $P(B/A) = P(A/B)$
17. The addition rule concerning probabilities applies to independent events.
18. Probability is a number between 0 and 1, exclusive.
19. If the two events, A and B are mutually exclusive then the probability that either one or the other will occur is $P(A) \cdot P(B)$.
20. A probability can be certainty.
21. In a classical probability, we can determine a prior probability based on a logical reasoning before any experiments take place.
22. An unconditional probability is also known as a marginal probability.
23. A subjective probability may be nothing more than an educated guess.
24. When using the relative frequency approach, probability figures become less accurate for large number of observations.
25. The relative frequency approach to probability will provide correct statistical probabilities after 100 trials.
26. A and B are independent events if $P(A/B) = P(B)$.
27. Symbolically, a marginal probability is $P(AB)$.
28. If A and B are independent events then $P(A \cap B) \neq P(A) \cdot P(B)$.

29. Using Baye's theorem, we may develop revised probabilities based on new information; these revised probabilities are also known as posterior probabilities.
30. Classical probability assumes that each of the possible outcomes of an experiment is independent to the other.



Chapter 3

Random Variables

Random Variable

A random variable is a real-valued function that associates a unique numerical value with every outcome of a random experiment. Let Ω be a sample space on which probability function defined. Let X be a real valued function defined on sample space, such that X transforms outcomes of the Ω into points on the real line. Then X is said to be random variable.

The outcome of an experiment need not be a number, for example, the outcome when a coin is tossed can be 'head' or 'tail'. However, we often want to represent outcomes as numbers.

Discrete Random Variable

A discrete random variable is one which may take on only a countable number of distinct values such as 0, 1, 2, 3,.... . Discrete random variables are usually (but not necessarily) counts. If a random variable can take only a finite number of distinct values, then it must be discrete. Examples of discrete random variables include the number of children in a family, the number of patients in a doctor's surgery, the number of defective light bulbs in a box of ten.

Continuous Random Variable

A continuous random variable is a random variable whose possible values from a continuous data set usually some interval of real numbers. Continuous random variables are usually measurements. Examples include height, weight, the amount of sugar in an orange, the time required to run a mile.

Probability Function

A probability function is a real valued function defined on the class of all subsets of the sample space Ω ; the value that is associated with a subset A is denoted by $P(A)$. The assignments of probability must satisfy the following three axioms:

- (i) $P(\Omega) = 1$
- (ii) $P(A) \geq 0$
- (iii) If $A_i (i = 1, 2, \dots)$ is a sequence of mutually exclusive events then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Probability Distribution

A table listing all possible values that a random variable can take on together with the associated probabilities is called probability distribution.

The probability distribution of a discrete random variable is a list of probabilities associated with each of its possible values. It is also sometimes called the probability function or the probability mass function.

More formally, the probability distribution of a discrete random variable X is a function which gives the probability $P(x_i)$ that the random variable equals x_i , for each value x_i :

$$P(x_i) = P(X=x_i)$$

It satisfies the following conditions:

- (i) $0 \leq P(x_i) \leq 1$
- (ii) $\sum P(x_i) = 1$

Probability Density Function (pdf)

The probability density function (pdf) of a continuous random variable is a function which can be integrated to obtain the probability that the random variable takes a value in a given interval.

More formally, the probability density function, $f(x)$, of a continuous random variable X is the derivative of the cumulative distribution function $F(x)$ (defined next):

$$f(x) = \frac{d}{dx} F(x).$$

If $f(x)$ is a probability density function then it must obey two conditions:

- (i) That the total probability for all possible values of the continuous random variable X is 1, i.e;

$$\int f(x) dx = 1$$

- (ii) That the probability density function can never be negative:
 $f(x) > 0$ for all x .

Cumulative Distribution Function (CDF)

All random variables (discrete and continuous) have a cumulative distribution function. It is a function giving the probability that the random variable X is less than or equal to x , for every value x .

Formally, the cumulative distribution function $F(x)$ is defined to be:

$$F(x) = P(X \leq x)$$

For a discrete random variable, the cumulative distribution function is found by summing up the probabilities as;

$$F(x) = P(X \leq x) = \sum P(X = x)$$

For a continuous random variable, the cumulative distribution function is the integral of its probability density function;

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

A cdf has the following properties:

- (i) $F(+\infty) = 1$ and $F(-\infty) = 0$
- (ii) $F(x)$ is non-decreasing function of X , i.e., $F(x_1) \leq F(x_2)$ if $x_1 \leq x_2$
- (iii) $F(x)$ is continuous at least on the right of each x

Expected Value

The expected value (or population mean) of a random variable indicates its average or central value. It is a useful summary value (a number) of the variable's distribution. Stating the expected value gives a general impression of the behavior of some random variable without giving full details of its probability distribution (if it is discrete) or its probability density function (if it is continuous). The expected value of a random variable X is symbolized by $E(X)$ or μ .

If X is a discrete random variable with possible values $x_1, x_2, x_3, \dots, x_n$, and $P(x_i)$ denotes $P(X = x_i)$, then the expected value of X is defined by:

$$\mu = E(X) = \sum x_i P(x_i), \text{ provided that series are convergent.}$$

If X is a continuous random variable with probability density function $f(x)$, then the expected value of X is defined by:

$$\mu = E(X) = \int x f(x) dx.$$

It is to be noted that

$$Var(X) = E(X^2) - \{E(X)\}^2$$

Also

$$E(X+C) = \mu + C,$$

where C is any constant.

Independent Random Variables

Two random variables X and Y (say), are said to be independent if and only if the value of X has no influence on the value of Y and vice versa.

Symbolically, X and Y are said to be independent if

$$f(x,y) = g(x)h(y),$$

where $g(x)$ and $h(y)$, are the marginal density functions of the random variables X and Y respectively and $f(x,y)$ is the joint density function of the both variables.

Moment-Generating Function (m.g.f)

The moment-generating function of a random variable X is

$$M_X(t) = E(e^{it}), \quad t \in \mathcal{R},$$

wherever this expectation exists. The moment-generating function generates the moments of the probability distribution, and thus uniquely defines the distribution of the random variable.

Provided the moment-generating function exists in an interval around $t = 0$, the r -th moment is given by

$$E(X^r) = M_X^{(r)}(0) = \left. \frac{d^r}{dt^r} M_X(t) \right|_{t=0}.$$

If X has a continuous probability density function $f(x)$ then the moment generating function is given by

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{itx} f(x) dx \\ &= \int_{-\infty}^{\infty} \left(1 + itx + \frac{t^2 x^2}{2!} + \dots \right) f(x) dx \\ &= 1 + t \mu'_1 + \frac{t^2 \mu'_2}{2!} + \dots, \end{aligned}$$

where μ'_i is the i -th moment about origin.

Cumulant-Generating Function (c.g.f)

The cumulant-generating function is the logarithm of the moment-generating function.

$$K(t) = \ln M_X(t) \\ = K_1 t + K_2 \frac{t^2}{2!} + K_3 \frac{t^3}{3!} + \dots,$$

We have following relations between cumulants and moments:

$$K_1 = \mu'_1$$

$$K_2 = \mu'_2 - \mu'^2_1 = \mu_2$$

$$K_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2\mu'^3_1 = \mu_3$$

$$K_4 = \mu'_4 - 4\mu'_1\mu'_3 + 3\mu'^2_2 + 12\mu'_1\mu'^2_2 - 6\mu'^2_1\mu'^2_2 = \mu_4$$

Characteristic Function

The characteristic function of any random variable completely defines its probability distribution. On the real line it is given by the following formula, where X is any random variable with the distribution $f(x)$:

$$\varphi_X(t) = E(e^{itX}),$$

where t is a real number.

The r -th moment about origin can be found as:

$$E(X^r) = -i^r \left[\frac{d^r}{dt^r} \varphi_X(t) \right]_{t=0}$$

Chebyshev's Inequality

Chebyshev's inequality (also known as Tchebysheff's inequality, Chebyshev's theorem, or the Bienaymé-Chebyshev inequality), states that in any data sample or probability distribution, nearly all the values are close to the mean value, and provides a quantitative description of "nearly all" and "close to". For example, no more than 1/4 of the values are more than 2 standard deviations away from the mean, no more than 1/9 are more than 3 standard deviations away, no more than 1/25 are more than 5 standard deviations away, and so on.

Generally, let X be a random variable with expected value μ and finite variance σ^2 . Then for any real number $k > 0$, the probability that a value of

X falls within k -standard deviation of the mean is at least $1 - \frac{1}{k^2}$ i.e.,

$$\Pr(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

Questions-Answers

Q.1 If a random variable (r.v) is a real valued function defined on sample space, what is domain and range for r.v?

Ans. The domain is sample space while the range is real line.

Q.2 If X is a r.v then whether any function of X , say, $g(X)$ is also a r.v?

Ans. Yes, a function of a r.v is also a r.v.

Q.3 Let X denotes the number of persons in a household selected at random, then what type of random variable X is?

Ans. Discrete.

Q.4 Give the examples for discrete and continuous random variable.

Ans. Discrete: Number of patients, daily coming in a clinic; Number on roadside accidents per day on a high way, etc.

Continuous: Weight, height, age, waiting time for a bus on a bus stop, etc.

Q.5 The CDF of a continuous variable T is defined as follows,

$$F(t) = \begin{cases} 0 & \text{if } t < 0 \\ t/12 & \text{if } 0 \leq t \leq 12 \\ 1 & \text{if } t > 12 \end{cases}$$

Determine,

a) $P(T \leq 6)$

b) $P(3 \leq T \leq 4.5)$

c) $P(-\infty \leq T \leq 3.75)$

d) $P(T > 9.45)$

e) $P(T \leq 15)$

f) $P(T = \pi)$

Ans. a) $P(T \leq 6) = F(6) = 6/12 = 0.5$

b) $P(3 \leq T \leq 4.5) = F(4.5) - F(3) = 4.5/12 - 3/12 = 0.125$

c) $P(-\infty \leq T \leq 3.75) = F(3.75) - F(-\infty) = 3.75/12 - 0 = 0.3125$

d) $P(T > 9.45) = 1 - F(9.45) = 1 - 9.45/12 = 0.2125$

e) $P(T \leq 15) = 1 - F(15) = 1 - 1 = 0$

f) $P(T = \pi) = P(\pi \leq T \leq \pi) = F(\pi) - F(\pi) = \pi/12 - \pi/12 = 0$

Q.6 In a given business venture, a man can make profit of Rs. 1000 or suffer a loss of Rs.500. If the probability of a profit is 0.6, what will be the expected profit in the venture?

Ans. $E(\text{profit}) = (1000)(0.6) + (-500)(0.4) = 400.$

Q.7 *If the variance of a random variable X is 0.8, what is the variance of the random variable $2X$, $X/2$ and $X + 2$?*

Ans. *We know that:*

$$\text{Var}(aX) = a^2 \text{Var}(X) \text{ and } \text{Var}(X) = 0.8$$

$$\therefore \text{Var}(2X) = 2^2 \text{Var}(X) \\ = 4(0.8) = 3.2$$

and

$$\text{Var}(X/2) = (1/2)^2 \text{Var}(X) \\ = 1/4 (0.8) = 0.2$$

$$\text{Var}(X + C) = \text{Var}(X) \text{ if } C \text{ is a constant so}$$

$$\text{Var}(X + 2) = \text{Var}(X) = 0.8.$$

Q.8 *If we want to get moments about mean rather than origin then what change we should make in computing mgf?*

Ans. *We should compute mgf taking mean as origin in place of zero i.e.,*

$$M_{X-\mu}(t) = E(e^{t(X-\mu)}) = e^{-t\mu} M_X(t).$$

Q.9 *What are the uses of characteristic function?*

Ans. *Characteristic function is particularly useful for dealing with functions of independent random variables. For example, if $X_1, X_2, \dots, X_n$ is a sequence of independent (and not necessarily identically distributed) random variables, and*

$$S_n = \sum_{i=1}^n a_i X_i,$$

where the a_i are constants, then the characteristic function for S_n is given by

$$\varphi_{S_n}(t) = \varphi_{X_1}(a_1 t) \varphi_{X_2}(a_2 t) \dots \varphi_{X_n}(a_n t).$$

Because of the continuity theorem, characteristic functions are used in the most frequently seen proof of the central limit theorem.

Characteristic functions can also be used to find moments of random variable. Provided that r -th moment exists, characteristic function can be differentiated r times.

Q.10 *What is the use Chebyshev's Inequality?*

Ans. *The inequality can be useful despite loose bounds because it applies to random variables of any distribution, and because these bounds can be calculated knowing no more about the distribution than the mean and variance.*

Exercise 3 (True/False)

Read the following statements carefully and indicate which statement is "True" or "False":

1. Since random variable is a real-valued function so the domain of this function is real line.
2. If a random variable X can take on only a finite number of values or an infinite number of values that are countable, then we call X a discrete random variable.
3. The value of distribution function approaches to 1 as a random variable X approaches to $-\infty$.
4. No two distinct distributions have the same density functions.
5. The expected value of an experiment is obtained by computing the arithmetic average value over all possible outcomes of the experiment.
6. If a man loses Rs. 100 if tail comes while tossing a fair coin otherwise he wins Rs. 100. The expected value for the person winning is Rs. 50.
7. If a constant is added to a variable then the variance changes by multiplying a constant with the variance of that variable.
8. The variance of difference of two independent random variables is equal to the sum of the variances of the both variables.
9. Chebyshev's inequality can only be applied to those variables that have finite variances.
10. The characteristic function is simply the logarithm of the *mgf*.



Chapter 4

Discrete Probability Distributions

Discrete Probability Distributions

Discrete probability distributions give the probability of every possible value of a discrete random variable (discussed in Chapter 3). In this chapter, we discuss some important discrete probability distributions; Binomial, Hypergeometric, Geometric, Negative Binomial, Poisson, and Multinomial distributions.

Bernoulli Trial

There are many practical and experimental situations where out come of each repeated random trail can result in just two categories, namely "success" and "failure", or dichotomy of results can be found. For example, two possible outcomes of each exam of a student can be passing or failing of that student, correct or wrong answer, catching or missing a bus on a stop, hitting or missing of a target, infected or non-infected from some disease after result of a test, etc. Such trials are called Bernoulli Trials. Simply speaking, a trial of a random experiment whose outcomes can be classified into two categories, "success" or "failure", is referred as Bernoulli trial.

Binomial Distribution

An experiment having n (say) Bernoulli trials with the following properties, is called Binomial Experiment:

- a) The results of each trial can be classified into one of two categories, say, "success" and "failure". The outcome of interest is usually referred as "success".
- b) The probability of success p is the same for each trial.

- c) Each trial is independent of all the others.
- d) The experiment can be performed a fixed number of times, say, n .

Distribution Summary:

Parameters: Two parameters, n and p ,
 n = Number of trials,
 p = Probability of success

pmf:
$$P(X = x) = \binom{n}{x} p^x q^{n-x}; \quad x = 0, 1, 2, \dots, n,$$

where

X = Number of successes in n trials,

$q = 1 - p$

Mean: np

Variance: npq

Skewness:
$$\frac{1 - 2p}{\sqrt{npq}}$$

Kurtosis:
$$\frac{1 - 6pq}{npq}$$

mgf: $(q + pe^t)^n$

Char. func.: $(q + pe^{it})^n$

Hypergeometric Distribution

There are many experiments in which the condition of independent trials is not met and resultantly, the probability of success does not remain constant from trial to trial. Such experiments are called hypergeometric experiments with the following properties:

- The results of each trial can be classified into one of two categories, say, "success" and "failure". The outcome of interest is usually referred as "success".
- The successive trials are dependent.
- The probability of success changes from trial to trial.
- The experiment can be performed a fixed number of times, say, n .

Distribution Summary:

Parameters: Three parameters, N , k and p ,
 N = Number of units in the set or population,
 k = Number of successes (units of interest) in the set or population,

$$p = \frac{k}{N} = \text{Probability of success.}$$

pmf:

$$P(X = x) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}},$$

where

X = Number of successes (units of interest)
in n (sample size of items selected
without replacement),

$$x = 0, 1, 2, \dots, n, \text{ and } x = 0, 1, 2, \dots, k.$$

Mean:

$$np = n \frac{k}{N}$$

Variance:

$$npq \left(\frac{N-n}{N-1} \right) = n \frac{k}{N} \left(1 - \frac{k}{N} \right) \left(\frac{N-n}{N-1} \right)$$

Geometric Distribution

In many practical situations, an experimenter is interested in the first success in the experiment. To obtain this, he repeats the experiment until he gets the first success. For example, a researcher keeps on taking blood samples until he gets O-ve blood group. To handle such situation, here a distribution presented called geometric distribution and an experiment in which trials are repeated until first success is obtained, is called geometric experiment. A geometric experiment has the following properties:

- a) The results of each trial can be classified into one of two categories, say, "success" and "failure". The outcome of interest is usually referred as "success".
- b) The successive trials are independent.
- c) The probability of success remains constant from trial to trial.
- d) The experiment is repeated a variable number of times until the first success is obtained.

Distribution Summary:

Parameters: One parameter, p ,
 p = Probability of success.

pmf: $P(X = x) = pq^x; \quad x = 0, 1, 2, \dots,$
 where
 X = Number of failures preceding first
 success,
 $q = 1 - p$.

Mean: $\frac{q}{p}$

Variance: $\frac{q}{p^2}$

mgf: $\frac{pe^t}{1 - qe^t}$

Note: Another function of geometric distribution is also used as follows:
Probability Function:

$$P(X = x) = pq^{x-1}; \quad x = 1, 2, \dots,$$

where

X = Number of trials to have first success,

p = Probability of success,

$q = 1 - p$,

also p is the only parameter of this distribution.

Mean: $\frac{1}{p}$

Variance: $\frac{1}{p^2}$.

Negative Binomial (Pascal) Distribution

In geometric experiment, an experimenter is interested in the first success and repeats the experiment until he gets the first success. But in many practical situations, this interest can be extended to a fixed number of success rather than just single success. For example, a researcher keeps on taking blood samples until he gets 10 donors having O-ve blood group. To handle such situation, here a distribution presented called negative binomial distribution and an experiment in which trials are repeated until a fixed number of successes, say k , is obtained, is called negative binomial having following properties:

- a) The results of each trial can be classified into one of two categories, say, "success" and "failure". The outcome of interest is usually referred as "success".
- b) The successive trials are independent.
- c) The probability of success remains constant from trial to trial.
- d) The experiment is repeated a variable number of times to obtain a fixed number of successes, say k .

Distribution Summary:

Parameters: Two parameters, p and k ,
 p = Probability of success.
 K = Number of successes.

pmf:
$$P(X = x) = \binom{x-1}{k-1} p^k q^{x-k}; \quad x = k, k+1, k+2, \dots$$

where

X = Number of trials to produce k successes,
 $q = 1 - p$.

Mean:
$$k \frac{q}{p}$$

Variance:
$$k \frac{q}{p^2}$$

mgf:
$$p^k (1 - qe^t)^{-k}$$

Note: Another function of negative binomial distribution is also used as follows:

$$Y = X - k,$$

$$P(Y = y) = \binom{y+k-1}{k-1} p^k q^y; \quad y = 0, 1, 2, \dots,$$

where

Y = Number of failures preceding k successes.

The mean and variance for Y will be changed accordingly.

Poisson Distribution

A distribution often used to compute probabilities for random variables distributed over time and space is the Poisson distribution. For example,

experience has shown that Poisson is an excellent model to use for computing probabilities associated with the number of calls coming into a telephone switchboard during a fixed period of time. Other examples are, the number of automobile deaths per month in a large city, the number of bacteria in a given culture, the number of atoms disintegrating per second from radioactive material etc. Poisson distribution is the premier probability model and perhaps, the second most frequently used discrete distribution after the binomial distribution. It is the limiting form of binomial distribution.

Following are the assumptions while using Poisson distribution:

- a) Events that occur in one time interval (or region or space) are independent of those in any other non-overlapping time interval (or region or space).
- b) For a small time interval (or region or space), the probability that an event occurs is proportional to the length of the time interval (or region or space).
- c) The probability that two or more events occur in a very small time interval (or region or space) is so small that it can be neglected.

Distribution Summary:

Parameters: One parameter, λ

pmf:
$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots, \infty$$

where

X = Number of events occurring per unit time
or per unit space.

Mean: λ

Variance: λ

mgf:
$$\text{Exp}\{\lambda(e' - 1)\}$$

Multinomial Distribution

If a trial's outcomes can be classified into more than two categories, a binomial experiment becomes multinomial experiment. For example, a finished product may be classified as excellent, good, average; a student's grade may be A, B, C, D, or F etc.

A multinomial experiment has the following properties:

- a) The results of each trial can be classified into one of k non-overlapping categories, say, $C_1, C_2, \dots, C_k$.
- b) The probability of i th outcome is p_i which remains constant from trial to trial and $\sum_{i=1}^k p_i = 1$.
- c) The successive trials are independent.
- d) The experiment can be performed a fixed number of times, say, n .

Distribution Summary:

Parameters: k (number of categories) parameters, $n, p_1, p_2, \dots, p_{k-1}$.

p_i is the probability of success of i th category.

pmf:
$$P(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k)$$
$$= \frac{n!}{x_1! x_2! \dots x_k!} (p_1)^{x_1} (p_2)^{x_2} \dots (p_k)^{x_k};$$

$$x_i = 0, 1, 2, \dots, n,$$

where

X_i = Number of successes in i th category,

$$\sum_{i=1}^k x_i = n, \sum_{i=1}^k p_i = 1.$$

Mean: np_i for X_i

Variance: $np_i q_i$ for X_i

Questions-Answers

Q.1 *What is the relation between Bernoulli trials and binomial experiment?*

Ans. *A binomial experiment has two or more Bernoulli trials.*

Q.2 *What is the distribution of the sum of independent binomial variates?*

Ans. *That distribution will also be binomial. If we have $X_1, X_2, \dots, X_m$ independent variates having binomial distribution with parameters n and p then $\sum_{i=1}^m X_i$ has also binomial with parameters mn and p .*

Q.3 *What is the reason that mean of binomial variates is always greater than its variance?*

Ans. *It is because mean of binomial variates is np while variance is npq . Since $q < 1$, the variance will be less than np (mean).*

Q.4 *Criticize the following statement:*

The mean of binomial distribution is 5 while standard deviation is 3.

Ans. *It is not possible. In above statement the variance is 9 which is greater than mean so it is not possible (see also Q.3 above).*

Q.5 *What is the main difference between a binomial and a hypergeometric experiment?*

Ans. *In binomial distribution, the successive trials are independent and the probability of success remains constant from trial to trial while, on the other hand, in hypergeometric distribution, the successive trials are dependent and the probability of success changes from trial to trial.*

Q.6 *What is the main difference between binomial experiment and negative binomial experiment?*

Ans. *In binomial experiment, the number of trials is fixed and the number of successes is variable while in negative binomial*

experiment, the number of trials is variable but the number of successes is fixed.

Q.7 What is the relation between negative binomial and geometric experiment?

Ans. If the number of successes is reduced to 1 to stop an experiment, the negative binomial experiment becomes geometric experiment. In other words, if $k = 1$, negative binomial distribution becomes geometric distribution.

Q.8 Criticize the following statement:

The mean of Poisson distribution is 3 while standard deviation is 3.

Ans. In Poisson distribution, mean and variance are equal. In the above statement, the variance is 9 that do not equate the mean.

Q.9 What is the distribution of the sum of independent Poisson variates?

Ans. That distribution will also be Poisson.

Q.10 Is it possible to apply binomial or Poisson to the same problem?

Ans. Yes, it is possible if the probability of success is very small and the number of trials is very large.

Q.11 Is it possible that if trial of a random experiment results in more than two outcomes but we can still apply binomial distribution?

Ans. Yes, it is quite possible because a trial can result in more than two possible outcomes but we should categorize them in just two categories.

Q.12 How many number of parameters in multinomial distribution?

Ans. As many as the number of categories into which outcomes of trial are divided.

Exercises

Exercise 4.1 (MCQs')

- Q.1** Which of the following could never be described by the binomial distribution?
- (a) The number of defective items produced by an assembly process.
 - (b) The amount of water used daily by a single household.
 - (c) The number of people in a class who can answer a particular question correctly.
 - (d) All of these.
- Q.2** Standard deviation of the binomial distribution depends upon;
- (a) Probability of success
 - (b) Number of trials
 - (c) both (a) & (b) above
- Q.3** For a Poisson distribution with standard deviation equals to 2 then mean of the Poisson distribution equals to;
- (a) 0
 - (b) 1
 - (c) 2
 - (d) 4
- Q.4** A binomial distribution with $n = 1000$ and $p = 0.5$ is;
- (a) Symmetrical
 - (b) Asymmetrical
 - (c) Skewed to right
 - (d) Skewed to left

- Q.5** If X has binomial distribution with parameter p and n then X/n has the variance
- (a) npq
 - (b) n^2pq
 - (c) pq/n
 - (d) pq/n^2
- Q.6** If X is number of trials for negative binomial distribution with parameters p and k then its minimum value is
- (a) 0
 - (b) k
 - (c) $k+1$
 - (d) $k-1$
- Q.7** For a given binomial distribution with n fixed if $p = 0.5$ then
- (a) Poisson distribution will provide a bad approximation.
 - (b) Poisson distribution will provide a good approximation.
 - (c) Binomial distribution will be skewed left.
 - (d) Binomial distribution will be skewed right.
- Q.8** Which of the following is necessary condition for use of a Poisson distribution:
- (a) Probability of one arrival per second is constant.
 - (b) The number of arrivals in any one second interval is independent of arrivals in other intervals.
 - (c) The probability of two or more arrivals in the same second is zero.
 - (d) (b) and (c) above.
- Q.9** The necessary and sufficient condition of the hypergeometric distribution is:
- (a) Sampling with replacement
 - (b) Sampling without replacement
 - (c) Trials are independent
 - (d) Probability of success remains constant

- Q.10** Which of the following is the most reasonable condition for the binomial approximation to the hypergeometric distribution?
- (a) $N = 200, n = 12$
 - (b) $N = 500, n = 20$
 - (c) $N = 640, n = 30$
 - (d) $N = 800, n = 50$
- Q.11** Suppose, we have a Poisson distribution with λ equals to 2 then the probability of having exactly 10 occurrences is;
- (a) $\frac{2^{-10} e^{-10}}{10!}$
 - (b) $\frac{2^{10} e^{-2}}{2!}$
 - (c) $\frac{10^2 e^{-10}}{10!}$
 - (d) $\frac{2^{10} e^{-2}}{10!}$
- Q.12** Which of the following is a characteristic of the probability distribution for any random variable:
- (a) A probability is provided for every possible value
 - (b) The sum of all probabilities is one
 - (c) No given probability occurs more than once
 - (d) (a) and (b) above
- Q.13** In what case would the Poisson distribution be a good approximation of the binomial distribution:
- (a) $n = 40, p = 0.32$
 - (b) $n = 40, q = 0.79$
 - (c) $n = 200, q = 0.98$
 - (d) $n = 10, p = 0.03$
- Q.14** The mode of the geometric distribution is:
- (a) 1
 - (b) 2
 - (c) 3

- (d) not possible

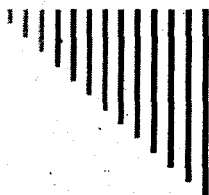
Q.15 A binomial distribution may be approximated by a Poisson distribution if:

- (a) n is large and p is large
- (b) n is small and p is large
- (c) n is small and p is small
- (d) none of these

Exercise 4.2 (True/False)

Read the following statements carefully and indicate which statement is "True" or "False":

- 1. Hypergeometric distribution has four parameters.
- 2. When the probability of success in a Bernoulli process is 0.5, its binomial distribution is symmetrical.
- 3. The variance of binomial distribution can be zero if the probability of success is one.
- 4. The binomial distribution is not really necessary because its values can always be approximated by another distribution.
- 5. The body temperatures of adult males can be described by Poisson distribution.
- 6. Tossing an unbiased coin 5000 times can be treated as binomial experiment.
- 7. Tossing a biased coin 50 times can be treated as hypergeometric experiment.
- 8. For Poisson distribution, mean and variance are equal.
- 9. The mean of the ratio of number of successes to the number of trials in binomial experiment is equal to probability of success.
- 10. Poisson distribution is also referred as distribution of rare events.



Chapter 5

Continuous Probability Distributions

Continuous Probability Distributions

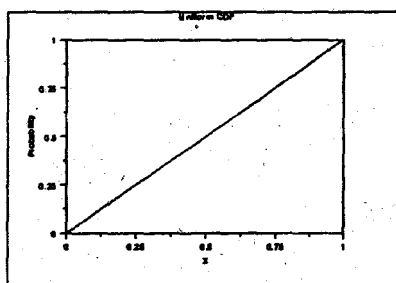
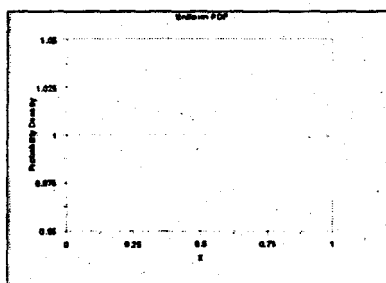
The probability distributions for continuous random variable, discussed in Chapter 3, are referred as continuous probability distributions. In this chapter, we discuss some important continuous probability distributions; Uniform, Exponential, Normal, Gamma, Chi-Square, t -, and F -distributions.

Uniform Distribution

A continuous random variable X is said to follow a Uniform distribution with parameters a and b , written $X \sim U(a,b)$, if its probability density function is constant within a finite interval $[a,b]$, and zero outside this interval (with a less than or equal to b). In other words, the values of a uniform random variable are uniformly distributed over an interval. For example, if buses arrive at a given bus stop every 15 minutes, and you arrive at the bus stop at a random time, the time you wait for the next bus to arrive could be described by a uniform distribution over the interval from 0 to 15 minutes.

One of the most important applications of the uniform distribution is in the generation of random numbers. That is, almost all random number generators generate random numbers on the $(0,1)$ interval. For other distributions, some transformation is applied to the uniform random numbers.

The following is the plots of the uniform probability density function and cumulative distribution function:



Distribution Summary:

Parameters: $a, b \in (-\infty, \infty)$

pdf: $\frac{1}{b-a}; \quad a < x < b$
 $0; \quad x < a \text{ or } x > b$

cdf: $\frac{x-a}{b-a}; \quad a \leq x < b$
 $1; \quad x \geq b$

Mean: $\frac{a+b}{2}$

Median: $\frac{a+b}{2}$

Mode: any value in $[a, b]$

Variance: $\frac{(b-a)^2}{12}$

Skewness: 0

Kurtosis: $-\frac{6}{5}$

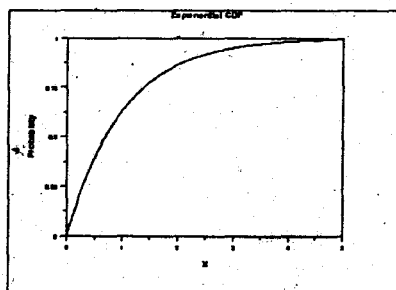
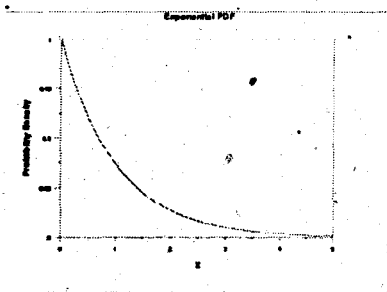
mgf: $\frac{e^{tb} - e^{ta}}{t(b-a)}$

Char. func.: $\frac{e^{itb} - e^{ita}}{it(b-a)}$

Exponential Distribution

The exponential distribution is used to model Poisson processes, which are situations in which an object initially in state A can change to state B with constant probability per unit time λ . The time at which the state actually changes is described by an exponential random variable with parameter λ . The exponential distribution is also known as the waiting-time distribution, describes the amount of time or distance between the occurrence of random events such as the time between major earthquakes or the time between two consecutive goals in a football match or the time until you get seat in a bus etc. This distribution is also used in connection with estimating the length of material life, or the length of time a process might take.

The following is the plot of the exponential probability density function and cumulative distribution function:



Distribution Summary:

Parameters: $\lambda > 0$; mean time (rate) of a process

pdf: $\lambda e^{-\lambda x}, \quad x \geq 0$
 $0, \quad x < 0$
 $X = \text{time elapsed}$

cdf: $1 - e^{-\lambda x}$

Mean: $\frac{1}{\lambda}$

Median: $\frac{\ln 2}{\lambda}$

Mode: 0

Variance: $\frac{1}{\lambda^2}$

Skewness: 2

Kurtosis: 6

mgf: $\left(1 - \frac{t}{\lambda}\right)^{-1}$

Char. func.: $\left(1 - \frac{it}{\lambda}\right)^{-1}$

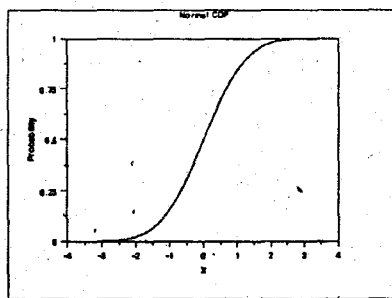
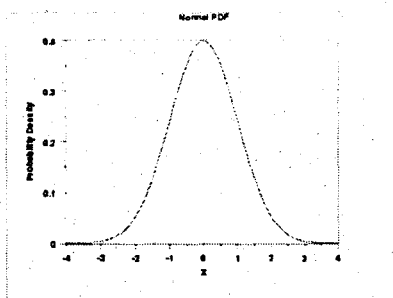
Normal Distribution

The Normal Distribution, also called Gaussian distribution, is an extremely important probability distribution in many fields. The fundamental importance of the normal distribution is its use as model of quantitative phenomena in the natural and behavioral sciences. A variety of psychological test scores and physical phenomena can be well approximated by a normal distribution. For example, height at a given age for a given gender in a given racial group is adequately described by a normal random variable even though heights must be positive.

For both theoretical and practical reasons, the normal distribution is probably the most important distribution in Statistics. For example, many classical Statistical tests are based on the assumption that the data follow a normal distribution. This assumption should be tested before applying these tests.

In modeling applications, such as linear and non-linear regression, the error term is often assumed to follow a normal distribution with fixed location and scale. The normal distribution is used to find significance levels in many hypothesis tests and confidence intervals. The normal distribution also arises in many areas of Statistics: for example, the sampling distribution of the mean is approximately normal, even if the distribution of the population from which the sample is taken is not normal. In probability theory, normal distribution arises as the limiting distribution of several continuous and discrete families of distributions.

The following are the plots of the normal probability density function and cumulative distribution function:



Distribution Summary:

Parameters: $-\infty \leq \mu \leq \infty; \sigma^2 > 0$

pdf:
$$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

$$0, \quad \text{otherwise}$$

cdf:
$$\frac{1}{2} \left(1 + \exp\left(\frac{x-\mu}{\sigma\sqrt{2}}\right) \right)$$

Mean: μ

Median: μ

Mode: μ

Variance: σ^2

Skewness: 0

Kurtosis: 3

mgf:
$$\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$$

Char. func.:
$$\exp\left(\mu it - \frac{\sigma^2 t^2}{2}\right)$$

Standard Normal Distribution

The Standard Normal Distribution is the normal distribution with a mean of zero and a standard deviation of one. It is the simplest case of the normal distribution. This is written as $N(0,1)$.

The pdf for standard Normal distribution is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

Since the general form of probability functions can be expressed in terms of the standard normal distribution, all subsequent formulas in the above table, can be obtained by plugging $\mu = 0$ and $\sigma^2 = 1$.

The standard normal distribution has wide use, especially, in the applications of Central Limit Theorem (CLT).

Central Limit Theorem (CLT)

The Central Limit Theorem states that whenever a random sample of size n is taken from any distribution with mean μ and variance σ^2 , then the sample mean $\bar{X}$ will be approximately normally distributed with mean μ and variance σ^2/n . The larger the value of the sample size n , the better the approximation to the normal. This is very useful when it comes to inference. For example, it allows us (if the sample size is fairly large) to use hypothesis tests which assume normality even if our data appear non-normal. This is because the tests use the sample mean $\bar{X}$, which the Central Limit Theorem tells us will be approximately normally distributed.

The normal distribution is widely used. Part of the appeal is that it is well behaved and mathematically tractable. However, the central limit theorem provides a theoretical basis for why it has wide applicability. The central limit theorem basically states that as the sample size n becomes large, the following occur:

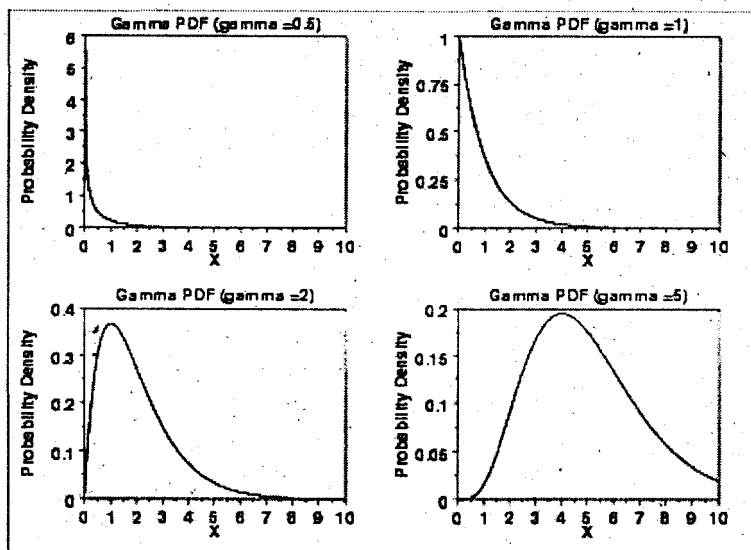
$$\lim_{n \rightarrow \infty} \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

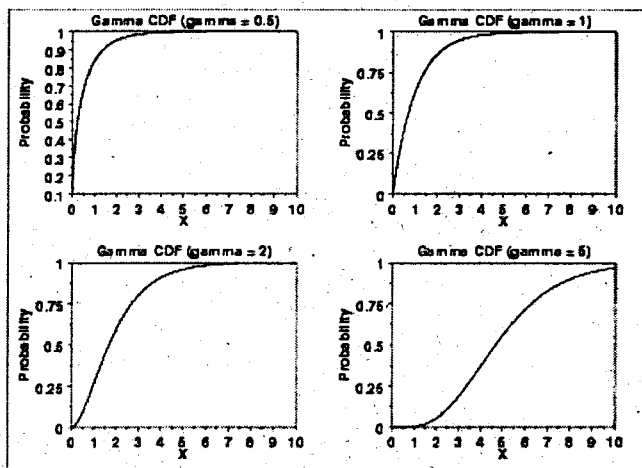
In other words, as sample size increases, the sampling distribution of the mean becomes approximately normal regardless of the distribution of the original variable.

Gamma Distribution

Gamma densities provide a fairly flexible class for modeling nonnegative random variables. The exponential distribution becomes gamma distribution, when we consider sum of independently identically distributed exponential variates. The gamma distribution can be used to describe the probability that α events will occur within a time period λ . Contrast it with the exponential distribution, which describes the probability that one event will occur.

The following are the plots of the gamma probability density function and cumulative distribution function:





Distribution Summary:

Parameters: $\alpha > 0$ shape (real)
 $\beta > 0$ scale (real)

pdf:
$$\frac{1}{\Gamma \alpha \beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x \geq 0$$

Mean: $\alpha\beta$

Mode: $(\alpha - 1)\beta$ for $\alpha \geq 1$

Variance: $\alpha\beta$

$$\text{Skewness: } \frac{2}{\sqrt{\alpha}}$$

$$\text{Kurtosis: } \frac{6}{\alpha}$$

$$\text{mgf: } (1 - \beta t)^{-\alpha} \text{ for } t < 1/\beta$$

$$\text{Char. func.: } (1 - \beta i t)^{-\alpha}$$

Chi-Square Distribution

In probability theory and Statistics, the Chi-square distribution (also Chi-squared or χ^2 distribution) is one of the theoretical probability distributions most widely used in inferential statistics, i.e. in statistical significance tests. It is useful because, under reasonable assumptions, easily calculated quantities can be proven to have distributions that approximate to the Chi-square distribution if the null hypothesis is true.

If X_i are k independent, normally distributed random variables with means μ_i and variances σ_i^2 , then the random variable

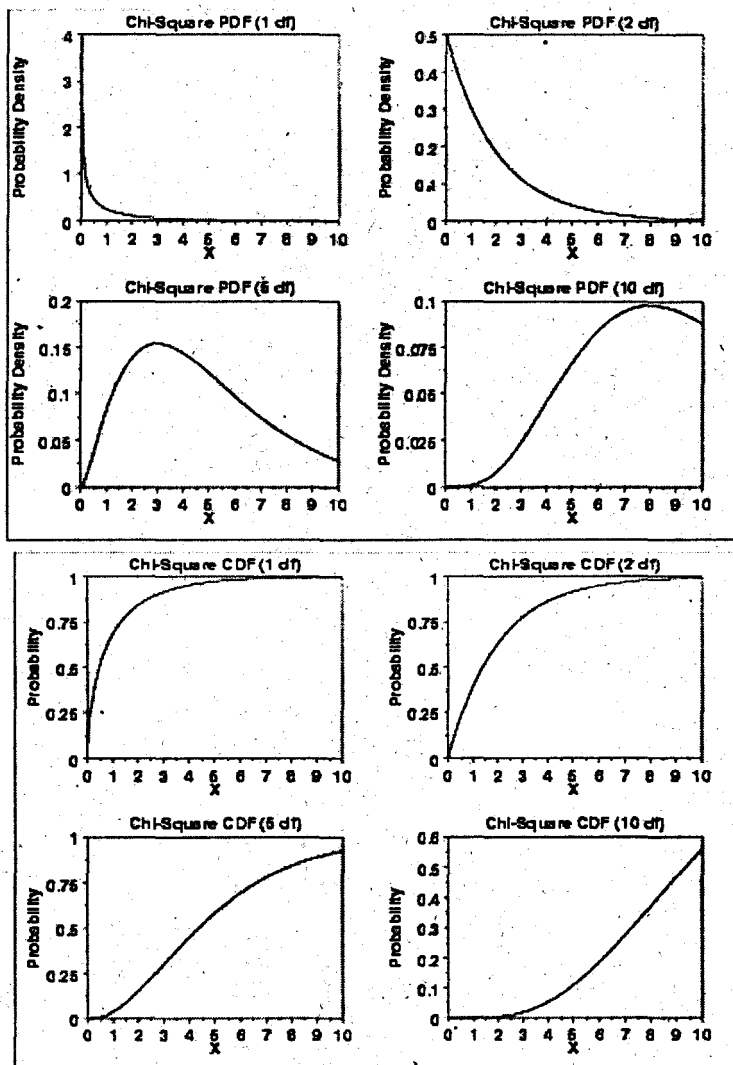
$$Z = \sum_{i=1}^k \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2$$

is distributed according to the Chi-square distribution.

The Chi-square distribution has one parameter: k - a positive integer which specifies the number of degrees of freedom (i.e. the number of X_i). The Chi-square distribution is a special case of the gamma distribution.

The best-known situations in which the Chi-square distribution is used are the common Chi-square tests for goodness of fit of an observed distribution to a theoretical one, and of the independence of two criteria of classification of qualitative data. Two common examples are the Chi-square test for independence in an $R \times C$ contingency table and the Chi-square test to determine if the standard deviation of a population is equal to a pre-specified value. However, many other statistical tests lead to a use of this distribution. One example is Friedman's analysis of variance by ranks.

The following are the plots of the Chi-Square probability density function and cumulative distribution function:



Distribution Summary:

Parameters: $k > 0$; degrees of freedom

$$\text{pdf:} \quad \begin{aligned} & \frac{1}{2^{k/2} \Gamma(k/2)} x^{\frac{k}{2}-1} e^{-\frac{x}{2}} & x \geq 0 \\ & 0 & \text{otherwise} \end{aligned}$$

Mean: k

Median: $k - \frac{2}{3}$

Mode: $k - 2$ if $k \geq 2$

Variance: $2k$

Skewness:

$$\sqrt{8/k}$$

Kurtosis: $12/k$

mgf: $(1 - 2t)^{-k/2}$ for $2t < 1$

Char. func.: $(1 - 2t)^{-k/2}$

Student's t -Distribution

The t -distribution or Student's t -distribution is a probability distribution that arises in the problem of estimating the mean of a normally distributed population when the sample size is small. It is the basis of the popular Student's t -test for the statistical significance of the difference between two sample means, and for confidence intervals for the difference between two population means. The Student's t -distribution is a special case of the generalized hyperbolic distribution.

The derivation of the t -distribution was first published in 1908 by William Sealy Gosset, while he worked at a Guinness brewery in Dublin. He was not allowed to publish under his own name, so the paper was written under the pseudonym Student. The t -test and the associated theory became well-known through the work of R.A. Fisher, who called the distribution "Student's t -distribution".

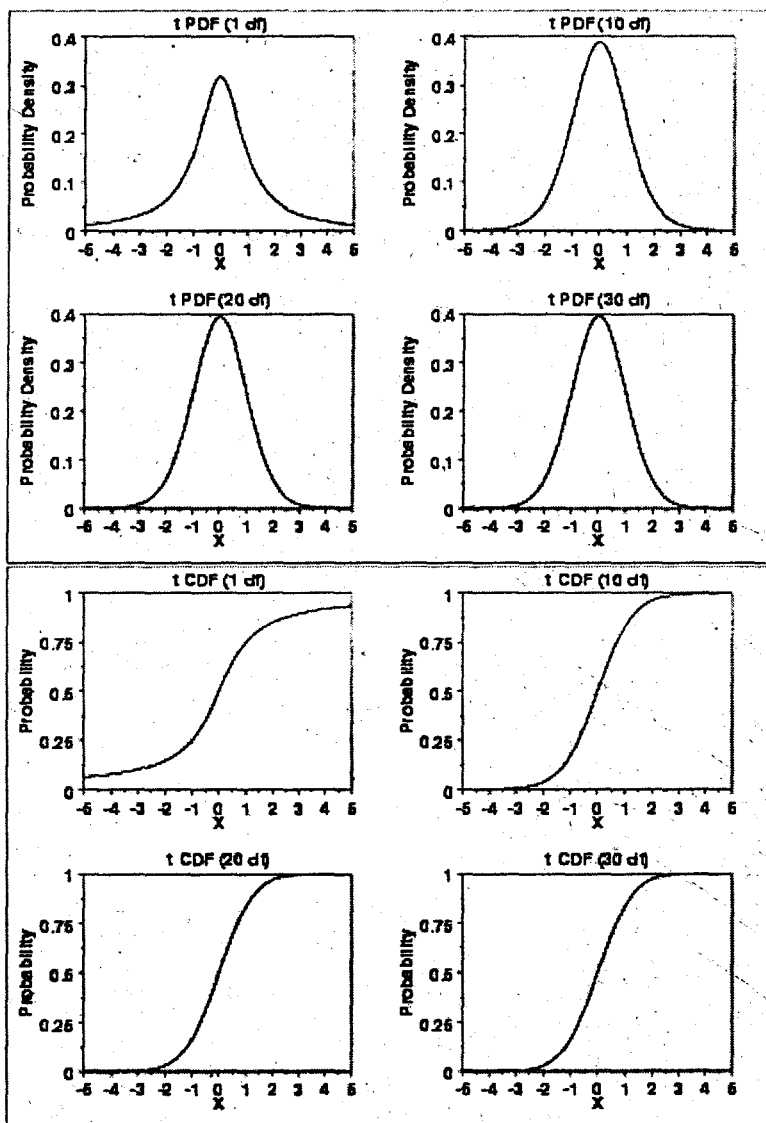
Student's t -distribution arises when (as in nearly all practical statistical work) the population standard deviation is unknown and has to be estimated from the sample data. A number of other statistics can be shown to have t -distribution for samples of moderate size under null hypotheses that are of interest, so that the t -distribution forms the basis for significance tests in other situations as well as when examining the differences between means. For example, the distribution of Spearman's rank correlation coefficient ρ in the null case (zero correlation) is well approximated by the t -distribution for sample sizes above or about 20.

Let V have a Chi-square distribution with ν degrees of freedom. Further suppose that Z is standard normal variate and, V and Z are independent then the ratio

$$\frac{Z}{\sqrt{V/\nu}}$$

has a t -distribution with ν degrees of freedom.

The following are the plots of the student's t probability density function and cumulative distribution function:



Distribution Summary:

Parameters: $\nu > 0$, degree of freedom (real)

pdf:
$$\frac{\Gamma((\nu + 1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)(1 + x^2/\nu)^{(\nu+1)/2}}$$
$$-\infty < x < \infty$$

Mean: 0 for $\nu > 1$, undefined for $\nu = 1$

Median: 0

Mode: 0

Variance: $\frac{\nu}{\nu - 2}$ for $\nu > 2$

Skewness: 0 for $\nu > 3$

Kurtosis: $\frac{3\nu - 6}{\nu - 4}$ for $\nu > 4$

***F*-Distribution**

The *F*-distribution is a continuous probability distribution. It is also known as Snedecor's *F*-distribution or the Fisher-Snedecor distribution (after R.A. Fisher and George W. Snedecor). *F* stands for Sir Ronald Fisher, English geneticist and statistician.

The distribution is used in the analysis of variance and is a function of the ratio of two independent random variables each of which has a Chi-square distribution and is divided by its number of degrees of freedom.

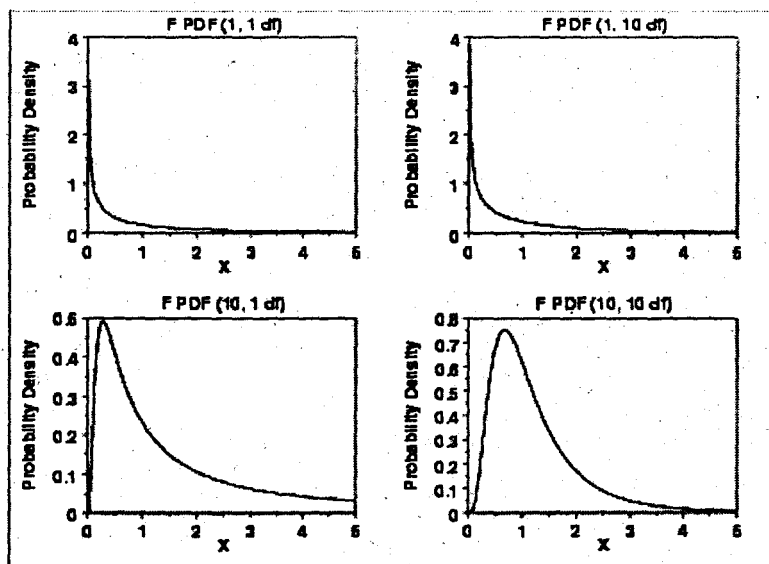
The *F* distribution is used in many cases for the critical regions for hypothesis tests and in determining confidence intervals. Two common examples are the analysis of variance and the *F*-test to determine if the variances of two populations are equal.

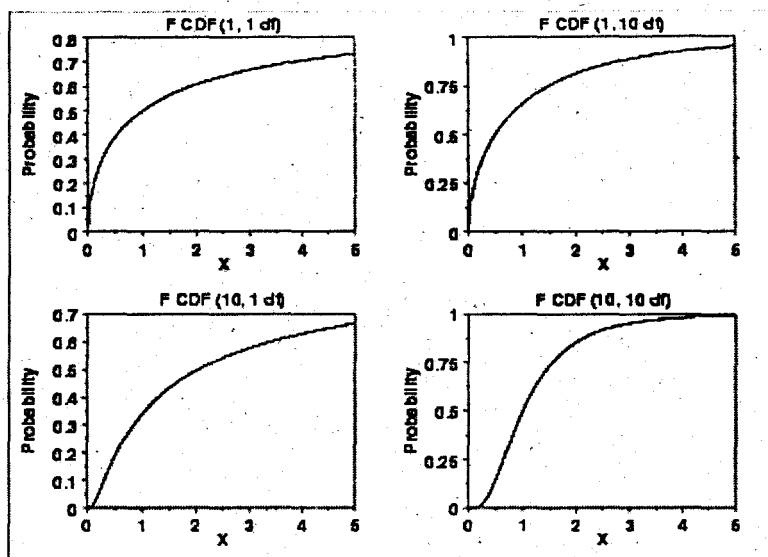
A random variate of the F -distribution arises as the ratio of two Chi-squared variates:

$$\frac{X_1/\nu_1}{X_2/\nu_2},$$

where X_1 and X_2 are independent and have Chi-square distributions with ν_1 and ν_2 degrees of freedom respectively.

The following are the plots of the F probability density function and cumulative distribution function:





Distribution Summary:

Parameters: $v_1 > 0, v_2 > 0$ degree of freedom

pdf:
$$\frac{\sqrt{\frac{(v_1 x)^{v_1} v_2^{v_2}}{(v_1 x + v_2)^{v_1 + v_2}}}}{x \beta \left(\frac{v_1}{2} + \frac{v_2}{2} \right)}$$

Mean:
$$\frac{v_2}{v_2 - 2} \text{ for } v_2 > 2$$

Mode:
$$\frac{v_1 - 2}{v_1} \frac{v_2}{v_2 + 2} \text{ for } v_1 > 2$$

Variance: $\frac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)}$ for $v_2 > 4$

Skewness: $\frac{(2v_1 + v_2 - 2)\sqrt{8(v_2 - 4)}}{(v_2 - 6)\sqrt{v_1(v_1 + v_2 - 2)}}$ for $v_2 > 6$

Questions-Answers

Q.1 *Why is the uniform distribution also called rectangular distribution?*

Ans. *The uniform distribution defines equal probability over a given range for a continuous distribution. For this reason, the shape of its probability density function is rectangular. It is parameterized by the smallest and largest values that the uniformly-distributed random variable can take, a and b . That is why uniform distribution is also known as rectangular distribution.*

Q.2 *What are the applications of uniform distribution?*

Ans. *The uniform distribution defines equal probability over a given range for a continuous distribution. For this reason, it is important as a reference distribution.*

One of the most important applications of the uniform distribution is in the generation of random numbers. That is, almost all random number generators generate random numbers on the $(0, 1)$ interval. For other distributions, some transformation is applied to the uniform random numbers.

Also when a p -value is used as a test statistic for a simple null hypothesis, and the distribution of the test statistic is continuous, then the test statistic is uniformly distributed between 0 and 1 if the null hypothesis is true.

Q.3 *What is the resemblance between exponential and geometric distribution?*

Ans. *The exponential distribution may be viewed as a continuous counterpart of the geometric distribution, which describes the number of Bernoulli trials necessary for a discrete process to change state. In contrast, the exponential distribution describes the time for a continuous process to change state.*

Q.4 *What are the applications of exponential distribution?*

Ans. *Exponential variables can be used to model situations where certain events occur with a constant probability per unit distance: the distance between mutations on a DNA strand;*

the distance between road-kill on a given street.

In queuing theory, the inter-arrival times (i.e. the times between customers entering the system) are often modeled as exponentially distributed variables. The length of a process that can be thought of as a sequence of several independent tasks is better modeled by a variable following the gamma distribution (which is a sum of several independent exponentially distributed variables).

Reliability theory and reliability engineering also make extensive use of the exponential distribution. Because of the memoryless property of this distribution, it is well-suited to model the constant hazard rate portion of the bathtub curve used in reliability theory. It is also very convenient because it is so easy to add failure rates in a reliability model. The exponential distribution is however not appropriate to model the overall lifetime of organisms or technical devices, because the "failure rates" here are not constant: more failures occur for very young and for very old systems.

In physics, if one observe a gas at a fixed temperature and pressure in a uniform gravitational field, the heights of the various molecules also follow an approximate exponential distribution.

Q.5 ***What is the importance of normal distribution in Statistics?***

Ans. *The fundamental importance of the normal distribution is its use as model of quantitative phenomena in the natural and behavioral sciences. For both theoretical and practical reasons, the normal distribution is probably the most important distribution in statistics. For example, many classical statistical tests are based on the assumption that the data follow a normal distribution. This assumption should be tested before applying these tests.*

In modeling applications, such as linear and non-linear regression, the error term is often assumed to follow a normal distribution with fixed location and scale.

The normal distribution is also used to find significance levels in many hypothesis tests and confidence intervals.

Q.6 ***Give the practical importance of the Central Limit Theorem (CLT).***

Ans. *The practical importance of the CLT is that the normal distribution can be used as an approximation to many other distributions. For example a binomial distribution with parameters n and p is*

approximately normal for large n and p not too close to 1 or 0 (see CLT above for more details).

Q.7 *What is the relation between exponential and gamma distribution?*

Ans. *Gamma variate is the sum of independent exponential variates.*

Q.8 *What is the relation between Chi-square distribution and gamma distribution?*

Ans. *Chi-square distribution is a special case of gamma distribution with $\alpha = k/2$ and $\beta = 2$, where k is the degree of freedom of Chi-square distribution.*

Q.9 *What are the applications of Chi-Square distribution?*

Ans. *The best-known situations in which the Chi-square distribution is used are the common Chi-square tests for goodness of fit of an observed distribution to a theoretical one, and of the independence of two criteria of classification of qualitative data. However, many other statistical tests lead to a use of this distribution. One example is Friedman's analysis of variance by ranks.*

Q.10 *What are the applications of t-distribution?*

Ans. *Student's t-distribution arises when (as in nearly all practical statistical work) the population standard deviation is unknown and has to be estimated from the sample data. A number of other statistics can be shown to have t-distributions for samples of moderate size under null hypotheses that are of interest, so that the t-distribution forms the basis for significance tests in other situations as well as when examining the differences between means. For example, the distribution of Spearman's rank correlation coefficient ρ in the null case (zero correlation) is well approximated by the t distribution for sample sizes above or about 20.*

Q.11 *What is the shape of t-distribution?*

Ans. *The overall shape of the probability density function of the t-distribution resembles the bell shape of a normally distributed variable with mean 0 and variance 1, except that it is a bit lower and wider.*

Q.12 *What are the uses of F-distribution?*

Ans. *The F-distribution is used in many cases for the critical regions for hypothesis tests and in determining confidence intervals. Two common examples are the analysis of variance (ANOVA) and the F-test to determine if the variances of two populations are equal.*

Exercises

Exercise 5.1 (MCQs')

- Q.1** A continuous random variable is a random variable that can:
- (a) Assess only countable values.
 - (b) Assess any value in one or more intervals.
 - (c) Have no random sample.
 - (d) Assume no continuous random frequency.
- Q.2** For a continuous random variable the area under the probability distribution curve between any two points is always:
- (a) Greater than one
 - (b) Less than zero
 - (c) Equal to one
 - (d) In the range zero and one
- Q.3** The probability that a continuous random variable assumes a single value is:
- (a) Less than one
 - (b) Greater than zero
 - (c) Equal to zero
 - (d) Between zero and one
- Q.4** For a continuous random variable X , the total probability of the mutually exclusive events (intervals) within which X can assume a value is:
- (a) Less than one
 - (b) Greater than one
 - (c) Equal to one
 - (d) Between zero and one

- Q.5** If X is a uniform variate $U(5, 10)$, then the mean of X is
(a) 5
(b) 7.5
(c) 10
(d) 15
- Q.6** If X is a uniform variate $U(5, 10)$, then the variance of X is
(a) 0.417
(b) 2.08
(c) 7.5
(d) 5
- Q.7** If mean of exponential distribution is 2 then its variance is
(a) 2
(b) 1
(c) 6
(d) 4
- Q.8** If mean of exponential distribution is 2 then sum of such 10 independent variates will follow gamma distribution with mean
(a) 2
(b) 10
(c) 5
(d) 20
- Q.9** If mean of exponential distribution is 2 then sum of such 10 independent variates will follow gamma distribution with variance
(a) 2
(b) 10
(c) 20
(d) 200
- Q.10** If $X \sim N(\mu, \sigma^2)$ and a and b are real numbers, then mean of $(aX + b)$ is
(a) $a + b$
(b) $a\mu + b$
(c) $a\mu$
(d) $a + b\mu$

- Q.11 If $X \sim N(\mu, \sigma^2)$ and a and b are real numbers, then variance of $(aX + b)$ is
- (a) $a + b$
 - (b) $a^2 \sigma^2 + b$
 - (c) $a^2 \sigma^2$
 - (d) $a + b \sigma^2$
- Q.12 Which of the following is not a characteristic of normal distribution?
- (a) The total area under the curve is equal to one.
 - (b) The curve is symmetric about the mean.
 - (c) The value of the mean is always greater than the value of the standard deviation.
 - (d) The two tails of the curve extend indefinitely.
- Q.13 The total area under a normal distribution curve to the left of the mean is always:
- (a) 1
 - (b) 0
 - (c) 0.5
 - (d) 0.9
- Q.14 The tails of the normal distribution:
- (a) Meet the horizontal axis at $z = 3.0$
 - (b) Never meet or cross the horizontal axis
 - (c) Cross the horizontal axis at $z = 4.0$
 - (d) Are asymmetric
- Q.15 For a normal distribution, the Z value for an X value is to the right of the mean is always:
- (a) Equal to zero
 - (b) Negative
 - (c) Greater than one
 - (d) Positive.

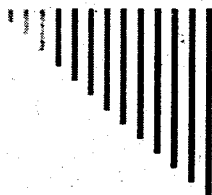
- Q.16** For a normal distribution, the mean of 40 and a standard deviation of 8. The value of Z for $X = 52$ is:
- (a) 2.00
 - (b) 1.50
 - (c) -1.75
 - (d) 0.80
- Q.17** If mean of the Chi-square distribution is 4 then its variance is:
- (a) 2
 - (b) 4
 - (c) 1
 - (d) 8
- Q.18** If X follows t -distribution with ν d.f. then the distribution of X^2 is:
- (a) Standard Normal
 - (b) Chi-square with ν d.f.
 - (c) F with 1 and ν d.f.
 - (d) F with ν and 1 d.f.
- Q.19** The area under the normal curve within two standard deviation of the mean is:
- (a) 68.26%
 - (b) 95.44%
 - (c) 99.73%
 - (d) 99.99%
- Q.20** The distribution of sample correlation is
- (a) Normal
 - (b) t
 - (c) Chi-square
 - (d) F

Exercise 5.2 (True/False)

Read the following statements carefully and indicate which statement is "True" or "False":

1. If X is a continuous uniform random variable $U(a, b)$ then $P(X = c) = 1/(b-a)$ where $a < c < b$.
2. The inflection points are the points where of a curve convexity changes.
3. The right and left tails of the normal curve extend indefinitely, never touching the horizontal axis.
4. For a normal distribution, the mean always lies between the mode and the median.
5. If $U_1, U_2, \dots, U_n$ are independent Chi-square variables with one degree of freedom, then the distribution of $U_1 + U_2 + \dots + U_n$ is Chi-square with n degrees of freedom.
6. If Z is a standard normal random variable, then the distribution of Z^2 is Chi-square with one degree of freedom.
7. As the number of degrees of freedom grows, the t -distribution approaches the normal distribution with mean 0 and variance 1.
8. The distribution of sample variance is F -distribution.
9. The mean of t -distribution is zero.
10. The median of t -distribution is zero.
11. The mode of t -distribution is its degree of freedom.
12. If the expected waiting time for certain service is 5 minutes then the variance of waiting time is also 5 minutes when we consider the waiting time to follow exponential distribution.

13. The variance of t -distribution does not exist if the degree of freedom is less than 2.
14. If X follows F -distribution with v_1 and v_2 d.f. then $1/X$ has standard normal distribution.
15. If X follows F -distribution with v_1 and v_2 d.f. then $1/X$ has also F -distribution but with v_2 and v_1 d.f.



Chapter 6

Regression & Correlation

Regression

Regression Analysis

Regression analysis refers to the methods of describing the functional relationship between the dependent and independent variable. It is used to predict value of one variable (dependent) on the basis of other (independent) variables.

Scatter Diagram

When we plot data in such a way that we obtain dots showing relation between dependent and independent variables, it is called scatter diagram.

Regression Line

A regression line is a line drawn through the points on a scatter plot to summarize the relationship between the variables being studied. When it slopes down (from top left to bottom right), this indicates a negative or inverse relationship between the variables; when it slopes up (from bottom right to top left), a positive or direct relationship is indicated. The regression line often represents the regression equation on a scatter plot.

If this line is a straight line then the regression is called linear regression while, on the other hands, it is referred as non-linear regression.

Regression Equation/Model

A regression equation (model) allows us to express the relationship between two (or more) variables algebraically. It indicates the nature of the

relationship between two (or more) variables. In particular, it indicates the extent to which you can predict some variables by knowing others, or the extent to which some are associated with others.

A simple linear regression equation is usually written;

$$Y = \beta_0 + \beta_1 X + \varepsilon,$$

where Y is the dependent variable, β_0 is the intercept; the average value of Y when there is no contribution of X , β_1 is the slope or regression coefficient; rate of change in Y when unit change in X , X is the independent variable (or covariate), ε is the random error term.

The equation will specify the average magnitude of the expected change in Y given a change in X . The regression equation is often represented on a scatter plot by a regression line.

Assumptions of Linear Regression Model

Some important assumptions are:

- (i) ε_i 's are normally distributed with zero mean and a constant variance σ^2 .
- (ii) $E(\varepsilon_i \varepsilon_j) = 0$ for all $i \neq j$ i.e., error terms are independent of each other.
- (iii) $E(X\varepsilon_i) = 0$ for all i , i.e., X and ε are independent of each other.

Simple Regression

Simple regression investigates the effect of one independent variable on the dependent variable.

Multiple Regression

Multiple regression investigates the effect of two or more independent variables on the dependent variable.

Method of Least Squares

The method of least squares is a criterion for fitting a specified model to observed data such that the sum of squares of the residuals (difference between observed and estimated value) is minimized.

The estimated model can be written as

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

Also $y - \hat{y} = e = \text{residual}.$

Standard Error of Estimate

A measure of the scatter of actual values of the dependent variable from their estimated values. It can be used to set up confidence intervals for the actual values of the dependent variables.

Total Variation

A measure of the variation of the actual values of the dependent variable from their mean.

Symbolically,

$$\text{Total Sum of Squares} = TSS = \sum (y - \bar{y})^2$$

$$\text{Total Variation} = \text{Explained Variation} + \text{Unexplained Variation.}$$

Explained Variation

A measure of the variation of the estimated values of the dependent variable from the mean of actual values.

Symbolically,

$$\text{Explained Sum of Squares} = ESS = \sum (\hat{y} - \bar{y})^2$$

Unexplained Variation

A measure of the variation of the actual values of the dependent variable from the estimated values of that variable.

Symbolically,

$$\text{Residual Sum of Squares} = RSS = \sum (y - \hat{y})^2$$

Coefficient of Determination

It measures the relative amount of variation in the dependent variable that has been explained by variation in the independent variable. It is the measure of strength of association that exists between variables.

It is denoted by R^2 .

$$R^2 = \frac{ESS}{TSS} = \frac{\sum (\hat{y} - \bar{y})^2}{\sum (y - \bar{y})^2}$$

Correlation

Correlation Analysis

Correlation analysis refers to the methodology of measuring the interdependence between two or more variables. For this measure, we compute coefficient of correlation whose value ranges from -1 to +1. The negative correlation shows that both the series under consideration are moving in different directions while the positive correlation shows that both the series jointly move in the same direction i.e., either going to increase or going to decrease. The zero correlation shows the statistical independence of the variables. Negative correlation shows indirect relation among the series while positive correlation shows a direct relation. Furthermore, -1 or +1 correlation shows a perfect relation among the series.

It is also to be noted that correlation coefficient is the square root of the coefficient of determination

Multiple Regression Correlation Coefficient

The multiple regression correlation coefficient, R^2 , is a measure of the proportion of variability explained by, or due to the regression (linear relationship) in a sample of paired data. It is a number between zero and one and a value close to zero suggests a poor model. A very high value of R^2 can arise even though the relationship between variables is non-linear. The fit of a model should never simply be judged from the R^2 value.

Partial Correlation

A partial correlation is used to measure the degree of linear relationship between any two variables in a multivariable problem by removing any common relationship or influence with all other variables.

Transformation to Linearity

Transformations allow us to change all the values of a variable by using some mathematical operation, for example, we can change a number, group of numbers, or an equation by multiplying or dividing by a constant or taking the square root. A transformation to linearity is a transformation of a response variable, or independent variable, or both, which produces an approximate linear relationship between the variables.

Questions-Answers

Q.1 *How would you decide to choose regression or correlation analysis for study of relationship?*

Ans. *If we want to measure of dependence of one variable on one or more variables, we use regression analysis. On the other hand, if we want to measure the interdependence among two or more variables, we use correlation analysis.*

Q.2 *Define Pearson's Product Moment Correlation Coefficient.*

Ans. *Pearson's product moment correlation coefficient, usually denoted by r , is one example of a correlation coefficient. It is a measure of the linear association between two variables that have been measured on interval or ratio scales, such as the relationship between height in inches and weight in pounds. However, it can be misleadingly small when there is a relationship between the variables but it is a non-linear one.*

Q.3 *Define Spearman's Rank Correlation Coefficient.*

Ans. *The Spearman rank correlation coefficient is usually calculated on occasions when it is not convenient, economic, or even possible to give actual values to variables, but only to assign a rank order to instances of each variable. It may also be a better indicator that a relationship exists between two variables when the relationship is non-linear.*

Q.4 *Before correlating two series, how can you determine which will be the dependent variable?*

Ans. *The dependent variable is the one for which estimates and forecasts will be made.*

Q.5 *Is the computation of coefficient of correlation is part of regression analysis?*

Ans. *No, although a correlation analysis usually makes a regression analysis more meaningful and useful.*

Q.6 *Can a coefficient of correlation be computed without computing a regression equation?*

Ans. *Yes.*

Q.7 *Is a coefficient of correlation, in any way, related to a regression equation even though a regression equation has not been computed?*

Ans. *A coefficient of correlation is always related to some regression equation. Closer the dots on the scatter diagram cluster about a regression line the higher the coefficient of correlation.*

Q.8 *Is it important to construct a scatter diagram before computing a regression line or coefficient of correlation, why or why not?*

Ans. *Yes, a scatter diagram is necessary in order to determine what type of regression line and coefficient of correlation to compute. The shape of the scatter of the dots on the diagram will be the determine factor.*

Q.9 *Under what conditions, can Spearman's rank correlation coefficient be used effectively?*

Ans. *The Spearman's rank correlation coefficient is used for measuring the closeness of the relationship of factors which can be ranked in order of importance or magnitude but for which exact measurement cannot be made. Also data for very small samples may be ranked and analyzed. A regular coefficient of correlation is not a very reliable measure for small samples because of sampling errors.*

Q.10 *Why do r (Coefficient of Correlation) and β (Coefficient of Regression) always have the same sign?*

Ans. *We know that:*

$$r_{XY} = r_{YX} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

and

$$\beta_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_Y^2},$$

$$\beta_{YX} = \frac{\text{Cov}(X, Y)}{\sigma_X^2}$$

Since $V(X)$ and $V(Y)$ both have positive values and only the quantity $\text{Cov}(X,Y)$ can have a positive or negative sign. According to the above equations if $\text{Cov}(X,Y)$ is positive then r and β are positive and vice versa. So it is clear that due to common numerator i.e. $\text{Cov}(X,Y)$ both r and β have the same sign.

Q.11 Define multiple regressions and multiple correlations.

Ans. Multiple regression is the term used concerning the combined effect of two or more independent variables on one dependent variable. Multiple correlation is the term used concerning the measurement the degree of closeness of the combined movement of two or more independent variables and the movement of the one dependent variable.

Q.12 What are the net regression coefficients?

Ans. A net regression coefficient measures the effect of one independent variable on the dependent variable when the effects of other independent variables are held constant.

Q.13 What does a coefficient of multiple determination measure?

Ans. A coefficient of multiple determination measures the proportion of the total variation in the dependent variable that has been explained by the combined movements of the several independent variables.

Q.14 What does a coefficient of multiple correlation measure?

Ans. The coefficient of multiple correlation is the square root of the coefficient of multiple determinations. It measures the degree to which the movement of the several independent variables is synchronized at the same time with the movement of the dependent variable.

Q.15 What does a coefficient of partial correlation measures?

Ans. A partial coefficient correlation measures the degree of the net influence of the movements of an independent variable on the movements of the dependent variable.

Q.16 *Of what use is a standard error of estimate in a multiple regression analysis?*

Ans. *The standard error of estimate indicates the reliability of the estimates made with the estimating equations. It can be used to calculate prediction intervals for actual values of the dependent variable.*

Q.17 *Under what condition will the net regression coefficient be smaller than their corresponding simple regression coefficients?*

Ans. *The net regression coefficients will tend to be smaller when there is a positive simple correlation between the two independent variable in a three variable multiple regression analysis.*

Q.18 *How is the sign (+ or -) determines for a coefficient of partial correlation?*

Ans. *The partial correlation coefficient receives the same sign as its corresponding net regression coefficient.*

Q.19 *A sign should not be affixed to a coefficient of multiple correlation. Why?*

Ans. *A sign should not be affixed because some independent variables may be positively correlated with the dependent variables while other are negatively correlated.*

Q.20 *Why are two variables are sometimes closely related?*

Ans. *The two variables may be closely related because changes in one causes in changes in another or because both variables are affected similarly by other forces. A close correlation can also occur by chance.*

Q.21 *If two variables are closely correlated, then the movements in one variable cause the movements in another. Why or why not?*

Ans. *No, there is no proof of causation in correlation theory. A correlation analysis never proves or disproves that there is relationship between two variables. All it does is measure the closeness of the relationship.*

Q.22 *What is the interpretation if coefficient of determination $R^2 = 0.7$?*

Ans. It means that 70% variation of the dependent variable can be explained by the regression. Hence it shows relatively good fit of regression model.

Q.23 What values does r assume if all the sample points fall on the same straight line and if the line has:

(a) a positive slope (b) a negative slope.
Ans. (a) $+1$ (b) -1 .

Q.24 A police research cell has shown that the crime rate is correlated with the number of unemployed people in Pakistan. Would you expect the correlation to be positive or negative?

Ans. Positive, because as number of unemployed people increases, the crime will also increase.

Q.25 Is the demand for a product correlated with price? If it is, would you expect the correlation to be positive or negative?

Ans. Yes; Negative.

Q.26 Do you believe that an exact relationship exists between two variables?

Ans. In reality, the answer is very definite no.

Q.27 What is the general form of a probabilistic model?

Ans. $Y = \text{Deterministic component} + \text{Random error}$. The mean value of the random error term equals zero. This is equivalent to assuming that the mean value of Y , i.e. $E(Y) = \text{Deterministic component}$.

Q.28 Give any example for the use of partial correlation.

Ans. Correlation between amount of additive oil and mileage covered by vehicle while removing the effect of age of the engine.

Q.29 Define the straight line probabilistic model.

Ans. The equation of the straight line is:

$$Y = a + bX + e,$$

Where Y = Variable to be predicted, called dependent variable and X = independent variable.

$E(Y) = a + bX$ is the deterministic portion of the model.

$a = Y$ - intercept of the line and $b =$ Slope of the line, i.e. rate of change of in the deterministic component of Y for every 1 unit increase in X .

Q.30 Show that the coefficient of correlation is the geometric mean of the coefficient of regression.

Ans. Regression coefficient of Y on X is: $b_{yx} = \frac{r\sigma_x}{\sigma_y}$

Regression coefficient of X on Y is: $b_{xy} = \frac{r\sigma_y}{\sigma_x}$

$$\text{G.M. of Regression coefficients} = \sqrt{b_{yx} \cdot b_{xy}} = \sqrt{\frac{r\sigma_y}{\sigma_x} \cdot \frac{r\sigma_x}{\sigma_y}} = r$$

Q.31 Is this statement correct? Give reasons, "The regression coefficients of X on Y is 3.2 and that of Y on X is 0.8.

Ans. No, it is not correct.

Here $b_{yx} = 0.8$ and $b_{xy} = 3.2$

$$\begin{aligned} r &= \sqrt{b_{yx} \cdot b_{xy}} \\ &= \sqrt{0.8 \times 3.2} \\ &= \sqrt{2.56} \\ &= \pm 1.6 \end{aligned}$$

Since $-1 \leq r \leq 1$. Therefore $|r| = 1.6$ is impossible, hence the given statement is wrong.

Q.32 Whether it is true that if one of the regression coefficient is greater than unity, the other must be less than unity?

Ans. Yes, it is true. Suppose that one of the regression coefficients say $b_{yx} > 1$ then $1/b_{yx} < 1$, also $r^2 \leq 1 \Rightarrow b_{yx} b_{xy} \leq 1 \Rightarrow b_{xy} \leq 1$.

Q.33 If two regression coefficients are 0.8 and 0.2. What would be the value of coefficient of correlation.

Ans. $r^2 = b_{yx} \cdot b_{xy} = (0.8) (0.2) = 0.16$,
 $r = 0.4$, positive since b_{yx} and b_{xy} both will be positive.

Exercises

Exercise 6 (MCQs')

- Q.1** The major difference between regression analysis and correlation analysis is that in regression analysis:
- (a) The independent variable is known without error.
 - (b) Both variables are random variables.
 - (c) The t -distribution is used for hypothesis testing.
 - (d) Only the slope is important.
- Q.2** The term homoscedasticity refers to:
- (a) Equal variances for X and Y .
 - (b) A normal bivariate distribution only.
 - (c) Equal variance of the dependent variable, Y , for any value of independent variable, X .
 - (d) Uniform variance of the independent variable.
- Q.3** The sum of squares of which type of deviations is minimized by the least square regression:
- (a) Deviation perpendicular to the line.
 - (b) Deviation parallel to the line.
 - (c) Deviations of the values of the dependent variable from the line.
 - (d) Deviations of values of the independent variable from the line.
- Q.4** A random sample of paired observations has been selected and the sample correlation coefficient has been found to be -1 . From this result we know that:
- (a) At least one sample observation does not lie on the sample regression line.
 - (b) All sample observations lie on the sample regression line

- (c) At least one sample observation lies on the population Regression.
(d) None of above
- Q.5** The output of a certain chemical-processing machine is linearly related to temperature. At -10°C the processor output is 200 Kgs. per hour and at 40°C the output is 220 Kgs. per hour. Calculate the linear equation for Kgs. per hour of output (Y) as a function of temperature in degree Celsius (X):
(a) $Y = 193.3 + 0.67X$ (b) $Y = 204.0 + 0.40X$
(c) $Y = -290.0 + 1.50X$ (d) $Y = -510.0 + 2.50X$
- Q.6** What information is given by a value of the coefficient of determination?
(a) Strength of relationship
(b) Both strength and direction of relationship
(c) Neither strength nor direction of relationship
(d) Direction of relationship only
- Q.7** For the regression equation $\hat{Y} = 10 + 2X$, the Y intercept is:
(a) 10 (b) 2 (c) 0 (d) -2
- Q.8** The correlation coefficient:
(a) Can be calculated only after regression analysis is performed.
(b) May be smaller than -1 only when X and Y are inversely related.
(c) Equals the positive or negative square root of the coefficient of determination.
(d) Equals the standard error of the estimate divided by the square root of the sample size
- Q.9** If the coefficient of determination is 0.49; the correlation coefficient may be:
(a) 0.51 (b) 0.49 (c) 0.24 (d) 0.70
- Q.10** The estimated regression line relating the market value of a person's stock portfolio to his annual income is $\hat{Y} = 5,000 + 0.10X$. This means that each additional rupee of income will increase the stock portfolio by:

- | | | | |
|-----|---------|-----|----------|
| (a) | Rs.0.50 | (b) | Rs.1.00 |
| (c) | Rs.0.10 | (d) | Rs.10.00 |

- Q.11** Which one of the following situations is inconsistent?
- (a) $\hat{Y} = 500 + 0.01 X$, and $r = 0.75$
(b) $\hat{Y} = -200 + 0.9 X$, and $r = -0.86$
(c) $\hat{Y} = 10 + 2 X$, and $r = 0.50$
(d) $\hat{Y} = -8 - 3 X$, and $r = -0.95$

- Q.12** Which one of the following statement is true?
- (a) The estimated and the true regression lines are always the same.
(b) The Y 's must be normally distributed about true regression line before the sample coefficient of determination can be calculated.
(c) The units in which X and Y are measured will not affect the value of r .
(d) The correlation coefficient can be calculated only after the estimated regression line has been found.

- Q.13** The true correlation coefficient ρ will be zero only if:
- (a) The Y intercept of the true regression line is equal to zero
(b) The slope of the true regression line is equal to zero
(c) $r = 0$
(d) $b = 0$

- Q.14** Whenever predictions are made from the estimated regression line, the relation between X and Y is assumed to be:
- (a) Direct (b) Inverse
(c) Linear (d) Perfect

- Q.15** The estimated coefficient of determination is equal to all except which one of the following?
- (a) The square of the correlation coefficient.
(b) The proportion of variation in Y explained by regression.
(c) 1 minus the proportion of variation in Y unexplained by regression.
(d) The slope of the estimated regression line.

- Q.16** The coefficient of partial determination differs from the coefficient of multiple determination in that:
- (a) One provides the slope of the regression plane and the other does not.
 - (b) One may be negative, but the other is always positive.
 - (c) Both of the above.
 - (d) None of the above.
- Q.17** The coefficient of multiple determination is 0.81. Thus, the multiple correlation coefficient is:
- (a) 0.19 (b) 0.9 (c) 0.6561 (d) 0.1
- Q.18** A larger sample size can be expected to achieve all but which one not of the following?
- (a) A smaller value for standard error of regression.
 - (b) Increase degrees of freedom.
 - (c) An estimated regression plane that is closer to the true regression plane.
 - (d) Increase in the value of coefficient of determination.
- Q.19** In multiple linear regression analysis, the square root of Mean Squared Error (MSE) is called the:
- (a) Multiple correlation coefficient
 - (b) Standard error of estimate
 - (c) Coefficient of determination
 - (d) Coefficient of non-determination
- Q.20** In multiple regression analysis, the purpose of solving the normal equations is to find:
- (a) The standard error of estimate.
 - (b) The constant and coefficients in the least squares relationship.
 - (c) The number of independent variables in the least squares relationship.
 - (d) The variance around the least squares relationship.



Chapter 7

Sampling

Basics of Sampling

Population

A group of individuals or entities about which you wish to know something and from which a sample will be taken, is called population.

Sampling Unit

This is a single member of a population; e.g., if the population is defined to be 100 trees on a lot, then the sampling unit is a single tree.

Sample

A sample is a sub-collection of elements drawn from a population; a subset of a population (a collection of Sampling Units), with the assumption that it represents the whole population.

Sampling

The procedure by which a few subjects are chosen from the universe (population) to be studied in such a way that the sample can be used to estimate the same characteristics in the total is referred to as sampling.

Sampling Frame

A list of all the sampling units. It includes lists that are available or that are constructed from different sources specifically for the study. Directories, membership or customer lists, even invoices or credit card receipts can serve as a sampling frame.

Sample Design

A program layout including all the procedures to take a sample.

Parameter

A summary, typically numerical, of a variable (or variables) over the entire population.

Statistic

A summary, typically numerical, of a variable (or variables) over a sample.

Statistical Inference

The process of making a statement about a population on the basis of sample information.

Example:

In an example to find out the average number of cups of tea that office worker take daily in a particular city:

<i>Population</i>	=	all the office workers in that city
<i>Sampling Unit</i>	=	an individual worker
<i>Sampling frame</i>	=	list of all the office workers obtained from all the offices of the said city
<i>Design</i>	=	probability sampling (discussed later)
<i>Sample</i>	=	200 (say) office workers
<i>Data gathered</i>	=	the daily tea consumption (number of cups) of each of the 200 workers selected in the sample
<i>Statistic</i>	=	the average number of tea cups/day of the 200 workers selected in the sample
<i>Parameter</i>	=	the average number of tea consumption (cups/day) the office workers obtained from all the offices of the said city

Census

The method that collects data from all members of the population, rather than from a selected subset of the population.

Sampling Techniques

Non-Probability Sampling Techniques

In non-probability sampling, the sample is selected in such a way that the chance of being selected of each unit within the population or universe is unknown. Indeed, the selection of the subjects is arbitrary or subjective, since the researcher relies on his/her experience and judgment.

There are five main types of non-probability sampling that we will review more closely:

- *Purposive Sampling*
- *Convenience Sampling*
- *Quota Sampling*
- *Judgment Sampling*
- *Snowball Sampling*

Purposive Sampling

In purposive sampling, the researcher selects the units with some purpose in mind, for example, students who live in dorms on campus, or experts on urban development. This is where the researcher targets a group of people believed to be typical or average, or a group of people specially picked for some unique purpose. The researcher never knows if the sample is representative of the population, and this method is largely limited to exploratory research. Sampling of cricket players, sampling of dresses in a market, and sampling of Jewelry etc., are few examples of purposive sampling.

Convenience Sampling

A convenience sample is used when a researcher simply stop anybody in the street who is prepared to stop, or when a researcher wander round a business, a shop, a restaurant, etc. and asking people he meet whether they

will answer his questions. In other words, the sample comprises subjects who are simply available in a convenient way to the researcher. There is no randomness and the likelihood of bias is high. One can't draw any meaningful conclusions from the results he obtains.

Quota Sampling

In quota sampling, the researcher constructs quotas for different types of units. For example, to interview a fixed number of shoppers at a mall, half of whom are male and half of whom are female. It is widely used in opinion polling and market research. Interviewers are each given a quota of subjects of specified type to attempt to recruit. e.g., an interviewer might be told to go out and select 20 working men and 20 working women, 10 school girls and 10 school boys so that they could interview them about their television viewing.

Judgement Sampling

In judgement sampling, the researcher or some other "expert" uses his/her judgement in selecting the units from the population for study based on the population parameters.

This type of sampling technique might be the most appropriate if the population to be studied is difficult to locate or if some members are thought to be better (more knowledgeable, more willing, etc.) than others to interview.

Snowball Sampling

It is also called network, chain, or reputational sampling. This method begins with a few people or cases and then gradually increases the sample size as new contacts are mentioned by the people you started out with.

Probability Sampling Techniques

In probability sampling, the sample is selected in such a way that each unit within the population or universe has a known chance of being selected. It is this concept of "known chance" that allows for the statistical projection of characteristics based on the sample to the population.

Following are the main types of probability or random sampling:

- *Simple Random Sampling*
- *Stratified Random Sampling*
- *Systematic Random Sampling*
- *Cluster (Area) Random Sampling*

Simple Random Sampling

A sampling procedure that assures that each element in the population has an equal chance of being selected is referred to as simple random sampling. Let us assume you had a school with a 1000 students, divided equally into boys and girls, and you wanted to select 100 of them for further study. You might put all their names in a drum and then pull 100 names out. Not only does each person have an equal chance of being selected, we can also easily calculate the probability of a given person being chosen, since we know the sample size (n) and the population size (N) and it becomes a simple matter of division:

$$n/N \times 100 \text{ or } 100/1000 \times 100 = 10\%.$$

This means that every student in the school has a 10% or 1 in 10 chance of being selected using this method.

Stratified Random Sampling

Stratified sampling techniques are generally used when the population is heterogeneous, or dissimilar, where certain homogeneous, or similar, sub-populations can be isolated (strata). In this random sampling technique, the whole population is divided first into mutually exclusive subgroups or strata and then units are selected randomly from each stratum. The segments are based on some predetermined criteria such as geographic location, size or demographic characteristic. This method is appropriate when you're interested in correcting for gender, race, or age disparities in your population. It is important that the segments be as heterogeneous as possible.

Systematic Random Sampling

In such sampling, each unit in the population is identified, and each unit has an equal chance of being selected in the sample. For example, to select a sample of 25 shops in a market, make a list of all the shops. Say there are 100 shops, number these shops from 1 – 100. Divide the total number of shops (100) by the number of shops you want in the sample (25). The answer is 4.

$$N = 100$$

$$n = 25$$

$$K = N / n = 100/25 = 4 \text{ (Sampling Interval).}$$

This means that you are going to select every fourth shop from the list. But you must first consult a table of random digits. Pick any point on the table, and read across or down until you come to a digit between 1 and 4. This is your random starting point. Say your random starting point is "3". This means you select shop 3 as your first shop, and then every fourth shop down the list (3, 7, 11, 15, 19, etc.) until you have 25 shop selected.

Cluster Sampling

Cluster sampling is a sampling technique where the entire population is divided into a number of heterogeneous groups, or clusters, and a random sample of these clusters is selected. All observations in the selected clusters are included in the sample.

Contrary to simple random sampling and stratified sampling, where single subjects are selected from the population, in cluster sampling the subjects are selected in groups or clusters. Suppose, you have a population that is dispersed across a wide geographic region. This method allows you to divide this population into clusters (usually counties, census tracts, or other boundaries) and then randomly sample everyone in those clusters. This approach allows to overcome the constraints of costs and time associated with a very dispersed population. Cluster sampling views the units in a population as not only being members of the total population but as members also of naturally-occurring in clusters within the population. For example, city residents are also residents of neighborhoods, blocks, and housing structures.

Bias and Errors

Bias and Sampling Bias

Bias is a term that refers to how far the average statistic lies from the parameter it is estimating, that is, the error that arises when estimating a quantity stated differently it is the difference between the expected value of an estimator and the true population value being estimated. It is also known as the systematic component of error. Errors from chance will cancel each other out in the long run, those from bias will not.

For a given estimator, the difference between the expected value of the estimator based on a sample and the estimate that would result if the sample were to include the entire population. i.e.,

If $\hat{\theta}$ is an estimator of θ then

$$\text{Bias} = E(\hat{\theta}) - \theta.$$

Standard Error

The standard deviation of the sampling distribution tells us something about how different samples would be distributed. It is referred to as the standard error.

Sampling Error

For a given estimator, the difference between an estimate based on a sample and population parameter.

Non-Sampling Error

For a given estimator, the difference between the estimate that would result if the sample were to include the entire population and the true population value being estimated.

Coverage Error

Error due to omissions, erroneous inclusions, and duplications of units in the frame used to conduct the survey; also, for household surveys, any omissions or duplicates within the households.

Non-Response Error

Error caused by survey failure to get a response to one or possibly all of the questions. Indirect measures include the detail disposition rates (un-weighted and weighted) of all the selected sample cases during data collection. Direct measures may require non-response follow-up.

Measurement Error

Error when the response received differs from the "true" value due to the respondent, the interviewer, the questionnaire, the mode of collection, or the respondent's record-keeping system(s).

Processing Error

Error during data editing, coding, capture (keying and scanning), imputation, and tabulation.

Estimation Error

For a given estimator, the difference between the value of the estimate and the true population value being estimated. Includes both sampling and non-sampling error.

Questions-Answers

Q.1 *What is the representative sample?*

Ans. A sample is "representative" if the distribution of the sample's characteristics is the same, on average, as the distribution of the characteristics of the population. The size of the sample and the type of sample (random or purposeful) are key decisions if you want to say that your sample represents the population.

Q.2 *Why are samples studied?*

Ans. Samples are studied to:

- (a) Save time and money
- (b) Save items that must be injured or destroyed in the process of studying their characteristics.
- (c) Samples are also studied because it is also the only practical way to obtain information about a universe because of its large size.

Q.3 *Explain the difference between a sample and a census.*

Ans. A census is a survey that attempts to include every element in the population while a sample is a partial enumeration of population.

Q.4 *What is target population?*

Ans. The target population is the entire group, a researcher is interested in; the group about which the researcher wishes to draw conclusions.

Q.5 *What is sampled population?*

Ans. A conceptual population that is a subset of the intersection of the Target Population and the Sample Frame. A population from which the sample is chosen.

Q.6 *Define sampling frame.*

Ans. The list of people from which the sample is taken. It should be comprehensive, complete and up-to-date. Examples of sampling frame are electoral register, postcode address file, telephone

directory etc. In other words the sampling frame is the list of ultimate sampling entities.

Q.7 *Differentiate sampling with replacement and sampling without replacement.*

Ans. Sampling with replacement is a method of sample selection in which a sample is obtained by first selecting one sampling unit from the sampling frame, replacing it, then making a second selection and replacing it before making a third selection, etc., until n selections have been made. A particular unit could be included more than once in the sample and possibly up to n times. Sampling without replacement is a method of sample selection in which a sample is obtained by selecting one sampling unit from the sampling frame and, without replacing it, selecting one of the remaining sampling units; then continuing this process until n different selections have been made. A unit can be included only once in any sample.

Q.8 *Is there any difference between the terms; Sampling Technique, Sampling Design and Sampling Model?*

Ans. There is no difference between all these terms. All these are conveying the idea of sampling strategy we are adopting. These techniques include simple random sampling, stratified sampling and all those discussed above.

Q.9 *What is sample size?*

Ans. A size of sample determined in relation to the required precision and available budget for observing the selected units.

Q.10 *Define an error.*

Ans. An error is the amount by which an observation differs from its expected value.

Q.11 *Distinguish clearly the difference between systematic errors and random errors. Explain which type of the error will decrease with the larger sample size, which will not, and why. Which error can and should be eliminated?*

Ans. Random error is attributable to chance fluctuations of sampling and tend to decrease, as sample size increases. On the other hand,

systematic errors are primarily attributable to faulty experimental design and measurement. Since systematic errors are not associated with the number of the elements sampled, they don't decrease as the sample size increase. Systematic error can and should be eliminated

Q.12 Define sampling variability?

Ans. Sampling variability refers to the different values, which a given function of the data takes when it is computed for two or more samples drawn from the same population.

Q.13 Establish the idea of precision.

Ans. Precision is a measure of how close an estimator is expected to be to the true value of a parameter. Precision is usually expressed in terms of imprecision and related to the standard error of the estimator. Less precision is reflected by a larger standard error.

Q.14 In what situations matched samples arise?

Ans. Matched samples can arise in the following situations:

- (a) Two samples in which the members are clearly paired, or are matched explicitly by the researcher. For example, IQ measurements on pairs of identical twins.
- (b) Those samples in which the same attribute, or variable, is measured twice on each subject, under different circumstances. Commonly called repeated measures. Examples include the times of a group of athletes for 1500m before and after a week of special training; or the milk yields of cows before and after being fed a particular diet.

Q.15 Define sampling distribution?

Ans. The distribution of an infinite number of samples of the same size as the sample in your study is known as the sampling distribution.

Q.16 What do you know about independent sampling?

Ans. Independent samples are those samples selected from the same population, or different populations, which have no effect on one another. That is, no correlation exists among the samples.

Q.17 *What are different types have non-probability sampling?*

Ans. *Purposive sampling, quota sampling, convenience sampling, snowball sampling, etc. are the main types of non-probability sampling.*

Q.18 *What are the advantages of non-probability sampling?*

Ans. *The main advantages of non-probability sampling are;*

- *Cheaper*
- *Used when sampling frame is not available*
- *Useful when population is so widely dispersed that cluster sampling would not be efficient*
- *Often used in exploratory studies, e.g. for hypothesis generation*
- *Some research not interested in working out what proportion of population gives a particular response but rather in obtaining an idea of the range of responses on ideas that people have.*

Q.19 *What is the major disadvantage of convenience sampling?*

Ans. *The major disadvantage of this technique is that we have no idea how representative the information collected about the sample is to the population as a whole. But the information could still provide some fairly significant insights, and be a good source of data in exploratory research.*

Q.20 *What is the main disadvantage of quota sampling?*

Ans. *The problem with it is that bias introduces on the sampling frame. Once the researcher identifies the people to be studied, they have to resort to haphazard or accidental sampling because no effort is usually made to contact people who are difficult to reach in the quota.*

Q.21 *What are different types have random or probability sampling?*

Ans. *There are some types of probability sample. The choice of these depends on nature of research problem, the availability of a good sampling frame, money, time, desired level of accuracy in the sample and data collection methods. Each has its advantages and*

disadvantages. They are: simple random sampling, stratified random sampling, systematic sampling, cluster sampling etc.

Q.22 *Why random samples are always preferred?*

Ans. *Random samples are always strongly preferred because only random samples permit statistical inference. That is, there is no way to assess the validity of results of non-random samples.*

Q.23 *Which would be the better sample of universe; a nonrandom sample of 50% of the universe or a random sample of 1% of the universe? Discuss and justify your answer.*

Ans. *A random sample of 1% of the universe would usually be a more representative sample. The more heterogeneous the universe, the more likely that this would be true. If a universe is quite homogeneous a 50% nonrandom sample may give representative results.*

Q.24 *What is the difference between the random selection and random assignment?*

Ans. *Random selection is how you can draw the sample of people for your study from a population. Random assignment is how you can assign the sample that you draw to different groups or treatments in your study.*

Q.25 *In what situations, there is the main application of simple random sampling?*

Ans. *Simple random sampling is most appropriate when the entire population from which the sample is taken is homogeneous.*

Q.26 *Define stratification.*

Ans. *A division of the sampling frames into subsets (called strata) before the selection of a sample within each of the subsets; for statistical efficiency, for production of estimates by stratum, or for convenience. Stratification is done such that each stratum contains units that are relatively homogeneous with respect to variables that are believed to be highly correlated with the information requested in the survey.*

Q.27 What are stratifying variables?

Ans. The variables whose joint values are used to classify a sampling frame into several classes (called strata) from each one of which a sample is drawn independently. Both numerical and non-numerical variables may be used for creating strata. Variables from outside sources, such as administrative records, censuses, etc., may be used as stratifying variables.

Q.28 What are the main reasons for using stratified sampling over simple random sampling?

Ans. Some reasons for using stratified sampling over simple random sampling are:

- (a) The cost per observation in the survey may be reduced
- (b) Estimates of the population parameters may be wanted for each sub-population; increased accuracy at given cost.

Q.29 What is sample size allocation?

Ans. The method in determining, how the sample should be distributed. In stratified sampling, it usually refers to the determination of the units selected from each stratum. In cluster sampling, it refers to the decision as to the number of clusters to be selected and the size of the sample in each cluster.

Q.30 What are different methods used for the allocation of sample size?

Ans. Proportional allocation, optimum allocation and Neyman allocation are used for sample size selection.

Q.31 Define proportional allocation, optimum allocation and Neyman allocation

Ans. Proportional Allocation:

Allocation in which the ratio of the number of sampled sampling units to the total number of sampling units is the same for each stratum.

Optimum Allocation:

In stratified sampling with a linear cost function, allocation in which the variance of the estimated mean or total is minimized for a specified cost or the cost is minimized for a specified variance of the estimated mean or total. For a given stratum, the sample size is

proportional to the product of the stratum's total number of sampling units and standard deviation, and it is inversely proportional to the square root of the cost per unit for the stratum.

Neyman Allocation:

A special case of optimum allocation in which the variance of an estimated mean or total is minimized for a specified total sample size. For a given stratum, the amount of the total sample size that is allocated depends on the relative size of the product of the stratum's total number of sampling units and standard deviation.

Q.32 *In what situations, there is the main application of systematic random sampling?*

Ans. This method is useful for selecting large samples, say 100 or more. It is less cumbersome than a 'simple random sample using either a table of random numbers or a lottery method. For example, you might have to sample files in a large filing cabinet. It is easier to select every 17th file than to pull out all the files and number them, etc.

Q.33 *In what situations, there is the main application of cluster sampling?*

Ans. Cluster sampling is typically used when the researcher cannot get a complete list of the members of a population they wish to study but can get a complete list of groups or 'clusters' of the population.

Q.34 *Why you might use a cluster sample of the households in an area rather than a simple random sample drawn from a directory giving addresses of households?*

Ans. A cluster sample of household might be used because directories with the address of the household are never 100% accurate. By the time the directory information is obtained and published, people have moved in an out of the area, which results in the population different from that listed in the directory.

Q.35 *Why you might use a cluster sample of the households in an area rather than a simple random sample drawn from a directory giving addresses of household?*

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time the directory information is obtained and published, people have moved in and out of the area, which results in the population different from that listed in the directory.

Q. 36 *How you can proceed while data collection using multistage sampling?*

Ans. *In multistage sampling the researcher divides the population into groups, samples the groups, then stratifies the samples, and then resample, repeating the process until the ultimate sampling units are selected at the last of the hierarchical levels. When the strata are geographic units, this method is sometimes called area sampling. For instance, at the top level, states may be sampled (with sampling proportionate to state population size); then cities may be sampled; then schools; then classes; and finally students. A sample of clusters is selected and then a sub sample of units is selected within each sample cluster. If the sub sample of units is the last stage of sample selection, it is called a two-stage design. If the sub sample is also a cluster from which units are again selected, it is called a three-stage design, etc.*

Q.37 *Differentiate cluster sampling and multistage sampling.*

Ans. *Cluster sampling is where all subjects at the lowest hierarchical level (e.g. all students in a school) are sampled for each primary sampling unit (PSU), which are the second-lowest hierarchical level, such as schools or census blocks), whereas multistage sampling is where only a random sample of lowest hierarchical level subjects are selected. The greater the heterogeneity of the group and the finer the grouping (that is, the smaller the clusters involved) depending on the objectivity of study, the more the precision of the results. For instance, grouping by gender at the highest level might well introduce bias in measuring opinions about an item known to be gender-related, whereas grouping by state would be less likely to introduce a bias since there are more categories (more states than genders) and there is less likely to be a correlation with the opinion item.*

Q.38 *Define primary sampling units.*

Ans. *Units that are selected first. Which are clusters of reporting units from which there is sub sampling to obtain reporting units in a multistage sample.*

Q.39 Define self-weighting sample.

Ans. A sample in which every sampling unit on the sampling frame has the same chance of selection, although unequal probabilities may have been used at various stages of sampling.

Q.40 Define double sampling.

Ans. A method of sample selection in which a sample is obtained by selecting a large sample in the first phase and then a sub sample of the first-phase is selected in the second phase. Information needed for sample design or estimation is collected from the large first-phase sample and then used in the design of the second-phase sample or in the final estimation.

Q.41 What do you know about spatial sampling?

Ans. A sampling technique concerned with sampling in two (or more) dimensions. For example, sampling of fields or other planar areas.

Q.42 On Friday 18th April, a personal interview of 1000 men, shopping in a mall during the hours at 5.00 p.m. to 9.00 p.m. showed that the 65% of these men thought there was too much news bulletins on the television. Do you think that the both sampling and systematic errors are present in the results of this study? Why or why not.

Ans. The systematic error is present because most the news bulletins are telecasted on Friday between 5.00 p.m. and 9.00 p.m. Therefore there is probably an over representation of the men who are not very much interested in the news bulletin in the survey. The random error is attributable to chance fluctuations in the men interviewed.

Q.43 Explain and criticize each part of the following statement: The frequency distributions of the family income, size of the business, and salaries of the skilled employees all tends to be skewed to the right.

Ans. Agree, since in all three cases it is quite logical to assume that there would be extreme values at the upper end of the scale that would tend to make the arithmetic mean larger than the median.

Q.44 What is survey design?

Ans. The process of selecting sites at which a response will be determined. Includes a probability model for inference based on the randomized selection process.

Q.45 Outline the steps involved in planning a sample survey.

Ans. The steps are:

- (i) Define and state the purpose of the study
- (ii) Determine the conditions under which the survey will be made
- (iii) Find out if any or part of the desired information is already available
- (iv) Choose the size and the type of the sample to be used and the method of selecting sample items and collecting data.
- (v) Prepare the questionnaire
- (vi) Collect the data
- (vii) Edit the returned questionnaire
- (viii) Tabulate and classify the information gathered
- (xi) Analyze the information
- (x) Interpret the findings

Q.46 What is meant by data collection?

Ans. Any process whose purpose is to acquire or assist in the acquisition of data. Collection of data is achieved by requesting and obtaining pertinent data from individuals or organizations via an appropriate vehicle.

Q.47 Explain the difference between an observational study and a controlled study.

Ans. An observational study essentially involves examination of the historical relationships that exist among the variables of interest; a controlled study involves controlling or fixing factors other than the variable of interest.

Q.48 What is Questionnaire?

Ans. A set of questions designed to collect information from a respondent. A questionnaire may be interviewer-administered or respondent-completed, using paper-and-pencil methods for data collection or computer-assisted modes of completion.

Q.49 *List ten rules that are useful for making out questionnaire.*

- Ans.**
- (i) Use items that can be easily understandable
 - (ii) Avoid ambiguous questions
 - (iii) Make sure that the questions asked can be accurately answered
 - (iv) Avoid double questions
 - (v) Avoid the direct embarrassing questions
 - (vi) Word the questions so that the answer can be easily tabulated and easily classified
 - (vii) Avoid the leading questions
 - (viii) List the question in a logical sequence
 - (ix) Make the questionnaire short and attractive
 - (x) Place the research organization's name and address on each questionnaire.

Q.50 *If a questionnaire asks questions that are understood by the respondents. The answers to the questions are considered reliable and useful? Explain.*

- Ans.** Not necessary, some people may fail to give correct answers even though they understand the questions. Also, the summarized answer may be of little value if the respondents are not from a probability sample of a well-defined universe.

Q.51 *What is response rate?*

- Ans.** The percentage of the intended sample from which one may actually be able to collect data. For example, if 100 health professionals are sampled and sent them all, questionnaires, but only 40 returned completed questionnaires, the response rate is 40%.

Q.52 *Define non-response.*

- Ans.** The failure to obtain responses or measurements for all sample elements.

Q.53 *What are different types of non-response?*

- Ans.** Item non-response, partial non-response and unit non-response are the main types of non-response.

Q.54 Define item non-response, partial non-response and unit non-response.

Ans. Item Non-response:

It occurs when a respondent provides some, but not all, of the requested information, or if the reported information is not useable.

Partial Non-response:

A partial interview is when some but not all items have responses. A partial interview is treated as a "unit response" when a sufficiently accurate response is obtained for only some of the data items required from a respondent and meets some minimum threshold level. A partial interview is treated as a "unit non-response" when this threshold is not met.

Unit Non-response:

It occurs when the sampled unit response does not meet a minimum threshold and is classified as not having responded at all; failure to make measurements or obtain observations on a listing unit selected for inclusion in a sample.

Q.55 What do you know about sub sampling non-respondents?

Ans. An economical method for reducing non response bias in which new attempts are made to obtain responses from a sub sample of sampling units that did not provide responses to the first attempt.

Q.56 Define coverage, over coverage and under coverage.

Ans. Coverage:

The extent to which a frame includes all the elements of the sampled population.

Over coverage:

The extent to which a frame includes more elements than the sampled population; including duplicate elements.

Under coverage:

The extent to which a frame includes fewer elements than the sampled population.

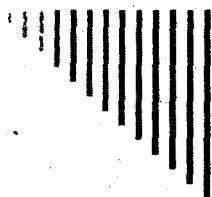
Exercises

Exercise 7 (True/False)

Read the following statements carefully and indicate which statement is "True" or "False":

1. There are many practical situations where sampling is inevitable that is there is no other way of data collection other than sampling.
2. There is no difference between random error and sampling bias.
3. When the items included in a sample are based on the judgment of the individual conducting the sample, the sample is said to be nonrandom sample.
4. A statistic is a characteristic of population.
5. A sampling plan that selects members from a population at uniform intervals in time, order, or space, is called stratified sampling.
6. As a general rule, it is not necessary to include a finite population multiplier in a computation for standard error of the mean when the size of the sample is greater than 50.
7. The probability distribution of all the possible means of samples is known as the sampling distribution of the mean.

8. The principles of simple random sampling are the theoretical foundations for statistical inference.
9. The standard error of the mean is as the standard deviation of the distribution of sample means.
10. A sampling plan that divides the population into well defined groups from which random samples are drawn is known as cluster sampling.
11. With increasing sample size, the sampling distribution of the mean approaches normality, regardless of the distribution of the population.
12. To perform a complete enumeration, one would need to examine every item in a population.
13. In everyday life, we see many examples of infinite populations of physical objects.
14. Large samples are always a good idea because they decrease the standard error.
15. The main problem of data collection through mail questionnaire is bias.



Chapter 8

Statistical Inference

Statistical Inference

Statistical Inference makes use of information from a sample to draw conclusions (inferences) about the population from which the sample was taken. It has two branches; Estimation and Testing of Hypothesis.

Estimator

An estimator is any quantity calculated from the sample data, which is used to give information about an unknown quantity in the population. In other words, any statistic that is used to estimate a population parameter is called estimator. For example, the sample mean is an estimator of the population mean.

Estimate

An estimate is a specific value or range of values used for indication of the value of an unknown quantity based on observed data. More formally, an estimate is the particular value of an estimator that is obtained from a particular sample of data and used to indicate the value of a parameter.

Estimation

Estimation is the process by which sample data are used to indicate the value of an unknown quantity in a population.

Results of estimation can be expressed as a single value, known as a Point Estimate, or a range of values, known as an Interval Estimate.

Properties of a Good Point Estimator

Unbiasedness:

An estimator $\hat{\theta}$ is said to be unbiased estimator of parameter θ if the mean of sampling distribution of the values of $\hat{\theta}$ is equal to θ i.e.

$$E(\hat{\theta}) = \theta.$$

Efficiency:

Efficiency refers to the size of standard error of the statistic. If we compare two statistics from a sample of the same size then the estimator with smaller standard error is said to be more efficient.

Particularly, an estimator $\hat{\theta}_1$ is said to be efficient estimator than $\hat{\theta}_2$ if

$\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2)$. The variance is calculated if the estimators are unbiased. If estimators are biased then mean squared errors will be compared.

Consistency:

An estimator is said to be consistent estimator of a population parameter if as the sample size increases, it becomes almost certain that the value of the statistic comes very close to the value of population parameter.

In other sense an estimator $\hat{\theta}_n$ (as n is the sample size) is a consistent estimator for parameter θ if and only if, for all $\varepsilon > 0$, no matter how small, we have;

$$P(|\hat{\theta}_n - \theta| < \varepsilon) = 1, \text{ when } n \rightarrow \infty.$$

Sufficiency:

An estimator is called sufficient estimator if it makes so much use of the sample information that no other estimator could extract from the sample additional information about the population parameter being estimated.

Hypothesis

It is a supposition or assumption, which acts as a foundation or as a starting point in an investigation, irrespective of its probable truth or falsity. For example, average body temperature of adults is 98.6°F, procedure *A* of cultivation is better than that of *B*, etc.

Statistical Hypothesis

A statistical hypothesis is a statement about parameter(s) of population(s). For example, average body temperature of adults is 98.6°F, more than 70% voters are in favour of a particular party, etc.

Testing of Hypothesis

Hypothesis testing begins with an assumption, called a hypothesis that we make about a population parameter. Then we collect sample data, produce sample statistic, and use this as information to decide how likely it is that our hypothesized population parameter is correct. The purpose of this type of inference to determine whether enough statistical evidence exists to enable us to conclude that a belief or hypothesis about a parameter is supported by the data.

Null Hypothesis

A hypothesis to be tested for possible rejection under the assumption that it is true, is called null hypothesis and is denoted by H_0 . For example, in a clinical trial of a new drug, the null hypothesis might be that the new drug is no better, on average, than the current drug. We would write H_0 : there is no difference between the two drugs on average.

We give special consideration to the null hypothesis. This is due to the fact that the null hypothesis relates to the statement being tested.

Alternative Hypothesis

The alternative hypothesis, denoted by H_1 , is to be considered as an alternate to the null hypothesis. It is also known as Research Hypothesis. For the above example, we would write

H_1 : the two drugs have different effects, on average.

The alternative hypothesis might also be that the new drug is better, on average, than the current drug. In this case we would write

H_1 : the new drug is better than the current drug, on average.

Simple Hypothesis

A simple hypothesis is a hypothesis, which specifies the population distribution completely.

For example,

1. $H_0: p = 0.5$, i.e., p is specified
2. $H_0: X \sim N(5, 20)$, i.e., μ and σ^2 are specified

Composite Hypothesis

A composite hypothesis is a hypothesis, which does not specify the population distribution completely.

For example,

1. $H_1: p > 0.5$, i.e., p is not completely specified
2. $H_1: X \sim N(5, \sigma^2)$, i.e., σ^2 is not completely specified

Type-I Error

In a hypothesis test, a type-I error occurs when the null hypothesis is rejected when it is in fact true; that is, H_0 is wrongly rejected. The probability of committing type-I error is denoted by α .

A type-I error is often considered to be more serious, and therefore more important to avoid, than a type II error. The hypothesis test procedure is therefore adjusted so that there is a guaranteed 'low' probability of rejecting the null hypothesis wrongly; this probability is never zero.

Type-II Error

In a hypothesis test, a type-II error occurs when the null hypothesis H_0 is not rejected when it is in fact false, H_0 is wrongly accepted. The probability of committing type-II error is denoted by β .

A type-II error would occur if it was concluded that the two drugs produced the same effect, i.e. there is no difference between the two drugs on average, when in fact they produced different ones. A type-II error is frequently due to sample sizes being too small.

Significance Level

The significance level of a statistical hypothesis test is a fixed probability of wrongly rejecting the null hypothesis H_0 , if it is in fact true. It is the probability of a type I error and is set by the investigator in relation to the consequences of such an error. That is, we want to make the significance level as small as possible in order to protect the null hypothesis and to prevent, as far as possible, the investigator from inadvertently making false claims. The significance level is usually denoted by α .

Test Statistic

A test statistic is a quantity calculated from the sample of data. Its value is used to decide whether or not the null hypothesis should be rejected in our

hypothesis test. The choice of a test statistic will depend on the assumed probability model and the hypotheses under question.

Critical Value

The critical value for a hypothesis test is a threshold to which the value of the test statistic in a sample is compared to determine whether or not the null hypothesis is rejected.

The critical value for any hypothesis test depends on the significance level at which the test is carried out, and whether the test is one-sided or two-sided (described below).

Critical Region

The critical region (CR), or rejection region (RR), is a set of values of the test statistic for which the null hypothesis is rejected in a hypothesis test. That is, the sample space for the test statistic is partitioned into two regions; one region (the critical region) will lead us to reject the null hypothesis H_0 , the other will not. So, if the observed value of the test statistic is a member of the critical region, we conclude "Reject H_0 "; if it is not a member of the critical region then we conclude, "Do not reject H_0 ".

One-Sided (One-Tailed) Test

The tails in a distribution are the extreme regions bounded by critical values. A test is said to be one-sided or one-tailed test if its entire critical region lies on just one (right or left) tail of the distribution under H_1 .

In other words, the critical region for a one-sided test is the set of values less than the critical value of the test, or the set of values greater than the critical value of the test.

Example:

If μ = average body temperature of adults;

$$H_0: \mu = 98.6,$$

Against;

$$H_1: \mu < 98.6 \text{ or } H_1: \mu > 98.6.$$

Two-Sided Test

A test is said to be two-sided or two-tailed test if its entire critical region lies on both tails of the distribution under H_1 . In other words, the critical region for a two-sided test is the set of values less than a first critical value of the test and the set of values greater than a second critical value of the test.

Example :

Suppose, we want to test a manufacturers claim that there are, on average, 50 sticks in a match-box. We could set up the following hypothesis

$$H_0: \mu = 50,$$

Against;

$$H_1: \mu \neq 50$$

The choice between a one-sided and a two-sided test is determined by the purpose of the investigation or prior reasons for using a one-sided test.

P-Value

The probability value (*p*-value) of a statistical hypothesis test is the probability of getting a value of the sample test statistic that is at least as extreme as the one found from the sample data assuming that the null hypothesis is true.

It is the probability of wrongly rejecting the null hypothesis if it is in fact true. It is, actually, observed significance level of the test for which we would only just reject the null hypothesis. The *p*-value is compared with the presumed significance level of our test and, if it is smaller, the result is significant. That is, if the null hypothesis were to be rejected at the 5% significance level, this would be reported as " $p < 0.05$ ".

Small *p*-values suggest that the null hypothesis is unlikely to be true. The smaller it is the more convincing is the rejection of the null hypothesis.

Power of the Test

The power of a statistical hypothesis test measures the test's ability to reject the null hypothesis when it is actually false, that is, to make a correct decision.

In other words, the power of a hypothesis test is the probability of not committing a type- II error. It is calculated by subtracting the probability of a type- II error from 1, usually expressed as:

$$\text{Power} = 1 - P(\text{type II error}) = 1 - \beta$$

The maximum power, a test can have is 1 and the minimum is 0. Ideally, we want a test to have high power, close to 1.

Questions-Answers

Q.1 *What are the major parts of the statistical inference?*

Ans. *Estimation of parameters and testing of hypothesis are the two major parts of the statistical inference.*

Q.2 *What is inductive inference?*

Ans. *The research worker performs an experiment and obtains some data. The conclusions usually go beyond the materials and operations of the particular experiment. This sort of the extension from the particular to the general is called inductive inference.*

Q.3 *What is deductive inference?*

Ans. *The conclusions based on "general to particular" is known as deductive inference. In other words when an argument claims that the truth of its premises guarantees the truth of its conclusion, it is said to involve a deductive inference.*

Q.4 *Differentiate positively and negatively biased estimator?*

Ans. *When the expected value of the estimator is greater than the corresponding parameter value, the estimator is positively biased. If the expected value of the estimator is smaller than the corresponding parameter value, the estimator is negatively biased.*

Q.5 *What are the methods used for parameter estimation?*

Ans. *Method of least square, method of maximum likelihood, method of moments etc. are used for parameter estimation.*

Q.6 *Define likelihood function.*

Ans. *A likelihood function $L(\theta)$ is the probability or probability density for the occurrence of a sample configuration $X_1, X_2, \dots, X_n$ given that the probability density function $f(x; \theta)$ with parameter θ is known. Symbolically it can be written as;*

$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \dots f(x_n; \theta).$$

Q.7 What is the principle of Maximum Likelihood (ML)?

Ans. The ML principle directs us to take as our estimator of θ that value within the admissible range of θ which make the likelihood function as large as possible. i.e. we choose $\hat{\theta}$ so that for any admissible value of θ ;

$$L(x/\hat{\theta}) \geq L(x/\theta)$$

Q.8 What is the method of least squares?

Ans. Method of least squares is a mathematical optimization technique that attempts to find a 'best fit' to a set of data by attempting to minimize the sum of the squares of the residuals between the fitted function and the data.

Q.9 What is the method of moments?

Ans. The method of moments is a method of estimation of parameters by equating sample moments with unobservable population moments and then solving those equations for the quantities to be estimated.

Q.10 What is confidence coefficient? How it can be interpreted?

Ans. The degree of confidence is the probability $(1 - \alpha)$ that the population parameter is contained in the confidence interval. This probability is often expressed as the equivalent percentage value. The degree of confidence is also referred to as the level of confidence or the confidence coefficient.

If we suppose $\alpha = 0.05$ then $100(1 - \alpha)\% = 95\%$ confidence coefficient means that we are 95% confident that the true value of population parameter will lie between the computed 95% interval estimate.

Q.11 Can every hypothesis be treated as statistical hypothesis? If not, then specify the difference between statistical and non-statistical hypothesis?

Ans. No, every hypothesis cannot be treated as statistical hypothesis. The kind of hypotheses, which we test in Statistics, is more restricted than the general scientific hypotheses. It is a scientific hypothesis that every particle of matter, in the universe, attracts every other particle, or the life exists on Mars; but these are not

hypotheses such as arise for testing from the statistical view point. Statistical hypothesis concerns the behavior of observable random variable. In other words statistical hypotheses are testable claims or assertions about one or more parameter of empirical distributions.

Q.12 *At what point in the research procedure should a hypothesis be stated?*

Ans. *Ideally, the hypothesis should be stated before the sample data are collected and analyzed.*

Q.13 *How does an analyst know which significance level is to be used in a test?*

Ans. *He must determine what risk he is willing to take in making a type-I error. In doing this, consideration is given to the consequences of making such error. If the decision is to be made about much critical situation the level of significance will be small, conversely if the decision to be made about less critical situation the level of significance will be large.*

Q.14 *Which type of testing error is more serious?*

Ans. *A type-I error is often considered to be more serious, and therefore more important to avoid, than a type-II error. The hypothesis test procedure is therefore adjusted so that there is a guaranteed 'low' probability of rejecting the null hypothesis wrongly.*

Q.15 *Define degree of freedom.*

Ans. *The number of independent groups or sub-categories into which sample or population may be divided is called degree of freedom. For example, in a sample of constant size n , grouped into k intervals there are ' $k-1$ ' degrees of freedom because if ' $k-1$ ' frequencies are specified the other is determined by the total size.*

Q.16 *Can the probability of making a type II error is determined in a life sampling problem?*

Ans. *No, because one does not know how false the hypothesis is.*

Q.17 How we minimize both type I error and type II error simultaneously?

Ans. If we increase the sample size both errors will be minimized.

Q.18 Criticize the following statements.

- (a) It is impossible to control both type-I and type-II error. Since decreasing one will increase the other.
- (b) A one sided alternative mean to test H_0 for only type-II error.
- (c) If a person has a choice of applying two different tests, to decide between a certain set of null and alternative hypotheses, assuming both have the same significance level. Then the person should use the test which has smaller probability of committing type- II error
- (d) It is important to select the significance level before choosing a decision rule.

- Ans.**
- (a) This would be true only for a fixed sample size. By increasing the sample size we can decrease the probabilities of both types of error.
 - (b) Disagree, a one sided alternative simply specifies the location of critical region on either side. It does not test null hypothesis for only type-II error, one may commit a type-I error.
 - (c) Agree.
 - (d) Agree.

Q.19 How we establish the conclusion of the hypothesis test?

Ans. The final conclusion once the test has been carried out is always given in terms of the null hypothesis. We either "Reject H_0 in favour of H_1 " or "Do not reject H_0 ". We never conclude "Reject H_1 ", or even "Accept H_1 ".

If we conclude "Do not reject H_0 ", this does not necessarily mean that the null hypothesis is true; it only suggests that there is not sufficient evidence against H_0 in favor of H_1 . Rejecting the null hypothesis then, suggests that the alternative hypothesis may be true.

Q.20 *Discuss the importance of normal distribution in testing of hypothesis.*

Ans. One reason the normal distribution is important is that many variables (e.g. psychological, educational etc.) are distributed approximately normal. Measures of reading ability, introversion, job satisfaction, and memory are among the many variables approximately normally distributed. Although the distributions are only approximately normal, they are usually quite close. A second reason the normal distribution is so important is that it is easy for mathematical statisticians to work with. This means that many kinds of statistical tests can be derived for normal distributions. Almost all statistical tests assume normal distribution. Fortunately, these tests work very well even if the distribution is only approximately normally distributed. Some tests work well even with very wide deviations from normality. Finally, if the mean and standard deviation of a normal distribution are known, it is easy to convert back and forth from raw scores to percentiles.

Q.21 *Discuss the role of Chi-square distribution in testing of hypothesis and confidence interval estimation.*

Ans. Let $Z \sim N(0,1)$ be a standard normal variable. If n random values $Z_1, Z_2, \dots, Z_n$ are drawn from this distribution, squared, and summed, the resultant statistic is said to have a χ^2 distribution with n degrees of freedom. It is right skewed distribution ranging from 0 to ∞ . Chi-square distribution has a wide range of application. In testing of hypothesis it is being used to test the variance, equality of variances. Moreover it is used to test the goodness of fit and also for testing association of attributes. With reference to confidence interval it is being used to construct the confidence interval for population variance.

Q.22 *What are the main uses of t-distribution in testing of hypothesis and construction of confidence intervals?*

Ans. The t-distribution may be defined in terms of normal and independent χ^2 variables.

Let $Z \sim N(0,1)$ and $Y \sim \chi^2(n)$ where Z and Y are independently distributed then;

$$t = \frac{Z\sqrt{n}}{\sqrt{Y}}$$

It is symmetrical about its mean point, zero, and tends asymptotically to the standard normal distribution. The standard normal distribution is replaced by t-distribution when population variance is estimated from sample data and central limit theorem is not applicable. t-distribution may be used for testing the means, equality of means (two), proportion and difference of proportions (two). All this will be done when the population variance is unknown.

The distribution is also applicable to construct the confidence interval for:

- i) Population mean and population proportion
- ii) Difference between two population means and population proportions either samples are independent or related.

Q.23 What type of hypotheses can be tested by using F-distribution?

Ans. The F-distribution is defined in terms of two independent χ^2 variables. Let U and V are independently distributed χ^2 variable with n_1 and n_2 degrees of freedom, respectively. Then the statistic;

$$F = \frac{U / n_1}{V / n_2};$$

is distributed as F with n_1 and n_2 degrees of freedom.

It is a continuous probability distribution, which has two parameters, n_1 and n_2 which are positive whole numbers. Like the Chi-square distribution it is right skewed ranging from 0 to ∞ . In testing of hypothesis it is being used to test the equality of two population variances, equality of more than two means (i.e. in ANOVA) and also to test the significance of overall regression.

Q.24 Define power and operation characteristic (OC) functions.

Ans. The function, which gives the probabilities of rejecting the null hypothesis for different values of alternative hypothesis, then, the function will be known as power function. Similarly the function, which gives probabilities of accepting the null hypothesis for different values of alternative hypothesis, is known as OC function.

The complement of the power function is the operating characteristic function.

Q.25 What are power and OC curves?

Ans. If the probabilities of rejecting the null hypothesis plotted for various alternative hypotheses then the obtained curve is power curve. Similarly if the probabilities of accepting null hypothesis are plotted for various alternative hypotheses then the operating characteristic curve is obtained.

Q.26 Describe the general procedure for testing of hypothesis.

Ans. Following are the desirable steps comprising the procedure for testing of hypothesis:

- i) Formulation of null and alternative hypothesis
- ii) Setting of level of significance
- iii) Specification of test statistic
- iv) Determination of critical region
- v) Computation
- vi) Results and conclusions.

Q.27 Define Best Critical Region (BCR).

Ans. If, among critical regions of fixed size there is one which minimizes the probability of type-II error, then it is known as the BCR.

Q.28 Define and explain sequential analysis.

Ans. In many circumstances, the observations to be used in evaluating models arrive sequentially rather than all at once. Sequential analysis is the procedure in which sample size is not fixed in advance. The decision is to be made after taking an observation. If the observation have a potency to make a decision about the truth or falsity of the hypothesis, than we stop, taking observations. If not so, then we take another observation and test for acceptance or rejection of hypothesis and so on until we reach at the result. In this type of testing we do not specify the sample size but we fix probability of type-I error (α) and probability of type-II error (β).

Exercises

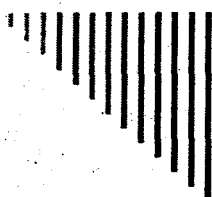
Exercise 8 (True/False)

Read the following statements carefully and indicate which statement is "True" or "False":

1. One estimator is more efficient than another if a large sample is used with the first.
2. A statistic will always be an unbiased estimator if the sample itself is chosen without bias.
3. A consistent estimator is one that tends to be more accurate with large samples than with small ones.
4. All efficient estimators are automatically unbiased.
5. The sample variance s^2 is calculated somewhat differently than the population variance σ^2 in order to make s^2 an unbiased estimator of σ^2 .
6. Confidence may always be increased by reducing the precision of the interval estimate.
7. The 95% confidence interval for the mean gasoline mileage for a certain make of a car is $15 \leq \mu \leq 17$. This means that there is a probability of 0.95 that μ actually falls between 15 and 17.
8. A 99% confidence interval will be wider than a 95% confidence interval constructed from the same data.

9. The student's t -distribution applies when $\bar{X}$ is used to estimate μ and when σ is an unknown value.
10. The more disperse of two populations tends to have a narrower confidence interval at the 95%.
11. Null hypothesis (H_0) always indicates no change.
12. Hypothesis testing models assume either that H_0 must be true or that it must be false.
13. The type-II error is equivalent to rejecting the alternative hypothesis when it is true.
14. The power curve relates the probability of rejecting H_0 to various possible population parameter values.
15. For an upper-tailed test, a reduction of the significance level from 0.05 to 0.01 implies that there must be a corresponding increase in the critical value.
16. An upper-tailed test occurs when $H_0: \mu \geq \mu_0$ and $H_1: \mu < \mu_0$.
17. Whenever the Type-I error does not occur, the Type-II error must occur.
18. H_0 is that $\mu \leq 10$. The critical value is $\bar{X} = 12$, and the observed sample mean is $\bar{X} = 15$. H_0 should therefore be accepted.
19. If α is the desired probability for incorrectly rejecting H_0 when in fact $\mu \geq 110$, then the null hypothesis is $H_0: \mu \geq 110$.
20. Hypothesis testing is a tool that statisticians use so that they never make incorrect decisions.
21. With a two sided hypothesis testing procedure, there are two ways in which the type-I error may occur.

22. In an upper tailed test, large values of the test statistic result in rejecting the null hypothesis.
23. For a prescribed decision rule, a larger sample size will result in a larger type-I error probability.
24. A contingency table can have only two rows and two columns.
25. The Chi-square test can be used to determine if there is a significant difference between two sample percentages.
26. In a Chi-square analysis the number of rows in the contingency table should equal the number of columns.
27. A Chi-square analysis should not ordinarily be made when there are fewer than five observations in one or more of the cells.
28. There are two main type of Chi-square tests i.e. goodness of fit and badness of fit.
29. A Chi-square test is actually a test to determine if the distribution of sample data conforms to some expected or actual distribution.
30. In a Chi-square test, it is assumed that the sample observations were selected from a normally distributed universe.



Chapter 9

Design and Analysis of Experiments

Experiment

An experiment is any process or study, which results in the collection of data, the outcome of which is unknown. In statistics, the term is usually restricted to situations in which the researcher has control over some of the conditions under which the experiment takes place.

Design of Experiment (DOE)

We are concerned with the analysis of data generated from an experiment. It is wise to take time and effort to organize the experiment properly to ensure that the right type of data, and enough of it, is available to answer the questions of interest as clearly and efficiently as possible. This process is called DOE. In other words a DOE is a structured, organized method for determining the relationship between factors (X 's) affecting a process and the output of that process (Y). DOE refers to experimental methods used to quantify indeterminate measurements of factors and interactions between factors statistically through observance of forced changes made methodically as directed by mathematically systematic tables.

Experimental Unit

Experimental units is the basic object upon which the study or experiment is carried out. The entity to which a specific treatment combination is applied. An experimental unit can be a individual agricultural plant, plot of land, PC board, silicon wafer, tray of components simultaneously treated, automotive transmissions etc.

Basic Principles of Experimental Design

Randomization, replication and local control are the basic principles of experimental design.

Randomization

Randomization is a schedule for allocating treatment material and for conducting treatment combinations in a DOE such that the conditions in one run neither depend on the conditions of the previous run nor predicts the conditions in the subsequent runs. The importance of randomization cannot be over stressed. Randomization is necessary for conclusions drawn from the experiment to be correct, unambiguous and defensible. Randomization is preferred since alternatives may lead to biased results. The main point is that randomization tends to produce groups for study that are comparable in unknown as well as known factors likely to influence the outcome, apart from the actual treatment under study. The analysis of variance F tests assume that treatments have been applied randomly.

Replication

The repetition of an experiment on a large group of subjects is known as replication. If a treatment is truly effective, the long-term averaging effect of replication will reflect its experimental worth. If it is not effective, then the few members of the experimental population who may have reacted to the treatment will be negated by the large numbers of subjects who were unaffected by it. Replication reduces variability in experimental results, increasing their significance and the confidence level with which a researcher can draw conclusions about an experimental factor.

Local Control

Local control refers to grouping of the experimental units in such a way that the units within a group (i.e., block) are more homogeneous than are units in different groups. The experimental materials or conditions are more alike within a group. Thus, the variation among experimental units within a group is less than the variation would have been without grouping. This leads to the comparison of treatment effects under more uniform conditions or on the more uniform materials. For example, the total variation in Randomized Complete Block Design (RCBD) is partitioned into variation due to two assignable causes, blocks and treatments, and variation due to a non-assignable cause or experimental error. This latter source of variation is reduced as the variation due to block is removed:

Experimental error = Total variation - Treatment variation - Block variation.

Treatment in Experimental Design

In experiments, a treatment is something that researchers manage to experimental units. For example, a corn field is divided into four, each part is 'treated' with a different fertilizer to see which produces the most corn; a teacher practices different teaching methods on different groups in his/her class to see which yields the best results; a doctor treats a patient with a skin condition with different creams to see which is most effective.

Treatments are administered to experimental units by 'level', where level implies amount or magnitude. For example, if the experimental units were given 5mg, 10mg, 15mg of a medication, those amounts would be three levels of the treatment. 'Level' is also used for categorical variables, such as Drugs A, B, and C, where the three are different kinds of drug, not different amounts of the same thing. In other words a treatment is a specific combination of factor levels whose effect is to be compared with other treatments.

Factor of an Experiment

A factor of an experiment is a controlled independent variable; a variable whose levels are set by the experimenter. A factor is a general type or category of treatments. Different treatments constitute different levels of a factor. For example, three different groups of runners are subjected to different training methods. The runners are the experimental units, the training methods, the treatments, where the three types of training methods constitute three levels of the factor 'type of training'.

Analysis of Variance (ANOVA)

A mathematical process for separating the variability of a group of observations into assignable causes and setting up various significance tests. In other words analysis of variance is a statistical technique for analyzing data that tests for a difference between two or more means by comparing the variances "within" groups and variances "between" groups.

Assumptions:

- Normality:* The populations have normal distributions.
- Homogeneity:* The populations have the same variances.
- Randomization:* The samples are random and independent of each other.

Questions-Answers

Q.1 *Define Model?*

Ans. *A Mathematical relationship, which relates changes in a given response to changes in one or more factors.*

Q.2 *Define the term Design.*

Ans. *A set of experimental runs, which allows us to fit a particular model and estimate our desired effects.*

Q.3 *What is meant by the term Treatment Combination?*

Ans. *The combination of the settings of several factors in a given experimental trial. Also known as a run.*

Q.4 *Define variance components.*

Ans. *Partitioning of the overall variation into assignable components.*

Q.5 *What is one-way ANOVA?*

Ans. *The one-way analysis of variance (ANOVA) allows us to compare several groups of observations, all of which are independent but possibly with a different mean for each group. A test of great importance is whether or not all the means are equal.*

The observations all arise from one of several different groups (or have been exposed to one of several different treatments in an experiment). We are classifying 'one-way' according to the group or treatment.

Q.6 *What is two-way ANOVA?*

Ans. *Two-way Analysis of Variance is a way of studying the effects of two factors separately (their main effects) and (sometimes) together (their interaction effect).*

Q.7 *What are the basic experimental designs?*

Ans. *Basic experimental designs are Completely Randomized Design (CRD), Randomized Complete Block Design (RCBD) and Latin Square Design (LSD).*

Q.8 *Explore the idea of completely randomized design.*

A. *The structure of the experiment in a completely randomized design is assumed to be such that the treatments are allocated to the experimental units completely at random.*

Q.9 *Explore the idea of randomized complete block design?*

Ans. *The randomized complete block design is a design in which the subjects are matched according to a variable, which the experimenter wishes to control. The subjects are put into groups (blocks) of the same size as the number of treatments. The members of each block are then randomly assigned to different treatment groups.*

Q.10 *Explore the idea of Latin square design.*

Ans. *Latin square (and related) designs are efficient designs to block from 2 to 4 nuisance factors.*

Q.12 *What is blocking?*

Ans. *This is the procedure by which experimental units are grouped into homogeneous clusters in an attempt to improve the comparison of treatments by randomly allocating the treatments within each cluster or 'block'.*

Q.13 *Explore the idea of factorial design.*

Ans. *A factorial design is used to evaluate two or more factors simultaneously. The treatments are combinations of levels of the factors. The advantages of factorial designs over one-factor-at-a-time experiments is that they are more efficient and they allow interactions to be detected.*

Q.14 *What is the effect?*

Ans. *Effect gives us how changing the settings of a factor changes the response.*

Q.15 *What is main effect?*

Ans. *The main effect is the average simple effect of a factor on a dependent variable. It is the effect of the factor alone averaged across the levels of other factors. In other words a main effect is a*

measurement of the average change in the output when a factor is changed from its low level to its high level.

Q.16 *What is interaction effect?*

Ans. *An interaction is the variation among the differences between means for different levels of one factor over different levels of the other factor.*

Q.17 *What is fixed effect?*

Ans. *An effect associated with an input variable that has a limited number of levels or in which only a limited number of levels are of interest to the experimenter.*

Q.18 *What is random effect?*

Ans. *The random effect model does not provide knowledge of the treatment effect at a particular level. It enables us to study the variability due to the effect of treatment. Therefore the random effect model is sometimes called the component of variance. Or an effect associated with input variables chosen at random from a population having a large or infinite number of possible values.*

Q.19 *What is interaction effect?*

Ans. *It occurs when the effect of one factor on a response depends on the level of another factor(s).*

Q.20 *What is the lack of fit error?*

Ans. *Error that occurs when the analysis omits one or more important terms or factors from the process model. Including replication in a DOE allows separation of experimental error into its components: lack of fit and random (pure) error.*

Q.21 *What is meant by the term confounding?*

Ans. *A confounding design is one where some treatment effects (main or interactions) are estimated by the same linear combination of the experimental observations as some blocking effects. In this case, the treatment effect and the blocking effect are said to be confounded. Confounding is also used as a general term to indicate that the value of a main effect estimate comes from both the main effect itself and also contamination or bias from higher*

order interactions. Confounding naturally arise when full factorial designs have to be run in blocks and the block size is smaller than the number of different treatment combinations. It also occurs whenever a fractional factorial design is chosen instead of a full factorial design.

Q.22 ***What are Taguchi methods?***

Ans. A technique for designing and performing experiments to investigate processes where the output depends on many factors (variables; inputs) without having to tediously and uneconomically run the process using all possible combinations of values of those variables. By systematically choosing certain combinations of variables it is possible to separate their individual effects.

Q.23 ***What is the balanced design?***

Ans. An experimental design where all cells (i.e. treatment combinations) have the same number of observations.

Q.24 ***What is meant by the comparative design?***

Ans. A design aimed at making conclusions about one a priori important factor, possibly in the presence of one or more other "nuisance" factors.

Q.25 ***What is meant by the term orthogonality?***

Ans. Two vectors of the same length are orthogonal if the sum of the products of their corresponding elements is 0. An experimental design is orthogonal if the effects of any factor balance out (sum to zero) across the effects of the other factors.

Q.26 ***Define responses.***

Ans. The output(s) of a process is known as response. Sometimes called dependent variable(s).

Q.27 ***What are response surface designs?***

Ans. A design of experiment that fully explores the process window and models the responses. These designs are most effective when there are less than 5 factors. Quadratic models are used for response surface designs and at least three levels of every factor are needed in the design.

Q.28 *Define scaling factor levels.*

Ans. *Transforming factor levels so that the high value becomes +1 and the low value becomes -1.*

Q.29 *What are screening designs?*

Ans. *A design of experiment that identifies which of many factors has a significant effect on the response. Typically screening designs have more than 5 factors.*

Q.30 *What are the tests used for mean comparison?*

Ans. *When treatment effect is declared significant, different tests are used for comparing the treatment means. These tests include Least Significance Difference (LSD) test, Duncan Multiple Range Test (DMRT), Tuckey's test, Scheffe's test, orthogonal contrasts, trend comparisons, etc.*

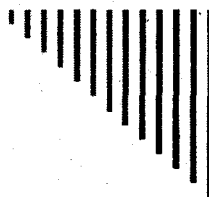
Exercises

Exercise 9 (True/False)

Read the following statements carefully and indicate which statement is "True" or "False":

1. The F -distribution is symmetrical.
2. The F -distribution is a distribution of ratios of sample variances.
3. An analysis of variance (ANOVA) is a useful tool for proving or disproving a null hypothesis of several means.
4. In an ANOVA an F -ratio of 1.00 means that a significant difference exists at both the 0.05 and 0.01 levels.
5. The F -ratio may be computed to determine if there is a significant difference among means.
6. In an ANOVA, the number of items in each column must be the same.
7. A one-way ANOVA is restricted to a maximum of seven columns of data.
8. In a two-way ANOVA the number of rows need not equal the number of columns.
9. The number of degrees of freedom lost when computing the variance within columns is equal to the total number of observations in all columns minus the number of columns.

10. In a two-way ANOVA the number of degrees of freedom for the residual variance is equal to $(r - 1)(c - 1)$. Where r denotes rows and c denotes columns.
11. In a one-way ANOVA, it is assumed that the residual variation is due to chance.
12. In a two-way ANOVA, it is assumed that the residual variation is due to chance.
13. In an ANOVA, it is assumed that the universes from which the samples were selected are normally distributed.
14. The number of degrees of freedom to be used in computing the variance between the columns is equal to the number of columns minus one $(c - 1)$.
15. In a two-way ANOVA, there are three possible sources of variation.



Chapter 10

Analysis of Time Series

Time Series

Any variable that is measured overtime in sequential order is called a time series or data recorded at regular intervals of time is called time series for example, daily city temperature, daily exchange rate of Pak. Rupee in US \$, monthly sale of a departmental store, annual yield of wheat of a country etc.

Analysis of Time Series

The statistical methods used to make an inference about the pattern of the time series data is known as the time series analysis. This analysis is used to detect patterns of change in statistical information over regular intervals of time.

Time Series Components

There are four components of time series:

1. *Secular Trend*
2. *Seasonal Variations*
3. *Cyclical Fluctuations*
4. *Irregular Variations*

Secular Trend

The basic long-term movement in a time series that can be described by a smooth line.

The trend is affected by changes in population, technology and productivity. Its duration is more than one year and in many cases it is often more than 10 years. The steady increase in the cost of living recorded by the consumer price index is an example of secular trend.

The methods of analyzing the secular trend are:

- The method of freehand curve
- The method of semi-averages
- The method of moving averages
- The method of least squares

Seasonal Variations

The seasonal variations are short-term movements occurring in a periodic manner. These variations are mainly caused by the changes in the season. These variations involves pattern of change within a year that tend to be repeated year to year. For example, increase in number of flu cases every winter, increase in use of electricity demand during summer, demand of liquid gas during winter, increase in sale of shoes near Eid, etc.

The methods of analyzing the seasonal variations by seasonal indices are:

- The Percentage-of-Annual Average Method
- The Raito-to-Moving Average Method
- The Ratio-to-Trend Method
- The Link Relative Method

Cyclical Fluctuations

Cyclical fluctuation is a wavelike pattern describing a long-term trend that is generally apparent over a number of years, resulting in a cyclical effect. A cycle is said to be completed when beginning with a peak, the declining curve reaches a low point, and then rising again reaching the next peak. By definition, it has duration of more than one year. Business cycle is the most common example of such variation.

Irregular Variations

Non-periodic or random fluctuations those are due to non-recurring or non-periodic events such as strikes, wars, elections, deaths, and weather changes etc.

Questions-Answers

Q.1 *How can you differentiate the cross-sectional and time series data sets?*

Ans. *Time series data are data collected over a periods of time. Such data are collected over different intervals of time, such as daily, weekly, monthly, quarterly, annually etc. While cross-sectional data are data on one or more variables collected at the same point in time such as individuals, consumers, firms, industry etc.*

Q.2 *What is the purpose of the time series analysis?*

Ans. *The basic purpose of the time series analysis is to use it for the forecasting. Therefore, it is a decision making tool. Time series analysis is used for many applications such as: Economic Forecasting, Sales Forecasting, Budgetary Analysis, Stock Market Analysis, Yield Projections, Process and Quality, Control, Inventory Studies, Workload Projections, Utility Studies, Census Analysis, etc.*

Q.3 *Why the term "Series" is applied in the time series analysis?*

Ans. *We use the term time series to refer to any group of statistical information accumulated at regular intervals.*

Q.4 *On what we frequently, base our predictions and decisions?*

Ans. *Past patterns of growth and changes are frequently used as a base for predications and decisions. The businessman may, for example, find that his sales have risen about 10% from April to May every year for the past 10-years, and use this to predict that his sales in the next year will increase from April to May too.*

- Q.5** *How can technological change affect the trend of a time series?*
Ans. *Technological change can cause upward or downward movement in the time series. For example, the development of the tractor caused a downward trend in the number of the mules, ox and camels on the farms. However, the development of the tractor helped produce an upward trend in the sales of the petrol.*
- Q.6** *What time of the time series have the composition $T \times C \times S \times I$?*
Ans. *Data classified by quarters, months and weeks have the composition $T \times C \times S \times I$.*
- Q.7** *What is the mathematical composition of time series?*
Ans. *Annual time series: $T \times C \times I$. A monthly and Quarterly time series: $T \times C \times S \times I$.*
- Q.8** *In which way you graphically represent the time series analysis?*
Ans. *Graphical representation of time series analysis is through the graph of time series; historigram.*
- Q.9** *Why it important to chart a time series before choosing the type of trend to fit to the series?*
Ans. *Charting the trend enables the one to make better choice of the equation or method use to describe it.*
- Q.10** *Give some rules for constructing time series line charts.*
Ans. *Some good rules are:*
- *Make a chart wider than high*
 - *Place time on the horizontal axis and a scale for the values on the vertical axis, show only a few scale values*
 - *Set the scale so that the line (or lines) will appear near the center of the chart*
 - *Start the scale with zero unless indexes are being charted*
 - *Make equal distance on the vertical scale represent equal absolute amounts*
 - *Plot points to the middle of the period for cumulative data and to the point of the time for non-cumulative data*
 - *Connect plotted points with the straight lines*

Q.11 *Why should the numerical scale of a time series line chart begin with the zero? Are there any exceptions?*

Ans. *The scale should begin with the zero because the base of the reference is zero. An exception is the scale for an index, which has the base of 100. Another exception is when a logarithmic scale (ratio scale) is used.*

Q.12 *What does the semi-logarithmic scales do?*

Ans. *A semi-logarithmic chart shows rates of the change. This chart is useful for comparing two series that are measured in the same units but different series that are measured in the same units but different magnitude, such as rupees and millions of rupees. It is also useful of comparing two series that are experienced in different units, such as tons of steels and number of autos.*

Q.13 *Define cumulative and non-cumulative data.*

Ans. *Cumulative data, often called period data, are those that cumulate from the beginning of a period to the end of a period. An example is annual sales for each year for 10 years. Non-cumulative data, often called point data, are those that refer to a specific point of time. An example would be the population of a province on July 1 of each year for 10 years.*

Q.14 *What is the basic reason for measuring the secular trend of a time series? Why is it important to know the reason?*

Ans. *The two basic reasons are:*

- *To project it and use it as a forecasting tool*
- *To eliminate it from the original data so that other movements can be studied independent from the trend's influence. When we know, the reasons for measuring the trend of a time series, it is better, able to choose an appropriate method.*

Q.14 *What type of trend is best one to use?*

Ans. *The type of trend that is best is the one that best fulfills the purpose of measuring it.*

Q.15 *In what direction does a secular trends move?*

Ans. *Secular trends can move up, down, or in both of the directions. i.e. value of the variable tends to increase or decrease over long period of the time.*

Q.16 *What is the most widely used trend line? What equation is generally used to describe a trend with one bend?*

Ans. *The least squares line is the mostly commonly used. The trend equation is;*

$$Y_t = a + bx + cx^2$$

is used more than another equation to describe a trend with one bend.

Q.17 *Explain the meaning of the coefficients of the equation*

$$Y_t = a + bx + cx^2$$

Ans. *The coefficient 'a' is the value of the trend as its origin, 'b' is the general basic slope of the line and 'c' is the typical change in the basic slope per unit time.*

Q.18 *What is the purpose of the measuring cycles of the time series?*

Ans. *Cycles of the time series usually measured so that they can be studied and analyzed in order to find a way to forecast turning point of the future cycles.*

Q.19 *Why are the turning points of the business cycles are different to predict?*

Ans. *Turning points of the business cycles are difficult to predict because turning points of past cycle have not occurred at regular interval of time.*

Q.20 *What are the four phases of a business cycle?*

Ans. *The four phases are the recovery, prosperity, recession and depression.*

Q.21 *Why it is difficult to measure, the typical business cycle by direct method?*

Ans. *The lack of regularity in timing and magnitude of cyclic fluctuations make it difficult to get a useful direct measure, such as an "average" cycle, which represent a typical business cycle.*

Q.22 *How is lag-lead relationship used?*

Ans. *Lag-lead relationships are sometimes used to predict the turning points of the cycles of one series based on the turning points of one or more series.*

Q.23 *How are the cycles isolated from the other components of a time series?*

Ans. *Cycles are isolated so that they must be analyzed and studied without being biased by the effect of other fluctuations.*

Q.24 *Give an example of an industry that experiences large irregular fluctuations in the activity. Tell why these fluctuations occur.*

Ans. *A good example would be construction because severe hot weather (June-July) changes can be abruptly stop/slow down construction activity. Another example is the stock market. The reason for these fluctuations is humorous and many are unknown. However, wars, strikes, political speeches, etc. will cause such fluctuations in prices of stock.*

Q.25 *Why would it be difficult to predict the point in time when a business cycle will reach a peak or trough?*

Ans. *The timing of cyclical peaks and troughs are difficult to predict because of the irregularity in the timing of the past business cycles.*

Q.26 *Show symbolically how the influence of seasonal variations may be removed from a monthly time series.*

Ans.
$$\frac{T \times C \times S \times I}{S} = T \times C \times I$$

Q.27 *If the direct measure of the cyclical and irregular movements of time series could be calculated, state symbolically how their combined influence could be removed from:*

- 1) *An annual time series*
- 2) *A monthly time series*
- 3) *A quarterly time series*

Ans. *Annual time series:* $T \times C \times I / C \times I = T$
Monthly time series and Quarterly time series
 $= T \times C \times S \times I / C \times I = T \times S$

Q.28 *Why do seasonal variations show a high degree of regularity in timing and amplitude?*

Ans. *A high degree of regularity in the seasonal variations resulted from the regularity of Holydays and seasonal buying habits of people and regularity of harvest, production and consumption that are caused by seasonal weather changes.*

Q.29 *What preliminary adjustment should sometime be made in data before measuring the typical seasonal pattern? Why?*

Ans. *Period data sometime need to be adjusted for the number calendar days or working days in a month. This would be done when one is interested in measuring the rate of seasonal activity rather than the total seasonal activity.*

Q.30 *Why might a seasonal pattern changes over a long period of time?*

Ans. *Changes in seasonal patterns over a long period can be due to technological improvements. e.g. improved transportation changed the seasonal consumption of certain fruits and vegetables in the cold climates.**

Q.31 *What is the chief advantage of using the percent of 12-months moving average method to compute a seasonal index?*

Ans. *The "percent of 12 month moving average" method enables the analyst to remove the effect of trend and cycles so that these movements will not bias the seasonal index.*

Q.32 *For what types of business decisions is a seasonal index of sales useful?*

Ans. Seasonal indices are useful for such decisions as diversification of products, personnel practices (for example, seasonal employment and vacation schedules), and advertising and promotions plans.

Q.33 Why are most important monthly economic indicators adjusted for seasonal variation?

Ans. The main purpose of the adjustments is to reveal the real basic directions of indicators so that the strength of the economy can be better evaluated.

Q.34 What are the two basic purposes for computing a seasonal index?

Ans. Seasonal indices are computed in order to use them (i) to make short-range forecasts and (ii) eliminate the effect of seasonal variations from the original data.

Q.35 What does the seasonal index measure?

Ans. It measures the average or typical seasonal pattern found in the time series.

Q.36 What is the difference between a stable seasonal index and a moving seasonal index?

Ans. A stable seasonal index has only 12 seasonal index numbers, one for each month. A moving seasonal index has several index numbers for each month of each year covered in the analysis. Moving seasonal indices are used to measure seasonal patterns that are changing.

Q.37 If the storm prevents people from shopping for two days, what type of fluctuation in the sales would this cause: trend, seasonal, cyclical, or irregular? Why?

Ans. "Irregular", because storms are irregular in the timings of the occurrence.

Q.38 Can we explain this factor mathematically?

Ans. As these factors are unpredictable we do not attempt to explain it mathematically.

Q.39 *Why an irregular fluctuation in the business activity impossible to forecast?*

Ans. *Irregular fluctuations are impossible to forecast the events cause them cannot be predicted with certainty.*

Q.40 *Define Smoothing.*

Ans. *Smoothing techniques are used to reduce irregularities (random fluctuations) in time series data. They provide a clearer view of the true underlying behavior of the series. In some time series, seasonal variation is so strong it obscures any trends or cycles, which are very important for the understanding of the process being observed. Smoothing can remove seasonality and makes long term fluctuations in the series stand out more clearly.*

Q.41 *What are different types of smoothing?*

Ans. *Moving average smoothing, exponential smoothing, running medians smoothing, etc.*

Q.42 *What is the most commonly used type of smoothing?*

Ans. *The most common type of smoothing technique is moving average smoothing. Since the type of seasonality will vary from series to series, so must the type of smoothing.*

Q.43 *Define moving average smoothing.*

Ans. *A moving average is a form of average, which has been adjusted to allow for seasonal or cyclical components of a time series. Moving average smoothing is a smoothing technique used to make the long-term trends of a time series clearer. When a variable, like the number of unemployed, or the cost of strawberries, is graphed against time, there are likely to be considerable seasonal or cyclical components in the variation. These may make it difficult to see the underlying trend. These components can be eliminated by taking a suitable moving average. By reducing random fluctuations, moving average smoothing makes long term trends clearer.*

Q.44 *What is forecasting?*

Ans. *In time series analysis, forecasting is a process of assessing the magnitude of a time series variable which it will assume at some future point of time. Forecasting is based on the assumption that the past pattern and behaviour of a variable will continue in the future.*

Exercises

Exercise 10 (MCQs')

- Q.1** Data concerning events over a period of time is called a:
- (a) Time Series
 - (b) Moving Average
 - (c) Frequency Distribution
 - (d) Random Sample
- Q.2** Which of the following is not a component of time series?
- (a) Seasonal Variation
 - (b) Cyclical Variation
 - (c) Variance
 - (d) Trend
- Q.3** In a time series, secular variation is:
- (a) A variation that occurs at regular intervals
 - (b) The difference between the largest and smallest item in any one-year
 - (c) The range of time to which the series applies
 - (d) The long-term trend
- Q.4** A cyclical variation is:
- (a) One that takes a number of years to complete
 - (b) One that goes round in circles
 - (c) One that occurs four times a year
 - (d) A regular change

- Q.5** The number of cars sold by a car-dealer during just 6 months in 2003 was as following;

January	February	March	April	May	June
18	16	28	51	47	55

What is the first 3 monthly moving average?

- (a) 20.67
 - (b) 31.67
 - (c) 42.00
 - (d) 51.00
- Q.6** A histogram is:
- (a) A frequency graph
 - (b) A time series graph
 - (c) A graph-plotting mean against standard deviation
 - (d) A correlative frequency chart
- Q.7** The weekly takings in a shop over 5 weeks were as follows:

Week	1	2	3	4	5
Taking (Rs.)	98	112	161	109	101

The first 4 weekly moving averages was:

- (a) Rs. 105.00
 - (b) Rs. 109.00
 - (c) Rs. 120.00
 - (d) Rs. 120.75
- Q.8** The following table shows (in thousands) the number of units of electricity used by a firm over a period of two years.

Quarter	1	2	3	4	1	2	3	4
No. of units (1000's)	85	49	25	87	89	53	29	86

What is the value of the second 4-quarterly moving average?

- (a) 61.50 thousand
- (b) 62.50 thousand

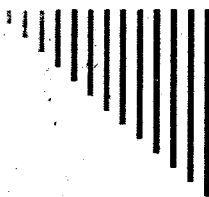
- (c) 86.50 thousand
- (d) 64.75 thousand

Q.9 Suppose you were considering a time series of data for the quarters of 1992 and 1993. The third quarter of 1993 would be coded as:

- (a) 2
- (b) 3
- (c) 5
- (d) 6

Q.10 Assume that you have been given quarterly sales data for a five-year period. To use the ratio-to-moving-average method of computing a seasonal index, your first step will be

- (a) Compute the four-quarter moving average
- (b) Discard highest and lowest value for each quarter
- (c) Calculate the four-quarter moving total
- (d) None of these



Chapter 11

Index Numbers

Index

A numerical scale used to compare variables with one another or with some reference number. In other words, a number or ratio (a value on a scale of measurement) derived from a series of observed facts.

Index Number

An index number measures how much a variable changes over time or space. This is a statistical measure to give average change in a variable or group of variables with respect to time or space.

We calculate an index number by finding the ratio of the current value to a base value then we multiply the resulting number by 100 to express the index as a percentage. This final value is the percentage relative. Note that the index number for the base point in time is always hundred.

Generally,

$$\text{Index Number} = \frac{\text{Current Value}}{\text{Base Value}} \times 100.$$

Base Period

The period of time for which data used as the base of an index number or other ratio. In other words, it is a period from which the changes are measured.

Types of Index Numbers

There are three principle types of indices:

- Price Index
- Quantity Index
- Value Index

Price Index

A price index is the one most frequently used. It compares levels of prices from one period to another.

$$\text{Price Index Number} = \frac{\text{Current Price}}{\text{Base Price}} \times 100 = \frac{P_n}{P_0} \times 100$$

For example, if the retail price of sugar is Rs. 18 in 1998 and Rs. 30 in 2005 from the following Table then the index number for 2005 on the base price in 1998 will be

$$P_{0,n} = P_{1998, 2005} = \frac{P_n}{P_0} \times 100 = \frac{P_{2005}}{P_{1998}} \times 100 = \frac{30}{18} \times 100 = 166.67.$$

It means that if the price of sugar/Kg. was Rs. 100 in 1998 then it becomes Rs. 166.67 in 2005 or the said price gets 66.67% increase in 2005 when we compare that in 1998.

The quantity $\frac{P_n}{P_0}$ is also called Price Relative.

Quantity Index

A quantity index measures how much the number or quantity of a variable changes over time.

From the following Table, the quantity index for year 2005 using year 1998 as base:

$$q_{0,n} = q_{1998, 2005} = \frac{q_n}{q_0} \times 100 = \frac{q_{2005}}{q_{1998}} \times 100 = \frac{25}{18} \times 100 = 186.67.$$

Value Index

The value index measures the changes in total monetary worth. In fact, the value index combines price and quantity changes to present a more informative index.

From the following Table, showing the current and base year prices and the quantities consumed (in 000' Kg.), the value index for year 2005 using year 1998 as base:

$$V_{0,n} = V_{1998, 2005} = \frac{V_n}{V_0} \times 100 = \frac{V_{2005}}{V_{1998}} \times 100 = \frac{750}{270} \times 100 = 277.78.$$

Table: Prices (Rs./Kg.), Quantities (consumed in 000' Kg.) and values for Sugar
(period '0' denotes 1998 while 'n' denotes 2005)

P_0	P_n	q_0	q_n	$v_0 = P_0 q_0$	$v_n = P_n q_n$
18	30	15	28	270	750

Simple and Aggregate Index Numbers

Simple index numbers calculate price changes for a single item over time. Index numbers are more accurate if they are constructed using actual prices paid for a single commodity, product or service rather than the more general aggregated index.

Aggregate index numbers calculate price changes for a group of related items over time. Aggregate indices permit analysis of price changes for the group of related products, such as price changes for apples, oranges, mangoes, etc.

Unweighted Aggregate Index Numbers

In unweighted index numbers, all the values considered are of equal importance or equally weighted. Aggregate means that we add, or sum, all the values.

It has two types:

- Simple Aggregate Index
- Simple Average of Relatives

Simple Aggregate Index

It measures the percentage change in the aggregate prices of a number of commodities, at different periods. It is computed by dividing the sum of the given year prices of all commodities by the sum of all the base year prices of the same commodities and expressing the results as a percentage.

$$p_{0..n} = \frac{\sum p_n}{\sum p_0} \times 100.$$

Simple Average of Relatives

It is the average of price relatives in an index obtained by taking the average of the price relatives of the given commodities for each year and expressing the results as a percentage.

$$p_{0..n} = \frac{1}{M} \sum \left(\frac{p_n}{p_0} \right) \times 100,$$

where M is the number of commodities under study.

Weighted Index Numbers

Sometimes, we have to attach greater importance to changes in some variables than to others when we compute an index. This weighting allows us to include more information than just the change in price over time. Generally, such index numbers measure the change in the prices of a group of commodities when the relative importance of the commodities has been taken into account.

It has two types:

- Weighted Aggregate Price Index Numbers
- Weighted Average of Relative Price Index Number

Weighted Aggregate Price Index Numbers

It is constructed for an aggregate of items (prices) that have been weighted. These weights are usually, corresponding quantities produced, consumed or sold.

Following are the different weighted price index numbers:

$$\text{Laspeyres' Price Index} = \frac{\sum p_n q_0}{\sum p_0 q_0} \times 100.$$

$$\text{Passche's Price Index} = \frac{\sum p_n q_n}{\sum p_0 q_n} \times 100$$

$$\text{Fisher's Ideal Price Index} = \sqrt{\frac{\sum p_n q_0 \times \sum p_n q_n}{\sum p_0 q_0 \times \sum p_0 q_n}} \times 100$$

$$\text{Marshall-Edgeworth's Price Index} = \frac{\sum p_n (q_0 + q_n)}{\sum p_0 (q_0 + q_n)} \times 100$$

$$\text{Walsh's Price Index} = \frac{\sum p_n \sqrt{q_0 q_n}}{\sum p_0 \sqrt{q_0 q_n}} \times 100$$

Weighted Average of Relative Price Index Number

It is computed by multiplying each price relative ($\frac{p_n}{p_0}$) by its weight, summing these products and dividing by the sum of the weights. The weights are always the total values of the commodities. Following are the different types of such indices:

$$\text{Laspeyres' Index} = \frac{\sum (p_n / p_0) p_0 q_0}{\sum p_0 q_0} \times 100$$

$$\text{Passche's Index} = \frac{\sum (p_n / p_0) p_0 q_n}{\sum p_0 q_n} \times 100$$

$$\text{Palgrave's Index} = \frac{\sum (p_n / p_0) p_n q_n}{\sum p_n q_n} \times 100$$

Plausibility (Tests) of Index Numbers

Following are the tests available to check the plausibility of index numbers:

- Time Reversal Test
- Circular Test
- Permutation or Price Bouncing
- Commensurability
- Factor Reversal Test

Time Reversal Test

This test requires that the index formula produced consistent results whether it is calculated going from period 0 to period n or from period n to period 0. More specifically, if the price observations for period 0 and n are interchanged then the resulting price index should be the reciprocal of the original index.

Symbolically,

$$P_{0n} = \frac{1}{P_{n0}} \text{ or } P_{0n} \times P_{n0} = 1.$$

Circular Test

This test often called transitivity is a multi-period test (essentially a test of chaining). It requires that the product of a price index obtained by going from period 0 to period 1 and from period 1 to 2 be the same as going directly from period 0 to 2.

Symbolically,

$$P_{01} \times P_{12} = P_{02}.$$

Generally,

$$P_{01} \times P_{12} \times \dots \times P_{n-1,n} \times P_{n0} = 1.$$

Permutation or Price Bouncing

This test requires that, if the order of the prices in either period 0 or period n (or both) is changed but not the individual prices, the index number should not change. This test is appropriate in situations where there is considerable volatility in prices, for example, because of seasonal factors or sales competitions.

Commensurability

This test requires that if the units of measurement of the items are changed (for example, from Kgs. to Tonnes), then the price index will not change.

Factor Reversal Test

This test requires that the product of the price index number for any period and an index of quantity obtained from the formula by interchanging the price and quantity terms should equal the ratio of expenditure in that period to the base period expenditure.

In other words, if the factor ' p ' is changed by the factor ' q ', index formula be interchanged (or reversed) so that a quantity (or price) index formula is obtained, then the product of the two index numbers should equal the value index number.

That is,

$$(\text{Price Index}) \times (\text{Quantity Index}) = \text{Value Index}.$$

Consumer Price Index (CPI)

Consumer Price Indices measure changes over time in the general level of prices of goods and services that a reference population acquires, uses or pays for consumption. A Consumer Price Index is estimated as a series of summary measures of the period-to-period proportional change in the prices of a fixed set of consumer goods and services of constant quantity and characteristics acquired, used or paid for by the reference population. Each of the elementary aggregate indices is estimated using a sample of prices for a defined set of goods and services obtained in, or by residents of, a specific region from a given set of outlets or other sources of consumption goods and services.

Questions-Answers

Q.1 *What is the main purpose for construction of index numbers? Explain with examples.*

Ans. *The main use of index number comes when we are interested to know that how much something has changed over a period of time. We may want to know how much the prices of certain daily used items have increased so we can adjust our budget accordingly. A factory manager may wish to compare this month's per unit production cost with that of past six months. Or a medical research team may wish to compare the number of malaria cases reported this year with the number reported in previous years. In each of these situations, we require the degree of change with respect to time. Typically, we use index numbers to measure such differences.*

Q.2 *Give the headlines/steps for the construction of price index numbers.*

Ans. *The following steps are involved in the construction of index numbers:*

- i. Selection of commodities to be included*
- ii. Collection of data*
- iii. Selection of the base period*
- iv. Selection of appropriate average*
- v. Selection of appropriate weights*

Q.3 *Differentiate between simple and composite index numbers.*

Ans. *An index number is called a simple index when it is computed for a single variable. For example, index numbers of registration of voters, index numbers of cement prices, etc. An index that is computed for two or more variables is called composite index. For example, wholesale price index numbers, CPI, etc.*

Q.4 *Differentiate between fixed and chain base method.*

Ans. *If the base period is fixed for the calculation of indices then it is known as fixed base method, otherwise the term chain base method*

is employed. In chain base method every period is a base period for its next period.

Q.5 *What are the main characteristics of the base period?*

Ans. The base period should be a normal period. It means that a period of economic stability and free from any major financial crisis caused by inflation, depression, wars, labour unrest, strikes, natural disasters etc. This period should not be too far distant in the past making the comparisons meaningless.

Q.6 *What is the most appropriate time duration of the base period?*

Ans. This period is frequently one year but it may be as short as one day or as long as the average of a group of years.

Q.7 *What the term "Weighting" reflects in measuring index numbers?*

Ans. Sometimes, we have to attach greater importance to changes in some variables than to others when we compute an index. This weighting allows us to include more information than just the change in price over time. The process which assigns numerical coefficients (weights or weighting factors) to each of the element in a data set, in order to provide them with a desired degree of importance relative to one another.

Q.8 *What is the most popular type of price index number?*

Ans. The CPI (given above) is the most common example of price index number.

Q.9 *What is the main purpose of construction of CPI?*

Ans. It measures overall price changes of a variety of consumer goods and services and is used to define cost of living.

Q.10 *What are the main methods to compute CPI?*

Ans. There are two methods used to construct CPI. They are:

- (i) Aggregate Expenditure Method
- (ii) Household Budget Method

Q.11 *What are the main shortcomings of CPI?*

Ans. *The construction of CPI involves the sampling of goods and services, the sampling errors and biases may affect indices and render them to suspect.*

In case of certain goods, it is difficult to collect prices actually needed. For example, the prices for clothing usually relate to cloth and not to tailored clothes.

Q.12. *What is Producer Price Index (PPI)?*

Ans. *PPI is a composite figure of producer's prices of representative commodities included in the market basket. It is used to measure monthly or yearly change in producer's prices of key commodities in the manufacturing sector. It is used to deflate production indicators and serves as the deflator in the estimation of manufacturing production in the system of national accounts.*

Q.13 *What are the main shortcomings in construction of index numbers?*

Ans. *The main shortcomings are:*

- (i) It is very difficult to collect prices of the commodities included in the construction of indices as well as to take into account all changes in quantity or product.*
- (ii) Sampling error is expected to be found in the calculation of indices.*
- (iii) The choice of a normal period is a difficult task as, at a time, more than one period can be considered as normal for all segments of the economy.*

Exercises

Exercise 11.1 (MCQs')

- Q.1** To measure changes in total monetary worth, one should calculate:
- (a) Price index
 - (b) Quantity index
 - (c) Value index
 - (d) CPI
- Q.2** If an index number calculations over 8 years with a base value of 100 gave an index for 2003 of 120, what would be the percentage relative for 2003?
- a) 100
 - b) 120
 - c) 880
 - d) 960
- Q.3** Which of the following describes an advantage of using the Laspeyres' method?
- (a) Many commonly used quantity measures are not tabulated for every period.
 - (b) Changes in consumption patterns are taken into account.
 - (c) One index can be easily compared with another.
 - (d) (a) and (c) but not (b)
- Q.4** When computing a weighted average of relative index, we would be best able to compare indices from various periods if:
- (a) Base values were used as $p_n q_n$
 - (b) Current values were used as $p_n q_n$
 - (c) Fixed values were used as $p_n q_n$
 - (d) Either base or fixed values were used as $p_n q_n$

- Q.5** Commodities subject to considerable price variations could best be measured by:
- (a) Price index
 - (b) Quantity index
 - (c) Value index
 - (d) CPI
- Q.6** A base period can be described as a normal period if:
- (a) It is neither the peak nor the trough of a fluctuation
 - (b) It must be the most recent period for which we have data
 - (c) There was no inflation or deflation of prices during the period
 - (d) (a) and (c) above
- Q.7** The weights used in a quantity index are:
- (a) Percentages of total quantity
 - (b) Prices
 - (c) Average of quantities
 - (d) None of these
- Q.8** To measure how much the cost of some variable changes over time, we would use:
- (a) Inflation index
 - (b) Quantity index
 - (c) Value index
 - (d) None of these
- Q.9** When the base year values are used as weights, the weighted average of relative price index is the same as:
- (a) The Paasche's index
 - (b) The Laspeyres' index
 - (c) The unweighted average of relative price index
 - (d) None of these

- Q.10** A primary difference between average of relatives and aggregate methods is that:
- (a) Aggregate methods sum all prices before finding the ratio
 - (b) Average of relatives methods sum all prices before finding the ratio
 - (c) Aggregates methods are useful only for price indices
 - (d) (a) and (c) but not (b)

Exercise 11.2 (True/False)

Read the following statements carefully and indicate which statement is "True" or "False":

1. The price index numbers are used to measure changes in a particular group of prices and help us in comparing the movement in prices of one commodity with another.
2. Index numbers give very valid comparison of changes in variable over long periods.
3. All index numbers are not suitable for all purposes.
4. Index numbers cannot be used to measure enrollment changes in an institution.
5. There are two ways for the selection of base period.
6. The arithmetic mean, median and geometric mean are the usual measures for the calculation of average in index numbers.
7. Weighted aggregate index number uses the relative importance of the index numbers.

8. Fisher ideal index has a theoretical advantage over other index numbers as it is the only index that follows the time reversal, factor reversal and circular test.
9. A CPI is used to measure changes in the composite price of a specified basket of goods and services during the given period as compared with the base period.
10. The indices calculated from different index numbers may not agree.
11. Fisher ideal index is the hybrid of two index numbers.
12. Index numbers cannot measure differences in a given variable in several locations.
13. A simplest form of a composite index is a weighted aggregate index.
14. An index number is always found by taking the ratio of current value to a base value and multiplying by 100.
15. The simple average of relatives method divides the weighted sum by the sum of weights.



Chapter 12

Nonparametric Statistics

Nonparametric Tests

Parametric tests require assumptions about the nature or shape of the populations involved; nonparametric tests do not require such assumptions. Consequently, nonparametric tests of hypotheses are often called Distribution Free Tests.

Some of these tests are Sign test, Wilcoxon Signed-Rank test, Runs test, Mann-Whitney U test, Kruskal Wallis test, etc.

Nonparametric tests may be, and often are, more powerful in detecting population differences when certain assumptions are not satisfied.

Nonparametric methods have a number of clear advantages over parametric methods. The advantages and disadvantages are as under;

Advantages:

1. They do not require us to make the assumption that a population is distributed in the shape of normal curve or another specific shape.
2. Generally, they are easier to handle and to understand.
3. Unlike the parametric methods, nonparametric methods can often be applied to non-numeric data such as the gender of survey respondents.
4. Some times, even formal ordering or ranking is not required.

Disadvantages:

1. They ignore a certain amount of information.
2. They are often not as efficient or sharp as the parametric tests.

3. Nonparametric tests typically require data to be rank-ordered and/or grouped according to nominal classifications.

Sign Test

The sign test is designed to test a hypothesis about the location of a population distribution. It is most often used to test the hypothesis about a population median, and often involves the use of matched pairs, for example, before and after data, in which case it tests for a median difference of zero.

We can use a signed test;

- (i) To test claims about the median of the paired differences for two dependent samples
- (ii) To test claims about certain types of nominal data
- (iii) To test the claim made about the median of a single population.

The Sign test does not require the assumption that the population is normally distributed. In many applications, this test is used in place of the one sample t -test when the normality assumption is questionable. This test can also be applied when the observations in a sample of data are ranks, that is, ordinal data rather than direct measurements.

Wilcoxon Signed Ranks Test

The signed test considers only whether each data value is above (+) or below (-) the value M given for the median in the null hypothesis. It does not take into account how much each data value is above or below M thus a large amount of pertinent information is ignored. The Wilcoxon Signed Ranks test incorporates these informations. The Wilcoxon Signed Ranks test is designed to test a hypothesis about the location (median) of a population distribution. It often involves the use of matched pairs, for example, before and after data, in which case it tests for a median difference of zero.

In many applications, this test is used in place of the one sample t -test when the normality assumption is questionable. It is a more powerful alternative to the sign test, but does assume that the population probability distribution is symmetric.

This test can also be applied when the observations in a sample of data are ranks, that is, ordinal data rather than direct measurements.

Mann-Whitney U Test

The Mann-Whitney U test is one of the most powerful nonparametric tests for comparing two populations. It is used as a test of comparison of medians or means of two populations using independent samples.

The Mann-Whitney test does not require the assumption that the differences between the two samples are normally distributed. In many applications, the Mann-Whitney test is used in place of the two sample t -test when the normality assumption is questionable.

This test can also be applied when the observations in a sample of data are ranks, that is, ordinal data rather than direct measurements. This test is also known as Mann-Whitney U test and Wilcoxon Rank-Sum test.

Kruskal-Wallis Test

The Kruskal-Wallis test is a nonparametric test used to compare three or more populations. It is used to test the null hypothesis that all populations have identical distribution functions against the alternative hypothesis that at least two of the samples differ only with respect to location (median), if at all.

It is the analogue to the F -test used in analysis of variance. While analysis of variance test depends on the assumption that all populations under comparison are normally distributed, the Kruskal-Wallis test places no such restriction on the comparison. It is a logical extension of the Mann-Whitney test.

It is also known as H -test. This H statistic has a distribution that can be approximated by the Chi-square distribution as long as each sample has at least five observations. It is therefore a right-tailed test.

Runs Test

In studies where measurements are made according to some well defined ordering, either in time or space, a frequent question is whether or not the average value of the measurement is different at different points in the sequence. The runs test provides a means of testing this. In other words, it is a procedure for testing the randomness of data.

Run:

A run is a sequence of data that exhibits the same characteristic; the sequence is preceded and followed by different data or no data at all. In other words, it is maximal sequence of similar elements.

The number of runs in the sequence, SS F S FFF SSS F, is 6.

Spearman's Rank Correlation Test

Spearman's Rank Correlation test is a nonparametric test for testing the significance of Spearman's Rank correlation that is the nonparametric counterpart of that parametric measure. Rank correlation is the measure of the correlation that exists between the two sets of ranks, a measure of the degree of association between the variables that we would not have been able to calculate otherwise.

Kolmogorov-Smirnov Test

For a single sample of data, the Kolmogorov-Smirnov test is used to test whether or not the sample of data is consistent with a specified distribution function. When there are two samples of data, it is used to test whether or not these two samples may reasonably be assumed to come from the same distribution. The Kolmogorov-Smirnov Test is therefore another measure of the goodness of fit of the theoretical frequency distribution as was the Chi-square test. The Kolmogorov-Smirnov test does not require the assumption that the population is normally distributed.

Questions-Answers

Q.1 *What are the usual normal-theory-based procedures for hypotheses testing?*

Ans. *For hypothesis testing z-test, t-test, Chi-square test, F-test are usual normal-theory-based procedures.*

Q.2 *When do we use nonparametric tests?*

Ans. *The following are some situations in which the use of nonparametric tests is appropriate;*

- (i) The hypothesis to be tested does not involve a population parameter.*
- (ii) The data have been measured on a scale weaker than that required for the parametric procedure that would otherwise be employed. For example, the data may consist of count or rank data.*
- (iii) The assumptions necessary for the valid use of parametric procedures are not met.*
- (iv) Results are needed in a hurry and calculations must be done by hand.*

Q.3 *What is the nonparametric alternative of the two sample t-test for independent samples?*

Ans. *Usually, when we have two samples that we want to compare concerning their mean value for some variable of interest, we would use the t-test for independent samples; nonparametric alternatives for this test is the Mann-Whitney U test.*

Q.4 *What is the nonparametric alternatives of the ANOVA, when samples are independent?*

Ans. *If we have multiple groups, we would use analysis of variance (ANOVA); the nonparametric equivalents to this method are the Kruskal-Wallis test.*

Q.5 *What are the nonparametric alternatives of the ANOVA, when samples are dependent?*

Ans. *If there are more than two variables that were measured in the same data, then we would customarily use repeated measures ANOVA. Nonparametric alternatives to this method are Friedman's two-way analysis of variance and Cochran Q test (if the variable was measured in terms of categories, e.g., "passed" vs. "failed"). Cochran Q is particularly useful for measuring changes in frequencies (proportions) across time.*

Q.6 *What are the nonparametric alternatives of the two sample t-test for dependent samples?*

Ans. *If we want to compare two variables measured in the same sample we would customarily use the matched pair t-test (or t-test for dependent samples). Nonparametric alternatives to this test are the Sign test and Wilcoxon's matched pairs test.*

Q.7 *What nonparametric tests are used to test the relationship between variables?*

Ans. *To express a relationship between two variables one usually computes the correlation coefficient. Nonparametric equivalents to the standard correlation coefficient are Spearman's R, Kendall Tau, and coefficient Gamma. If the two variables of interest are categorical in nature (e.g., "passed" vs. "failed" by "male" vs. "female") appropriate nonparametric statistics for testing the relationship between the two variables are the Chi-square test, the Phi coefficient, and the Fisher exact test. In addition, a simultaneous test for relationships between multiple cases is available i.e Kendall coefficient of concordance.*

Q.8 *For what purpose Kendall coefficient of concordance is used?*

Ans. *Kendall coefficient of concordance is often used for expressing inter-rater agreement among independent judges who are rating (ranking) the same stimuli. It is extension of rank correlation coefficient.*

Q.9 *What can be determined with the Mann-Whitney test? What kind of data are used for this test?*

Ans. *The t-test and z-test are useful for testing whether two samples have been drawn from populations that are assumed to be normally distributed and which have equal means and variances. However, the Mann-Whitney test enables us to make a significance test without assumptions, although the data for the Mann-Whitney test are assumed to be continuous and which must be ranked.*

Q.10 *What is the purpose of the Kruskal-Wallis H test? What type of data are used for this test?*

Ans. *This test is used to test the hypothesis that the several samples come from populations with the same means. It requires no assumptions about the distributions of the populations from which the samples were drawn except that the population data are continuous. To make this test the data for all samples must be pooled and ranked in order of magnitude.*

Q.11 *What is the null hypothesis when we use the Spearman Rank correlation coefficient test?*

Ans. *The null hypothesis is that there is no significant correlation between the two rankings.*

Q.12 *Discuss the merits and demerits of Kolmogorov-Smirnov(K-S) goodness of fit test with comparison of Chi-square goodness of fit test.*

Ans. *There are some merits and demerits of K-S goodness of fit test with comparison of Chi-square goodness of fit test:*

- (i) The K-S test does not require that the observations be grouped as in the case with the Chi-Square test. The consequence of this difference is that the K-S test makes use of all the information present in a set of table.*
- (ii) The K-S test can be used with any size of sample. It will be recalled that certain minimum sample sizes are required for the use of the Chi-square test.*
- (iii) K-S test is not applicable when parameters have to be submitted from the sample. The Chi-square test may be used in these situations by reducing the degrees of freedom by 1 for each parameter estimated.*

Q.13 *Define non-randomness.*

Ans. *A very large or very small number of runs in a sequence show non-randomness.*

Q.14 *What are the major nonparametric tests used to test the randomness in the data?*

Ans. *Runs test for randomness and Kendal Tau test are the major tests used to test the randomness in the data.*

Q.15 *Give any example for runs test?*

Ans. *Suppose that, as part of a screening program for heart disease, men aged 45-65 years have their blood cholesterol level measured on entry to the study. After many months it is noticed that cholesterol levels in this population appear somewhat higher in the Winter than in the Summer. This could be tested formally using a Runs test on the recorded data, first arranging the measurements in the date order in which they were collected.*

Exercises

Exercise 12.1 (MCQs')

- Q.1** The sign test is:
- (a) Less powerful than that of the Wilcoxon signed rank test
 - (b) More powerful than the paired sample t -test
 - (c) More powerful than the Wilcoxon signed rank test.
 - (d) Equivalent to Mann-Whitney test
- Q.2** The nonparametric equivalent of an unpaired samples t -test is the:
- (a) Sign test
 - (b) Wilcoxon signed rank test
 - (c) Mann-Whitney U test.
 - (d) Kruskal-Wallis Test
- Q.3** The Mann-Whitney U test is preferred to a t -test when:
- (a) Data are paired
 - (b) Sample sizes are small
 - (c) The assumption of normality is not met.
 - (d) Samples are dependent
- Q.4** When using the sign test, if two scores are tied, then we:
- (a) Count them
 - (b) Discard them
 - (c) Depends upon the scores
 - (d) None of these
- Q.5** The sign test assumes that the:
- (a) Samples are independent
 - (b) Samples are dependent
 - (c) Samples have the same mean
 - (d) None of these

- Q.6 When testing for randomness, we can use:
- (a) Mann-Whitney U test
 - (b) Sign test
 - (c) Runs test
 - (d) None of these
- Q.7 The Runs test results in rejecting the null hypothesis of randomness when:
- (a) There is an unusually large number of runs
 - (b) There is an unusually small number of runs
 - (c) Either of the above
 - (d) None of the above
- Q.8 The Wilcoxon rank-sum test can be:
- (a) Upper tailed.
 - (b) Lower tailed
 - (c) Either of the above
 - (d) None of the above
- Q.9 The Wilcoxon rank-sum test compares:
- (a) Two populations
 - (b) Three populations
 - (c) A sample mean to the population mean.
 - (d) Any number of populations
- Q.10 The Wilcoxon signed rank is used:
- (a) Only with independent samples
 - (b) Only in matched pairs samples
 - (c) As an alternative to the Kruskal-Wallis test
 - (d) To test for randomness
- Q.11 Which of the following test use rank sums?
- (a) F test
 - (b) Chi-square and Sign tests
 - (c) Runs test
 - (d) Kruskal-Wallis and Wilcoxon tests

- Q.12** Which of the following tests must be two-sided?
- (a) Kruskal-Wallis
 - (b) Wilcoxon Signed rank
 - (c) Runs test
 - (d) Sign test
- Q.13** In testing for the difference between two populations, it is possible to use:
- (a) The Wilcoxon rank-sum test
 - (b) The sign test
 - (c) Either of the above
 - (d) None of the above
- Q.14** In a Wilcoxon rank-sum test:
- (a) Ties never affect the decision
 - (b) Ties always affect the decision
 - (c) Ties within one sample may affect the decision
 - (d) Ties between the two samples may affect the decision
- Q.15** The Spearman rank-correlation test requires that the:
- (a) Data must be measured on the same scale
 - (b) Data at least ordinal scaled
 - (c) Data must be from two independent samples
 - (d) Data must be distributed at least approximately as a t -distribution
- Q.16** To perform a runs test for randomness the data must be:
- (a) Qualitative
 - (b) Quantitative
 - (c) Divided into at least two classifications
 - (d) Divided into exactly two classifications
- Q.17** To compare the annual income of engineers with those of clerks, two random samples are obtained $n_1 = 20$ and $n_2 = 20$. A decision was made to use the Wilcoxon rank-sum test to determine if clerks earn more than engineers. Which of the following can be a valid reason for this decision.
- (a) The sample are too small to estimate μ and σ^2
 - (b) The samples do not adequately represent the populations

- (c) The samples are from non-normal distributions
 - (d) The samples are not of equal size
- Q.18** Three brands of coffee are rated for taste on a scale of 1 to 10. Six persons are asked to rate each brand so that there is a total of 18 observations. The appropriate test to determine if three brands taste equally good is:
- (a) One way analysis of variance
 - (b) Wilcoxon rank-sum test
 - (c) Spearman rank difference
 - (d) Kruskal-Wallis test
- Q.19** If a Chi-square goodness of fit test has 6 categories and an $N=30$, then the correct number of degrees of freedom is:
- (a) 5
 - (b) 6
 - (c) 28
 - (d) 29
- Q.20** Comparing the times-to-failure of radar transponders made by firms *A*, *B*, and *C*, based on an airline's sample experience with the three types of instruments, one may well call for:
- (a) A Kolmogorov-Smirnov test
 - (b) A Kruskal-Wallis test
 - (c) A Wilcoxon rank-sum test
 - (d) A Spearman rank correlation test
- Q.21** Which of the following tests is most likely assessing this null hypothesis: H_0 : The number of violations per apartment in the population of all city apartments is binomially distributed with a probability of success in any one trial of $p=0.3$:
- (a) The Kolmogorov-Smirnov test
 - (b) The Kruskal-Wallis test
 - (c) The Mann-Whitney test
 - (d) The Wilcoxon signed-rank test

- Q.22 In the Kruskal-Wallis test of k samples, the appropriate number of degrees of freedom is
- (a) k
 - (b) $k-1$
 - (c) n_k-1
 - (d) $n-k$
- Q.23 When compare to parametric methods, nonparametric methods are
- (a) Less accurate
 - (b) Less efficient
 - (c) Computationally easier
 - (d) (b) and (c) but not (a)

Exercise 12.2 (True/False)

Read the following statements carefully and indicate which statement is "True" or "False":

1. Ties between paired observations should be deleted before a sign test is made.
2. The Mann-Whitney U test is a two tailed test.
3. The Mann-Whitney U test can be used to test if two samples came from the same population or identical populations.
4. If six or more observations are in each sample, the Kruskal-Wallis H test is distributed as Chi-square.
5. The Kruskal-Wallis H test can be used to determine if several samples came from populations with equal means.
6. A Chi-square analysis should not be originally made when there are fewer than 5 observations in one or more of the cells.
7. The runs test is upper tailed.

8. The Wilcoxon signed rank test may be used whenever the sign test is applicable.
9. The binomial distribution can be used to provide probabilities for outcomes when the sign test is used.
10. One of the main advantage of the nonparametric tests is that the underlying assumptions are often less restrictive than those of parametric tests.
11. Parametric tests are easier to compute and therefore more desirable than nonparametric tests.
12. Nonparametric tests are more powerful than parametric tests because they are not bound by population assumptions.
13. Parametric tests are appropriate when the dependent variable is the number of correct trials out of a fixed number of trials.
14. The nonparametric tests apply only to populations that have no parameters.
15. A nonparametric test is often less efficient than its parametric counterpart, because a nonparametric test utilizes less of the information contained in the sample.



Answers to Exercises

Chapter 1: Introductory Statistics

Exercise 1.1 (MCQs')

1. d 2. d 3. b 4. a 5. a

Exercise 1.2 (True/False)

- | | | | | |
|-----------|-----------|-----------|-----------|-----------|
| 1. False | 2. True | 3. True | 4. False | 5. False |
| 6. True | 7. False | 8. True | 9. True | 10. True |
| 11. True | 12. False | 13. True | 14. False | 15. False |
| 16. False | 17. False | 18. True | 19. True | 20. False |
| 21. False | 22. True | 23. False | 24. False | 25. True |
| 26. False | 27. True | 28. False | 29. True | 30. False |
| 31. True | 32. False | 33. False | 34. True | 35. True |
| 36. False | 37. False | 38. False | 39. True | 40. True |

Chapter 2: Basic Probability Theory

Exercise 2.1 (MCQs')

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. d | 2. a | 3. d | 4. b | 5. a |
| 6. d | 7. c | 8. c | 9. d | 10. c |
| 11. d | 12. c | 13. a | 14. b | 15. a |
| 16. b | 17. a | 18. c | 19. b | 20. d |

Exercise 2.2 (True/False)

- | | | | | |
|-----------|-----------|-----------|-----------|-----------|
| 1. True | 2. True | 3. False | 4. True | 5. False |
| 6. True | 7. True | 8. False | 9. True | 10. False |
| 11. True | 12. True | 13. True | 14. True | 15. False |
| 16. False | 17. False | 18. False | 19. False | 20. True |
| 21. True | 22. True | 23. True | 24. False | 25. False |
| 26. False | 27. False | 28. False | 29. True | 30. False |

Chapter 3: Random Variables

Exercise 3 (True/False)

- | | | | | |
|----------|----------|----------|---------|-----------|
| 1. False | 2. True | 3. False | 4. True | 5. True |
| 6. False | 7. False | 8. True | 9. True | 10. False |

Chapter 4: Discrete Probability Distributions

Exercise 4.1 (MCQs')

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. b | 2. c | 3. d | 4. a | 5. c |
| 6. b | 7. a | 8. d | 9. b | 10. d |
| 11. d | 12. d | 13. c | 14. a | 15. d |

Exercise 4.2 (True/False)

- | | | | | |
|----------|----------|---------|----------|----------|
| 1. False | 2. True | 3. True | 4. False | 5. False |
| 6. False | 7. False | 8. True | 9. True | 10. True |

Chapter 5: Continuous Probability Distributions

Exercise 5.1 (MCQs')

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. b | 2. d | 3. c | 4. c | 5. b |
| 6. b | 7. d | 8. d | 9. d | 10. b |
| 11. c | 12. c | 13. c | 14. b | 15. d |
| 16. b | 17. d | 18. c | 19. b | 20. b |

Exercise 5.2 (True/False)

- | | | | | |
|-----------|-----------|----------|-----------|----------|
| 1. False | 2. False | 3. True | 4. False | 5. True |
| 6. True | 7. True | 8. False | 9. True | 10. True |
| 11. False | 12. False | 13. True | 14. False | 15. True |

Chapter 6: Regression & Correlation

Exercise 6 (MCQs')

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. a | 2. c | 3. c | 4. b | 5. b |
| 6. a | 7. a | 8. c | 9. d | 10. c |
| 11. b | 12. c | 13. b | 14. c | 15. d |
| 16. b | 17. b | 18. d | 19. b | 20. b |

Chapter 7: Sampling

Exercise 7 (True/False)

- | | | | | |
|----------|----------|----------|-----------|-----------|
| 1. True | 2. False | 3. True | 4. False | 5. False |
| 6. False | 7. True | 8. True | 9. True | 10. True |
| 11. True | 12. True | 13. True | 14. False | 15. False |

Chapter 8: Statistical Inference

Exercise 8 (True/False)

- | | | | | |
|-----------|-----------|-----------|-----------|-----------|
| 1. False | 2. False | 3. True | 4. False | 5. True |
| 6. False | 7. True | 8. True | 9. True | 10. False |
| 11. True | 12. False | 13. True | 14. True | 15. True |
| 16. False | 17. False | 18. False | 19. True | 20. False |
| 21. True | 22. True | 23. False | 24. False | 25. False |
| 26. False | 27. True | 28. False | 29. True | 30. True |

Chapter 9: Design and Analysis of Experiments

Exercise 9 (True/False)

- | | | | | |
|----------|----------|----------|----------|----------|
| 1. False | 2. True | 3. True | 4. False | 5. True |
| 6. False | 7. False | 8. True | 9. False | 10. True |
| 11. True | 12. True | 13. True | 14. True | 15. True |

Chapter 10: Analysis of Time Series

Exercise 10 (MCQs')

- | | | | | |
|------|------|------|------|-------|
| 1. a | 2. c | 3. d | 4. b | 5. a |
| 6. a | 7. c | 8. b | 9. c | 10. c |

Chapter 11: Index Numbers

Exercise 11.1 (MCQs')

- | | | | | |
|------|------|------|------|-------|
| 1. c | 2. b | 3. d | 4. d | 5. b |
| 6. d | 7. b | 8. d | 9. b | 10. a |

Exercise 11.2 (True/False)

- | | | | | |
|----------|-----------|-----------|----------|-----------|
| 1. True | 2. False | 3. True | 4. False | 5. True |
| 6. True | 7. True | 8. True | 9. True | 10. True |
| 11. True | 12. False | 13. False | 14. True | 15. False |

Chapter 12: Nonparametric Statistics

Exercise 12.1 (MCQs')

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. a | 2. c | 3. c | 4. b | 5. b |
| 6. c | 7. c | 8. c | 9. a | 10. b |
| 11. d | 12. d | 13. c | 14. b | 15. b |
| 16. d | 17. b | 18. d | 19. a | 20. b |
| 21. a | 22. b | 23. d | | |

Exercise 12.2 (True/False)

- | | | | | |
|----------|-----------|-----------|-----------|----------|
| 1. True | 2. False | 3. True | 4. True | 5. True |
| 6. True | 7. False | 8. True | 9. True | 10. True |
| 11. True | 12. False | 13. False | 14. False | 15. True |



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